Western and Central Pacific Fisheries Commission

Workflow demonstration

Assessment workflow demo: ALB

WORKFLOW DEMO

2026-09-06

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1 Executive summary

This is a workflow demonstration, not a stock assessment. It uses ALB outputs to demonstrate the model-to-report tools. The diagnostic-model figures use the archived MFCL-compatible KIMSA fit from job 24781. Stepwise compares completed native MFCL runs (INI 1003 and INI 1007) with the same KIMSA baseline. Starting-value, self-test, retrospective, sensitivity, ensemble, profile, projection, hindcast and additional-output figures use illustrative display values to demonstrate report construction. They are not new assessment estimates.

Table 1 links the report and all interactive supplements. Replace the illustrative study bundles with fitted results, write the assessment interpretation in the section files, and update the report cover before submission.

2 Introduction

3 Data compilation

The diagnostic model retains the ALB observation timing and two-region structure. Catch, abundance-index, length-composition and conditional age-at-length observations are available in the interactive diagnostic supplement.

Region boundaries follow the 2024 South Pacific albacore assessment, Rev.03, Section 4.1 and Figure 2. The spatial comparison also shows its Figure 1 reproduction of the 2021 structure.

Table 2 describes the fishery roles and observation periods. Figure 6 shows data availability. Index observations and predictions follow the model likelihood's geometric-mean centering convention.

4 Model description

The archived ALB fit uses the MFCL-compatible age-region state, age-based selectivity and five-pass hybrid F, without tagging. Its original F upper bound of 3 is retained for reproduction; new model specifications default to 6.

Length compositions use square_fita with fishery divisors and a raw-record sample-size cap of 1000.

Figure 10, Figure 11 and Figure 12 describe the biological inputs and estimates; Figure 39 shows fishery selectivity. Additional-format panels illustrate output types that are not estimated by this tag-free ALB reference model.

5 Model development

Figure 13 compares the latest completed matching ALB baselines found in Kflow: native MFCL 2.2.7.0 / INI 1003 (job 24514), native MFCL 2.2.7.9 / INI 1007 (job 24515), then KIMSA / hybrid F (job 24781). All used TunaFlow 2.5. These panels use the archived annual report censuses; they are distinct from the time-weighted annual summaries of the detailed diagnostic snapshot.

6 Results

Figure 38 shows annual reference-model trajectories. Stocks use time-weighted annual means; recruitment and catches use annual totals. Depletion is the ratio of annual spawning-potential summaries. Index points and predictions use annual geometric means. The interactive viewer opens annual summaries; select Original to inspect the fitted time steps.

Length-composition fits (Figure 16; Figure 19) show observed sample counts as bars and predictions as lines, pooled over years on the original full-bin support. Length-residual panels retain the fitted records and compressed-tail support. Conditional age-at-length fits are pooled by region using raw sample counts, preserving the model's prediction scale.

Index fits and residuals (Figure 15), length-composition residuals and conditional age-at-length fits (Figure 25) are generated from the same result bundle used by the interactive viewer.

The following studies demonstrate the intended diagnostic presentation: starting-value trials (Figure 26), simulation-estimation recovery (Figure 29), retrospective analysis (Figure 31), objective profiles (Figure 32) and sensitivity comparisons. Their displayed values are illustrative. Hindcasting (Figure 72; Table 14) uses annual observed indices, five annual rolling training cutoffs, and forecasts up to three years ahead. The demonstration fits log-linear index trends independently at each cutoff; exponentiated predictions are conditional medians. MASE uses only training naive errors for scaling. These are not refitted stock-assessment models.

7 Reporting examples

The ensemble and management-quantity panels demonstrate the reporting formats. Projection panels compare fixed-catch and F scenarios using a simple surplus-production kernel initialized at the ALB biomass scale; its dynamics and variability are illustrative. They do not establish ALB stock status. Supply actual reference points, model retention decisions, uncertainty draws and projection results before writing management conclusions.

8 Discussion

9 Tables

Table 1: Public report, interactive supplements and published outputs for this workflow demonstration.

Resource	Figures and tables	Interactive output
Workflow demonstration	PDF report	HTML report
Diagnostic fit	HTML supplement	Model viewer
Stepwise development	HTML supplement	Model viewer
Jitter	HTML supplement	Model viewer
Self-test	HTML supplement	Model viewer
Retrospective analysis	HTML supplement	Model viewer
Sensitivities	HTML supplement	Model viewer
Uncertainty ensemble	HTML supplement	Model viewer
Objective profiles	HTML supplement	Model viewer
Additional output examples	HTML supplement	Model viewer
Hindcasting	HTML supplement	Model viewer
Projection scenarios	HTML supplement	Model viewer
Published outputs	Public repository	Rendered artifacts

Table 2: Fishery definitions and observation units.

Fishery	Region	Role	First year	Last year	Unit
1	1	Catch	1954	2022	fish
2	1	Catch	1954	2022	fish
3	1	Catch	1954	2022	fish
4	1	Catch	1954	2022	fish
5	1	Catch	1992	2022	fish
6	1	Catch	1982	2022	fish
7	1	Catch	1987	2022	fish
8	1	Catch	1954	2022	fish
9	1	Catch	1984	2022	fish

Fishery	Region	Role	First year	Last year	Unit
10	1	Catch	1985	2022	fish
11	1	Catch	1982	2022	t
12	1	Catch	1987	2022	t
13	1	Catch	1983	1990	t
14	2	Catch	1957	2022	fish
15	2	Catch	1961	2021	fish
16	2	Catch	1963	2021	fish
17	2	Catch	1986	2006	t
18	1	Index	1954	2022	relative index
19	1	Index	1992	2022	relative index
20	2	Index	1954	2022	relative index

Table 3: Illustrative output. Dirichlet–multinomial composition settings.

Fishery	Likelihood	Group	Theta
Fishery A	dirichlet_multinomial	1	20

Table 4: Illustrative output. Groups sharing composition-weight parameters.

Group	Fishery
1	Fishery A

Table 5: Illustrative output. Tag-release and reporting groups.

Programme	Region	Mixing
Programme A	Region 1	Two supplied model periods

Table 6: Illustrative output. Parameters on or near their declared bounds.

Parameter	Estimate	Lower	Upper
Steepness	0.99	0.2	0.99

Table 7: Illustrative output. Parameter correlations from the supplied covariance estimate.

Parameter	Partner	Correlation
log R0	Index q	-0.97
Initial abundance	Recruitment scale	0.96

Table 8: Management quantities and supplied uncertainty summaries.

Quantity	Period	Median	50% interval	80% interval	95% interval
$SB/SB_{F=0}$	2016–2020	0.45	0.41–0.49	0.37–0.53	0.33–0.57
F/F_{MSY}	2016–2020	0.68	0.61–0.75	0.54–0.82	0.46–0.90
SB/SB_{MSY}	2016–2020	1.35	1.22–1.48	1.08–1.62	0.94–1.76

Table 9: Spawning-depletion comparisons for the declared reference periods.

Period	Reference	Ratio
2016–2020	2012–2015	0.95

Table 10: Stock-status probabilities for the declared reference points.

Condition	Probability
$SB/SB_{F=0} < 0.20$	0.01
$F/F_{MSY} > 1$	0.05
$SB/SB_{MSY} < 1$	0.04

Table 11: Model reference points and their definitions.

Quantity	Median	Lower	Upper	Unit
MSY	60000	48000	72000	t/year
F_{MSY}	0.3	0.24	0.36	per year
SB_{MSY}	1000000	800000	1200000	t
SB_{MSY}/SB_0	0.28	0.22	0.34	ratio

Table 12: Comparison of supplied uncertainty summaries.

Source	Quantity	Median	Lower	Upper
Structural	$SB/SB_{F=0}$	0.45	0.39	0.51
Structural	F/F_{MSY}	0.68	0.57	0.79
Structural and estimation	$SB/SB_{F=0}$	0.45	0.35	0.55
Structural and estimation	F/F_{MSY}	0.68	0.51	0.85

Table 13: Illustrative projected quantities in the final projection year; medians and central 80% intervals.

Scenario	Quantity	Year	Median	Lower	Upper	Unit
fixed-catch	biomass depletion	2052	0.9499	0.9473	0.952	ratio
fixed-catch	catch	2052	45150	45150	45150	t
fixed-catch	fishing mortality	2052	0.01518	0.01514	0.01522	per year
fixed-catch	total biomass	2052	2997000	2989000	3004000	t
higher-F	biomass depletion	2052	0.7427	0.7378	0.7519	ratio
higher-F	catch	2052	180200	179000	182400	t
higher-F	fishing mortality	2052	0.08	0.08	0.08	per year
higher-F	total biomass	2052	2343000	2328000	2372000	t
lower-F	biomass depletion	2052	0.8678	0.8609	0.8719	ratio
lower-F	catch	2052	107400	106500	107900	t
lower-F	fishing mortality	2052	0.04	0.04	0.04	per year
lower-F	total biomass	2052	2738000	2716000	2751000	t

Table 14: Held-out prediction bias (predicted minus observed), RMSE and MASE by series and forecast lead. N counts scored forecasts; MASE uses each cutoff's training naive-error scale.

Series	Lead	N	Bias	RMSE	MASE	Unit
Index 18 (Region 1)	1	5	-0.08847	0.10778	0.39718	relative
Index 18 (Region 1)	2	4	-0.095363	0.117	0.42491	relative
Index 18 (Region 1)	3	3	-0.093805	0.12075	0.41723	relative
Index 19 (Region 1)	1	5	-0.0691	0.18227	0.52469	relative
Index 19 (Region 1)	2	4	-0.0245	0.13014	0.41413	relative
Index 19 (Region 1)	3	3	-0.026698	0.1508	0.49854	relative

Series	Lead	N	Bias	RMSE	MASE	Unit
Index 20 (Region 2)	1	5	-0.2195	0.25108	0.84494	relative
Index 20 (Region 2)	2	4	-0.26202	0.28574	0.99863	relative
Index 20 (Region 2)	3	3	-0.30353	0.32467	1.1481	relative

10 Figures

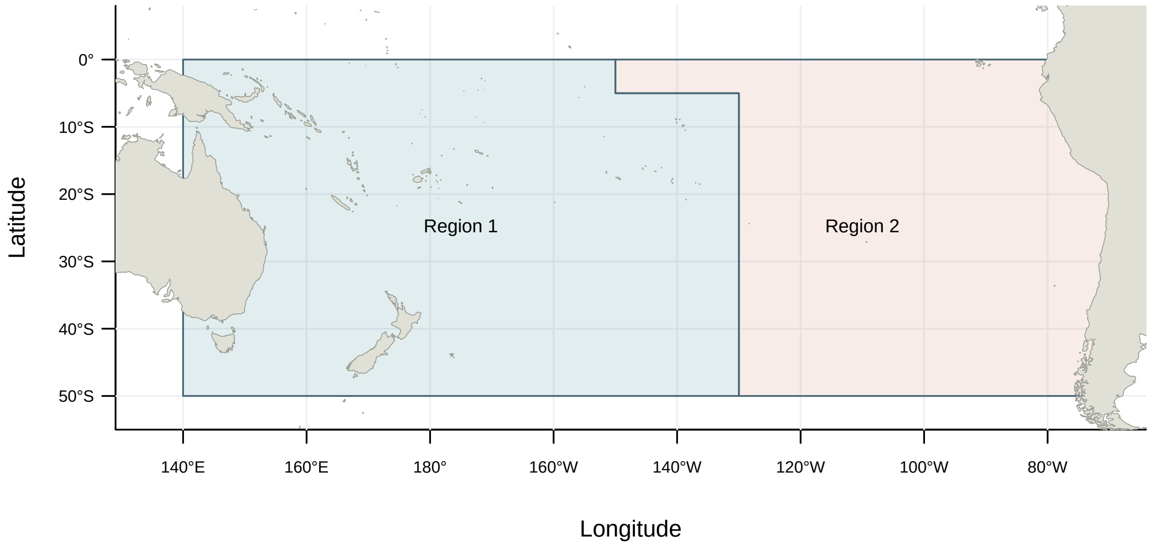

Figure 1: Model regions for the 2024 South Pacific albacore assessment (WCPFC-SC20-SA-WP-02, Rev.03, Figure 2).

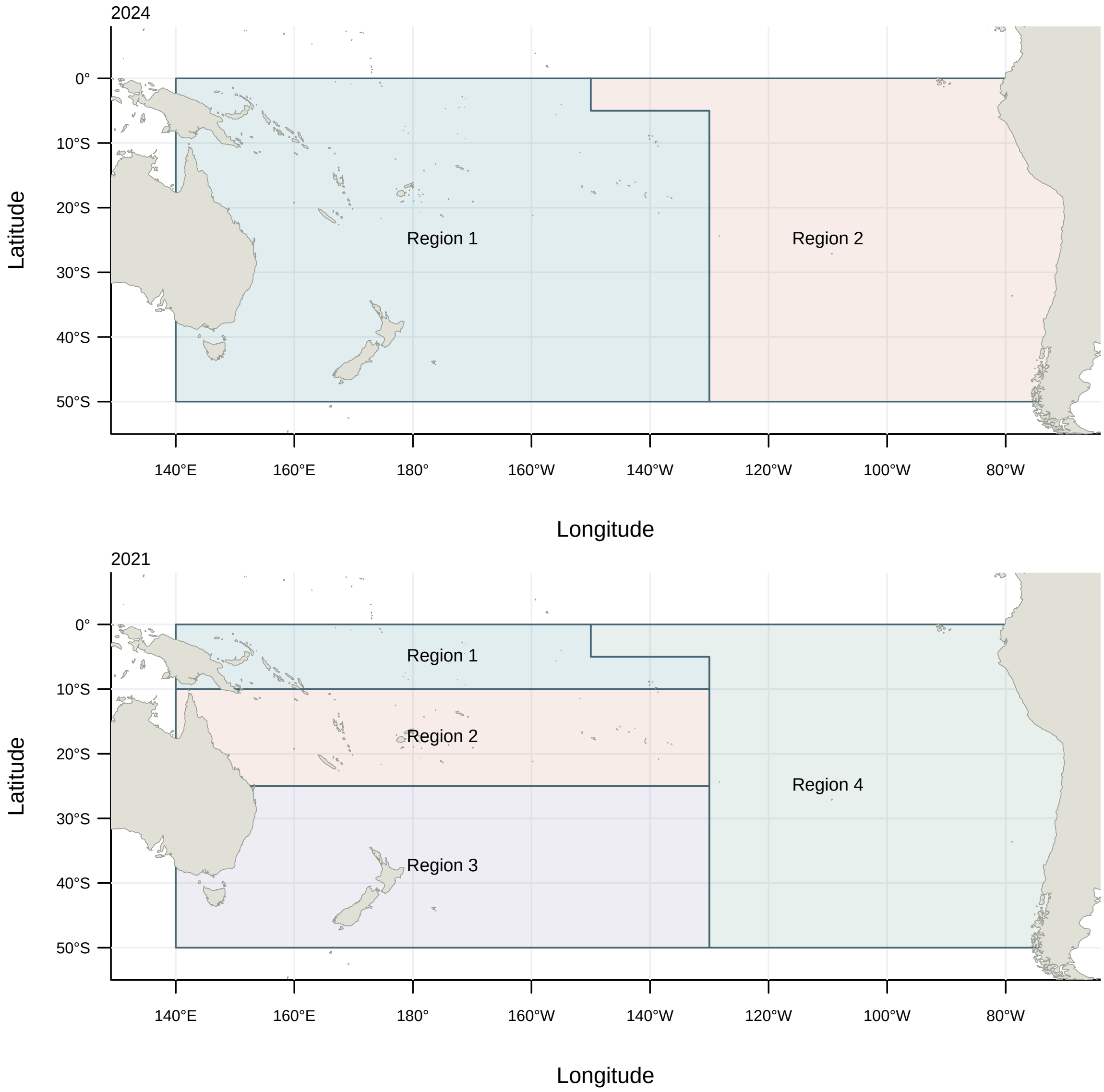

Figure 2: Spatial structures of the 2024 and 2021 South Pacific albacore assessments (WCPFC-SC20-SA-WP-02, Rev.03, Figures 1–2).

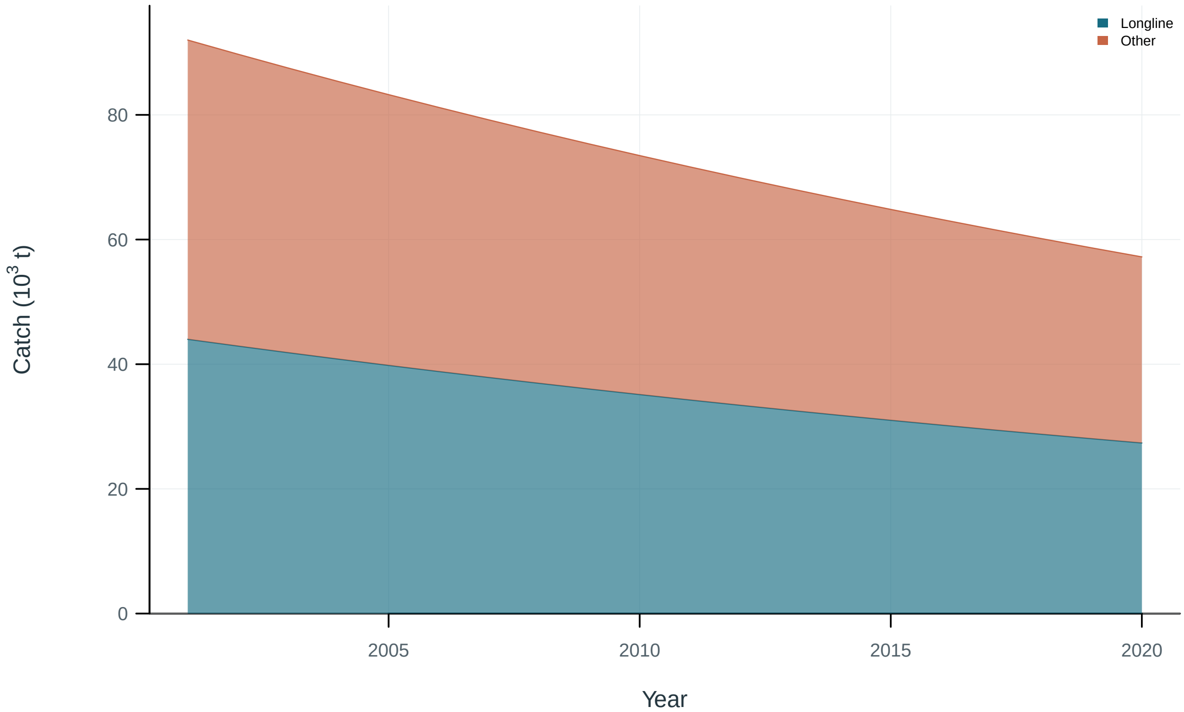

Figure 3: Illustrative output. Catch by fishing group.

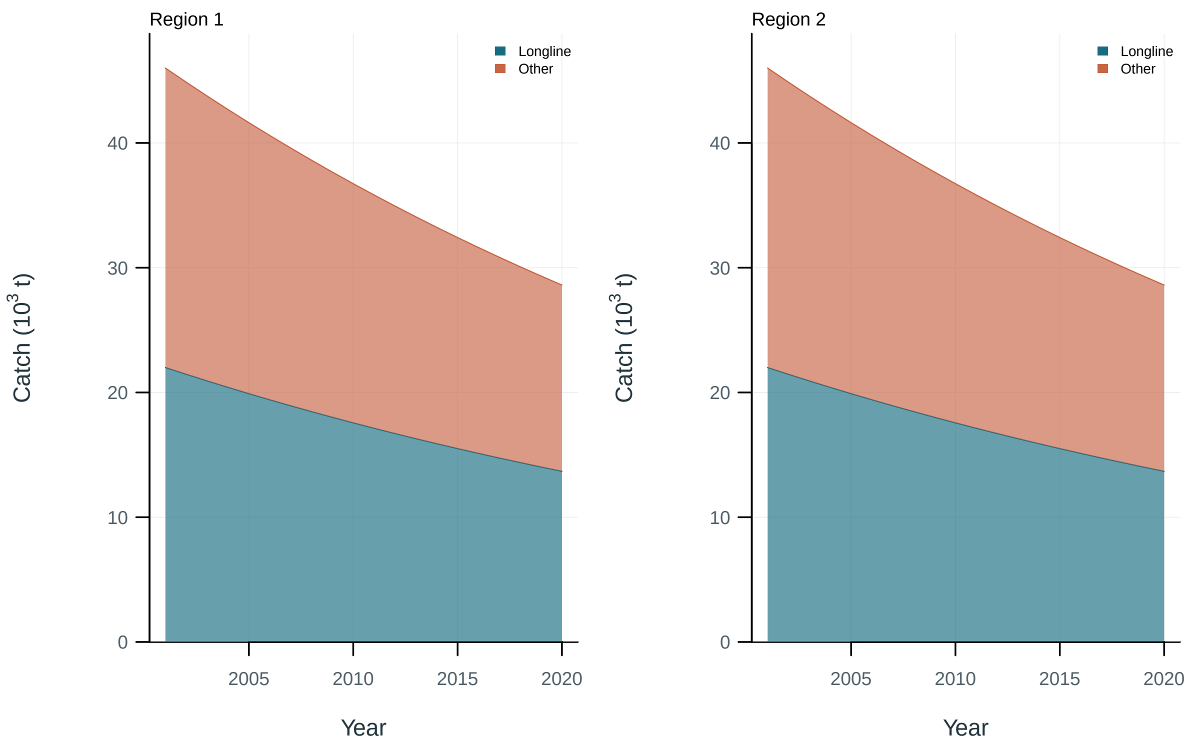

Figure 4: Illustrative output. Catch by fishing group and region.

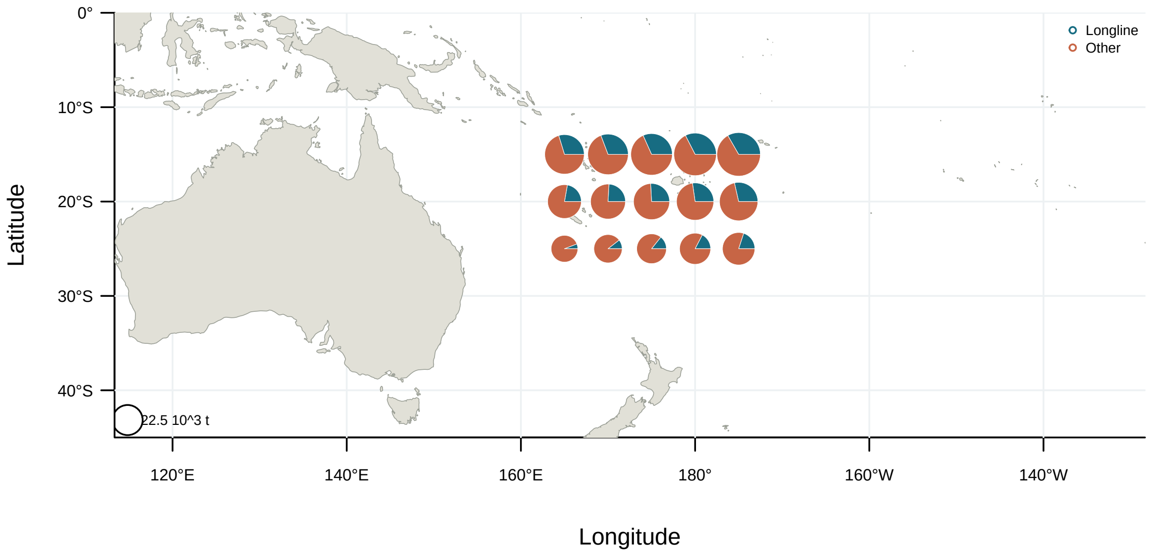

Figure 5: Illustrative output. Spatial catch distribution by fishing group.

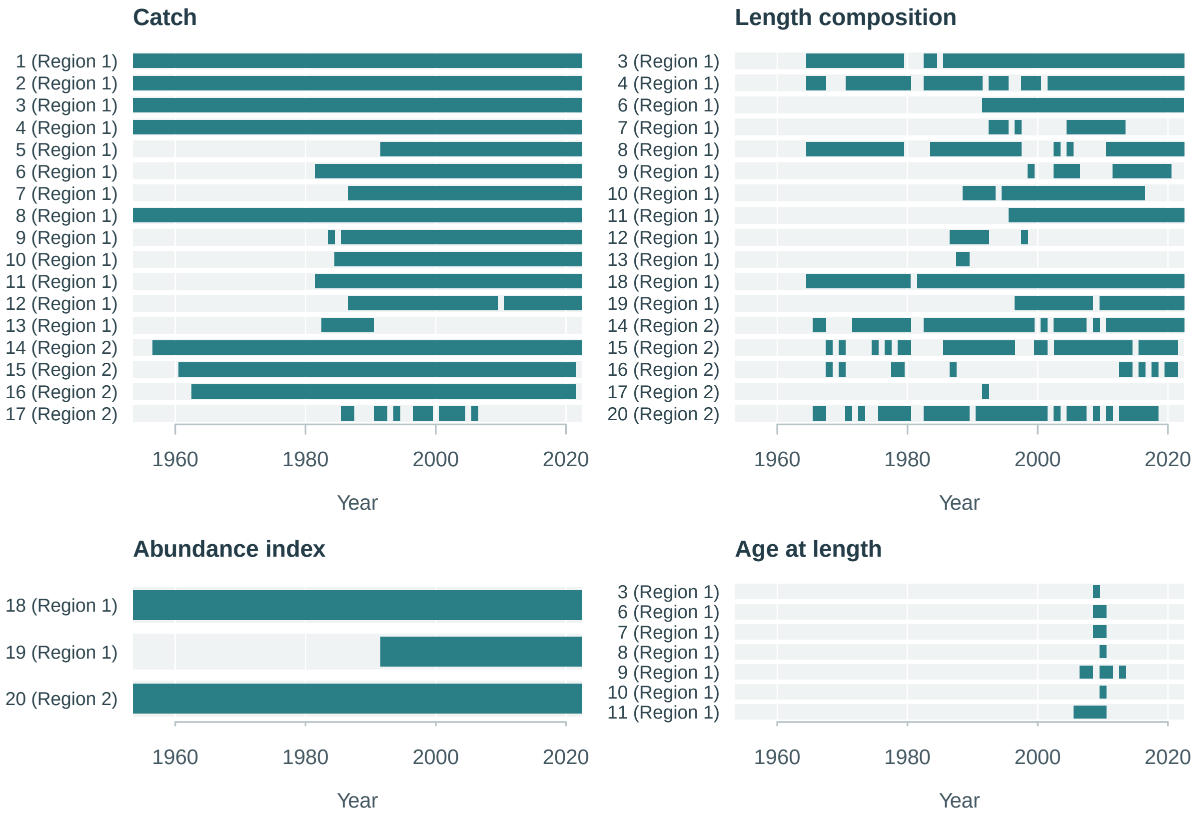

Figure 6: Observation availability by data type and fishery. Coloured cells indicate years with observations.

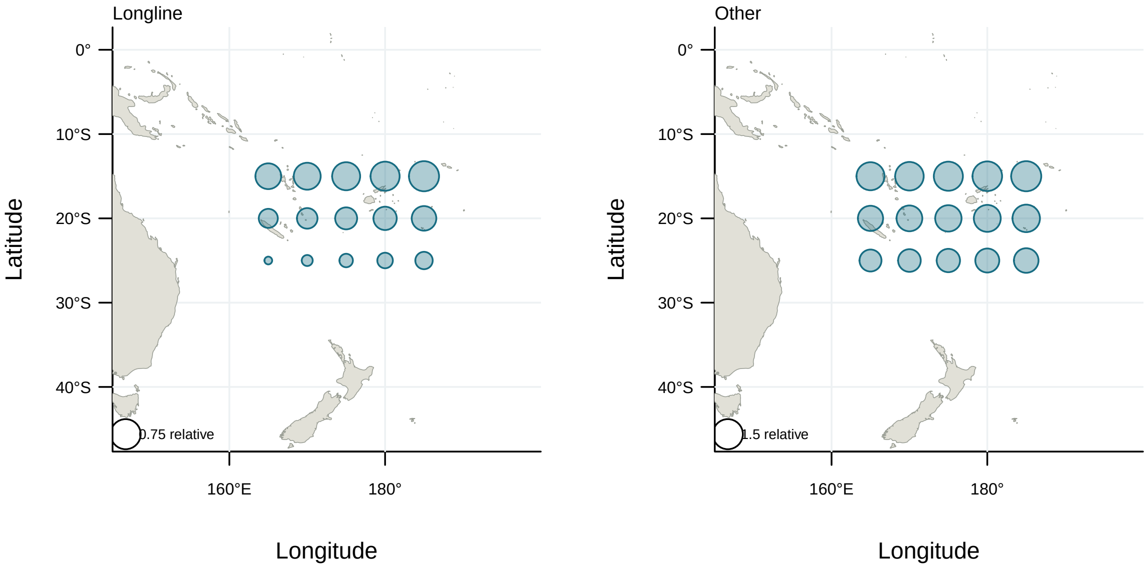

Figure 7: Illustrative output. Spatial distribution of nominal abundance indices.

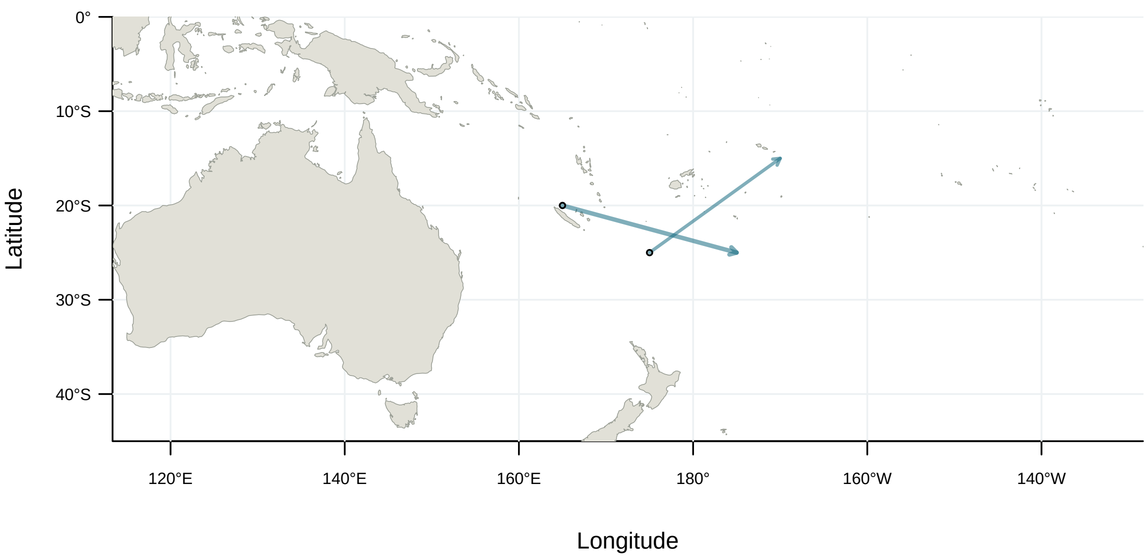

Figure 8: Illustrative output. Tag-release and recapture displacements by programme.

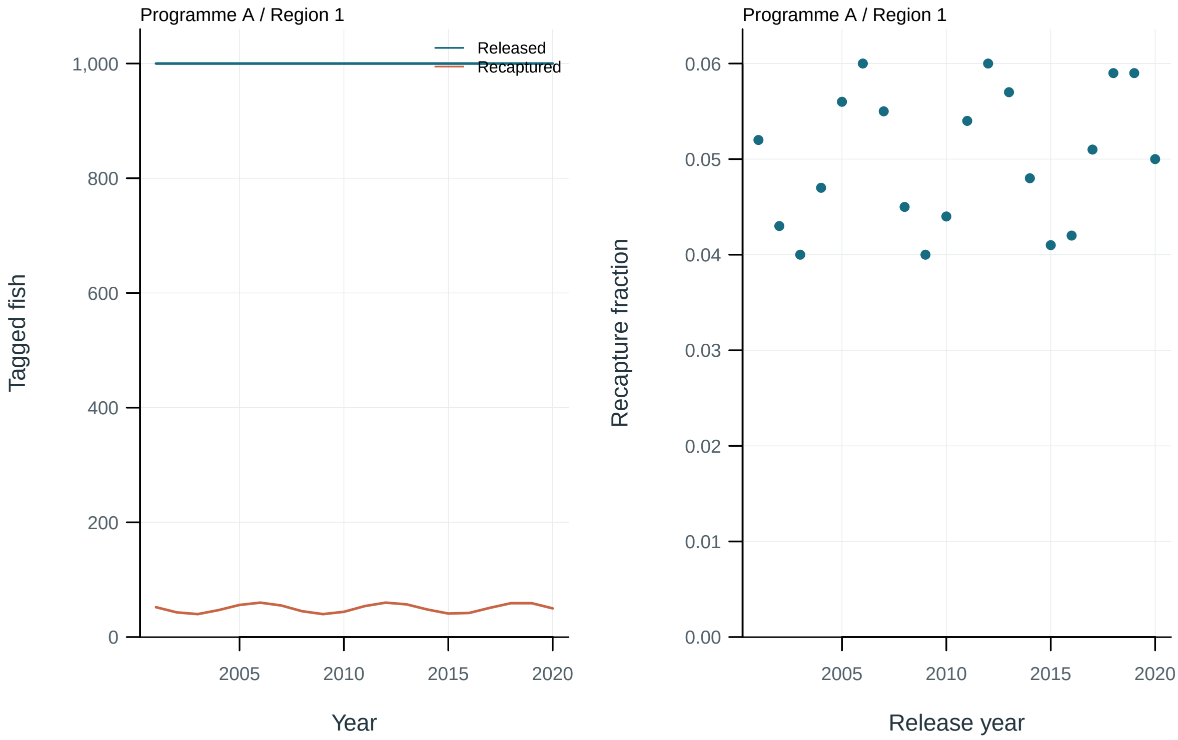

Figure 9: Illustrative output. Tag releases, recaptures and recapture fractions by cohort.

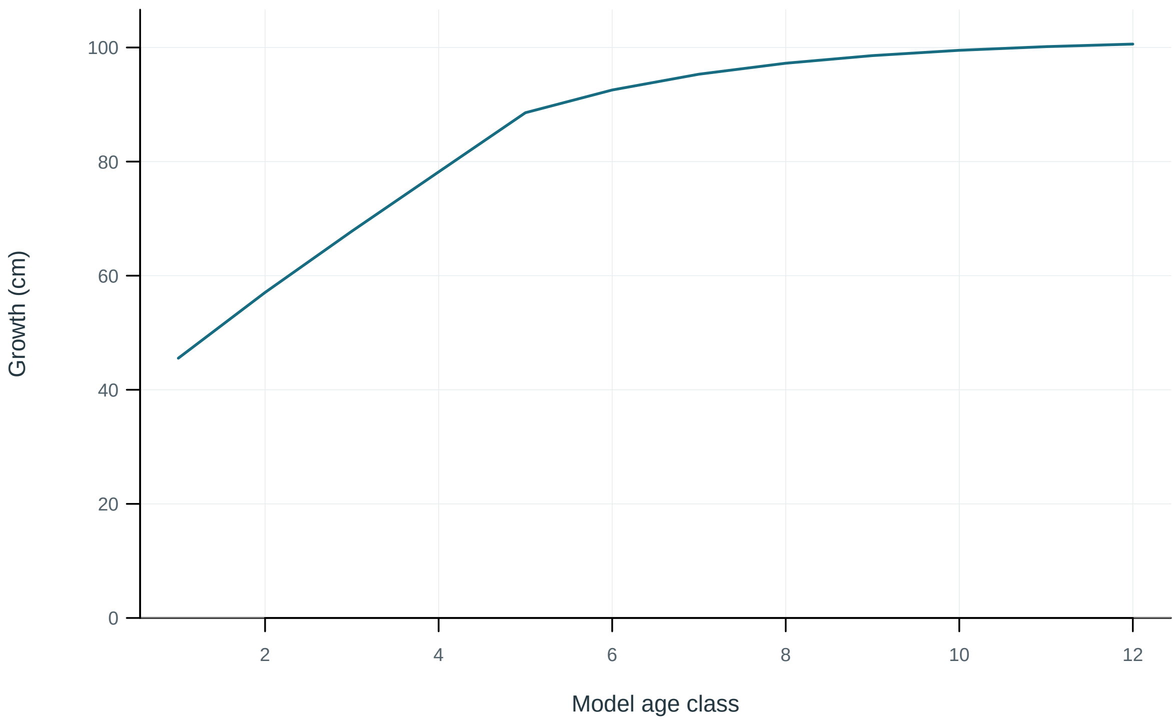

Figure 10: Mean length at the start of each model age class.

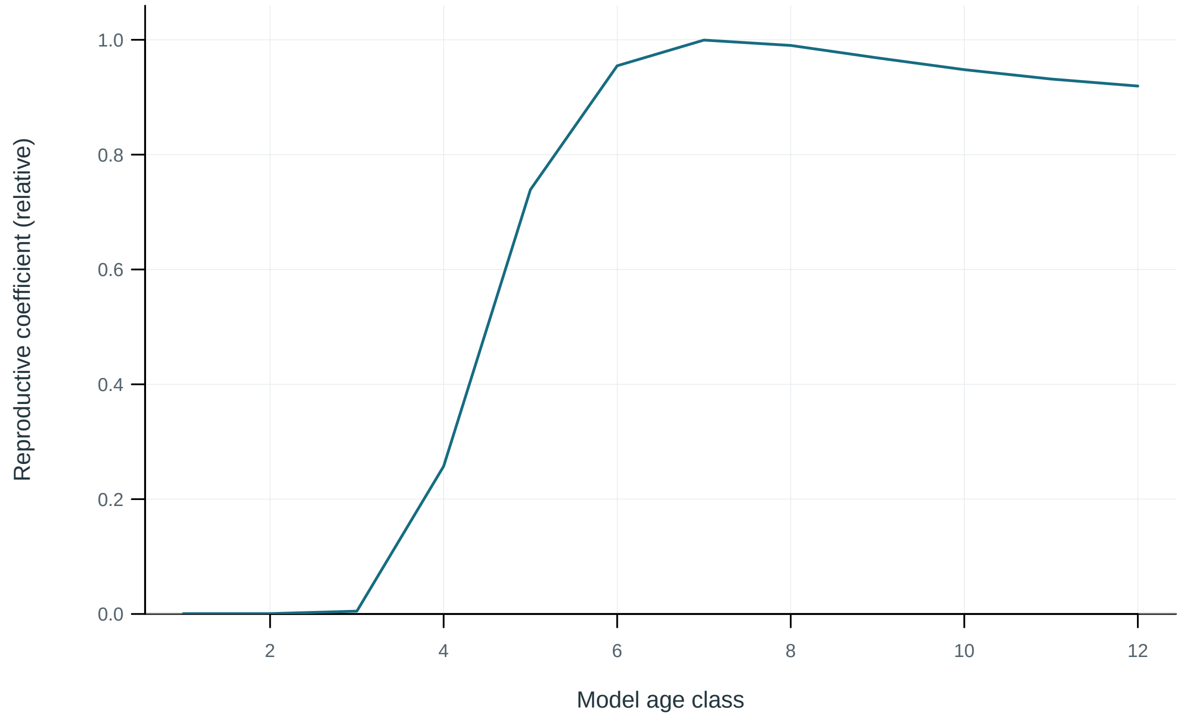

Figure 11: Model reproductive coefficient at age.

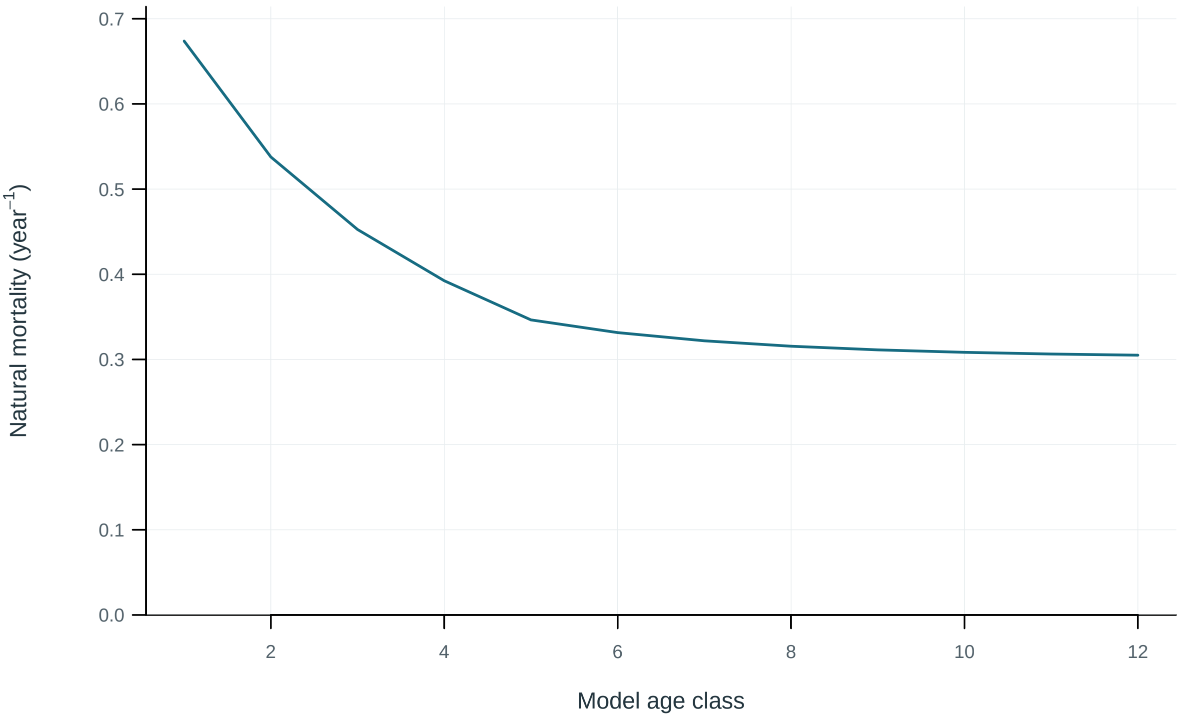

Figure 12: Annual instantaneous natural mortality at age.

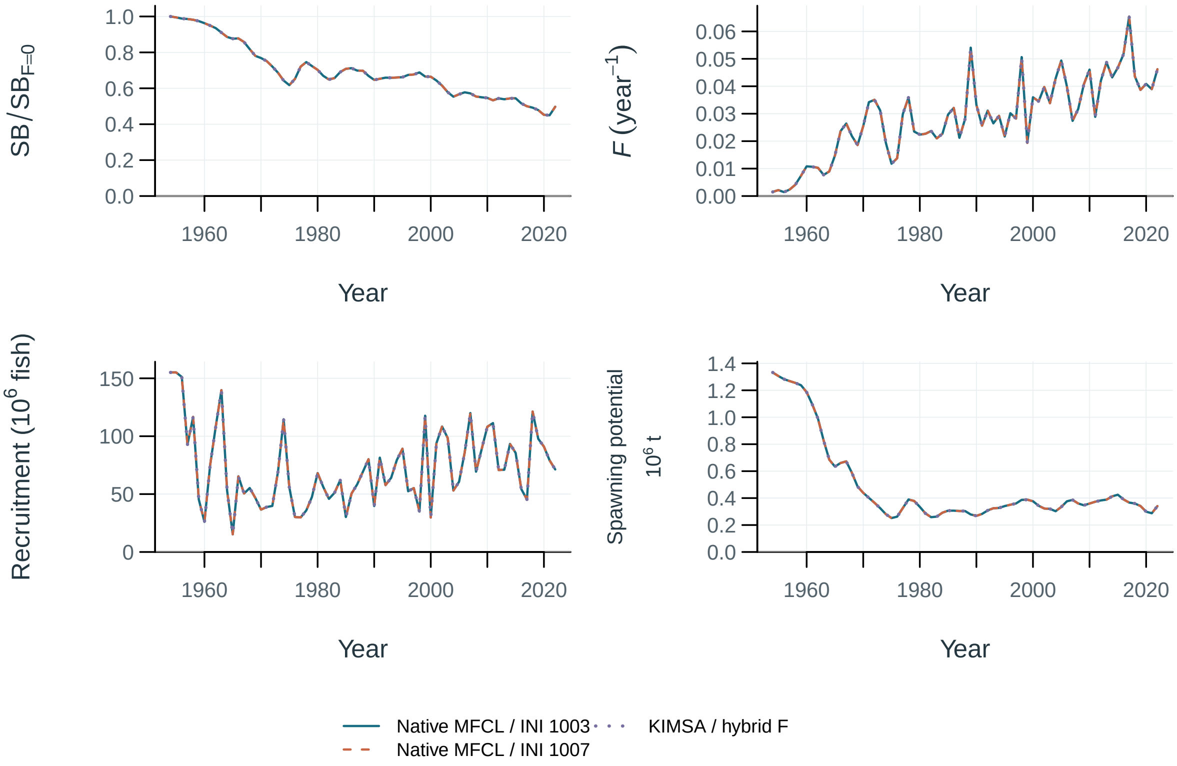

Figure 13: Stepwise model trajectories.

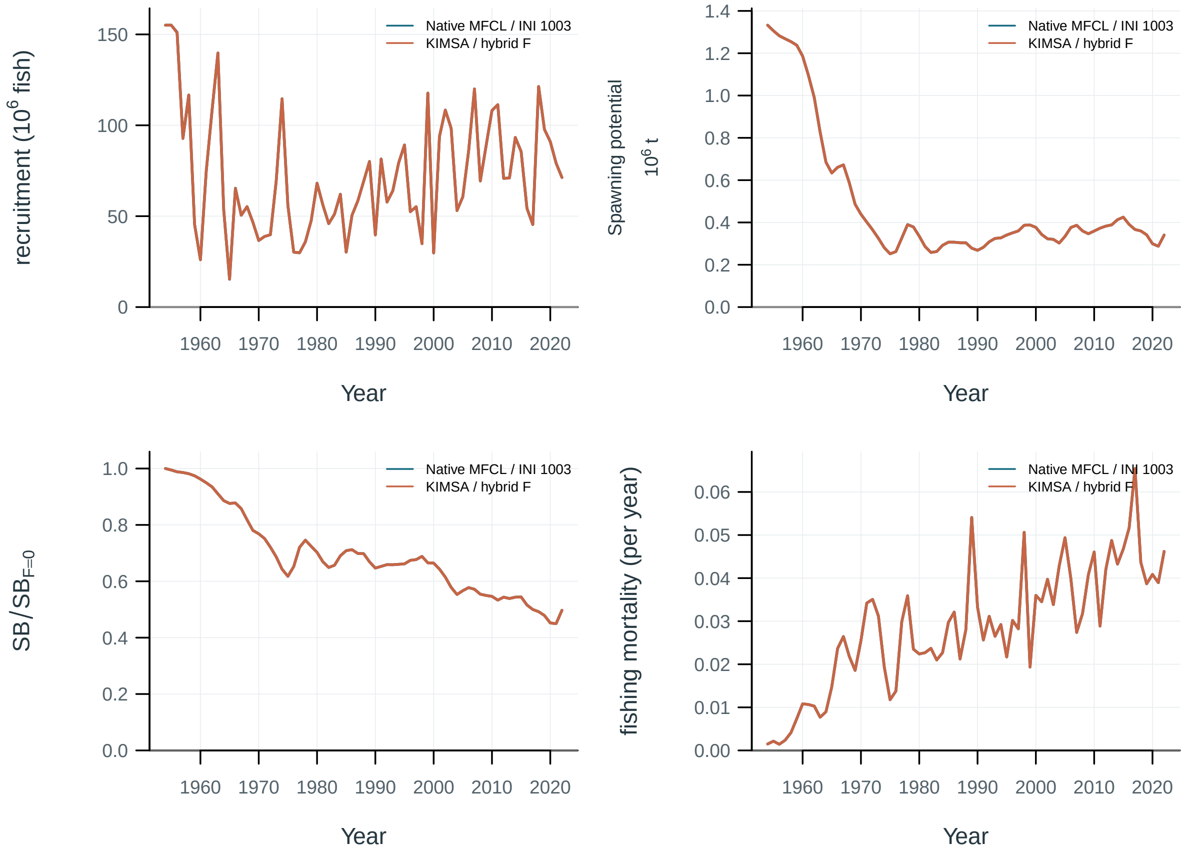

Figure 14: Selected model-development trajectories.

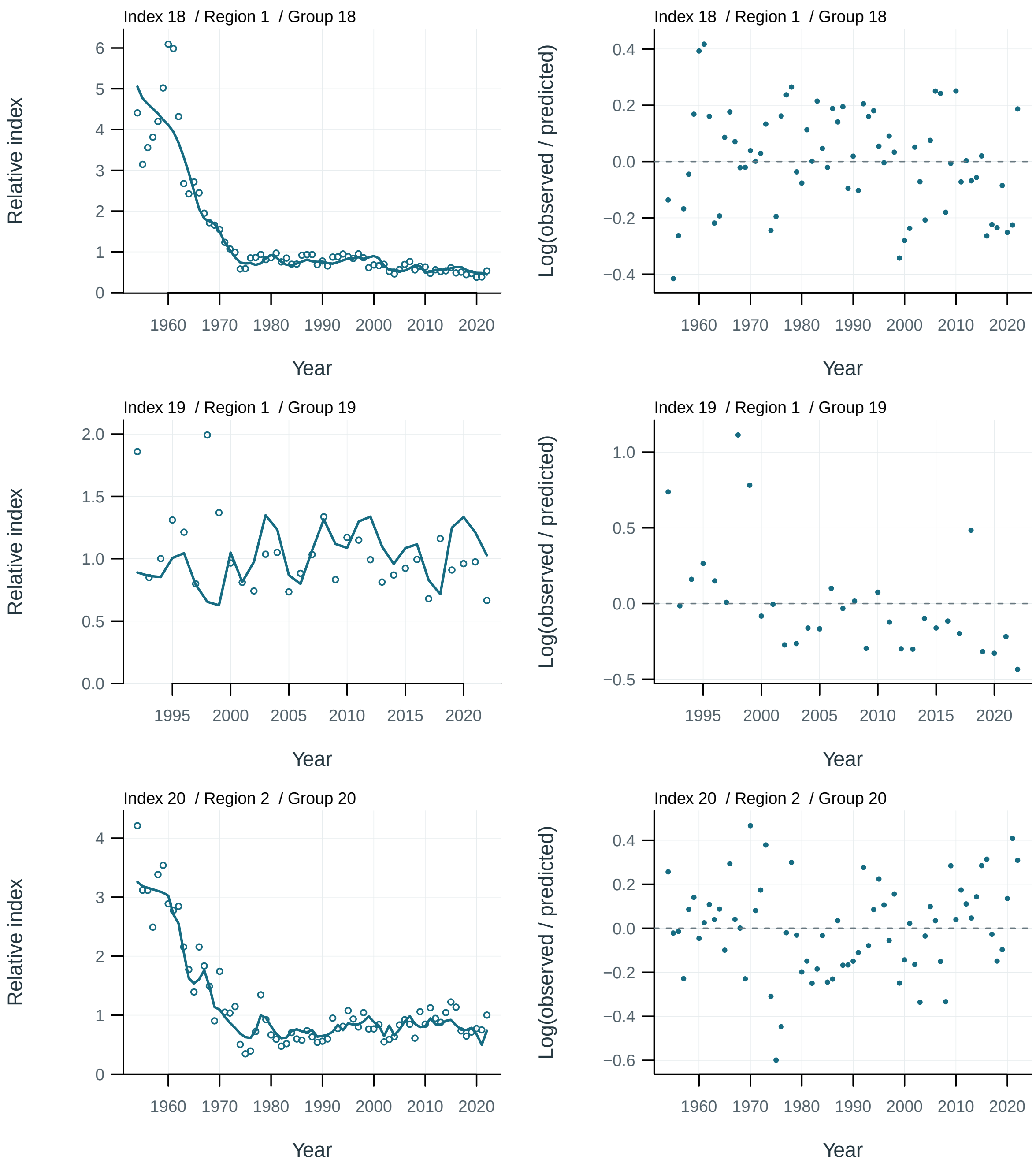

Figure 15: Abundance-index fits and log residuals by series. Residuals are log(observed/predicted).

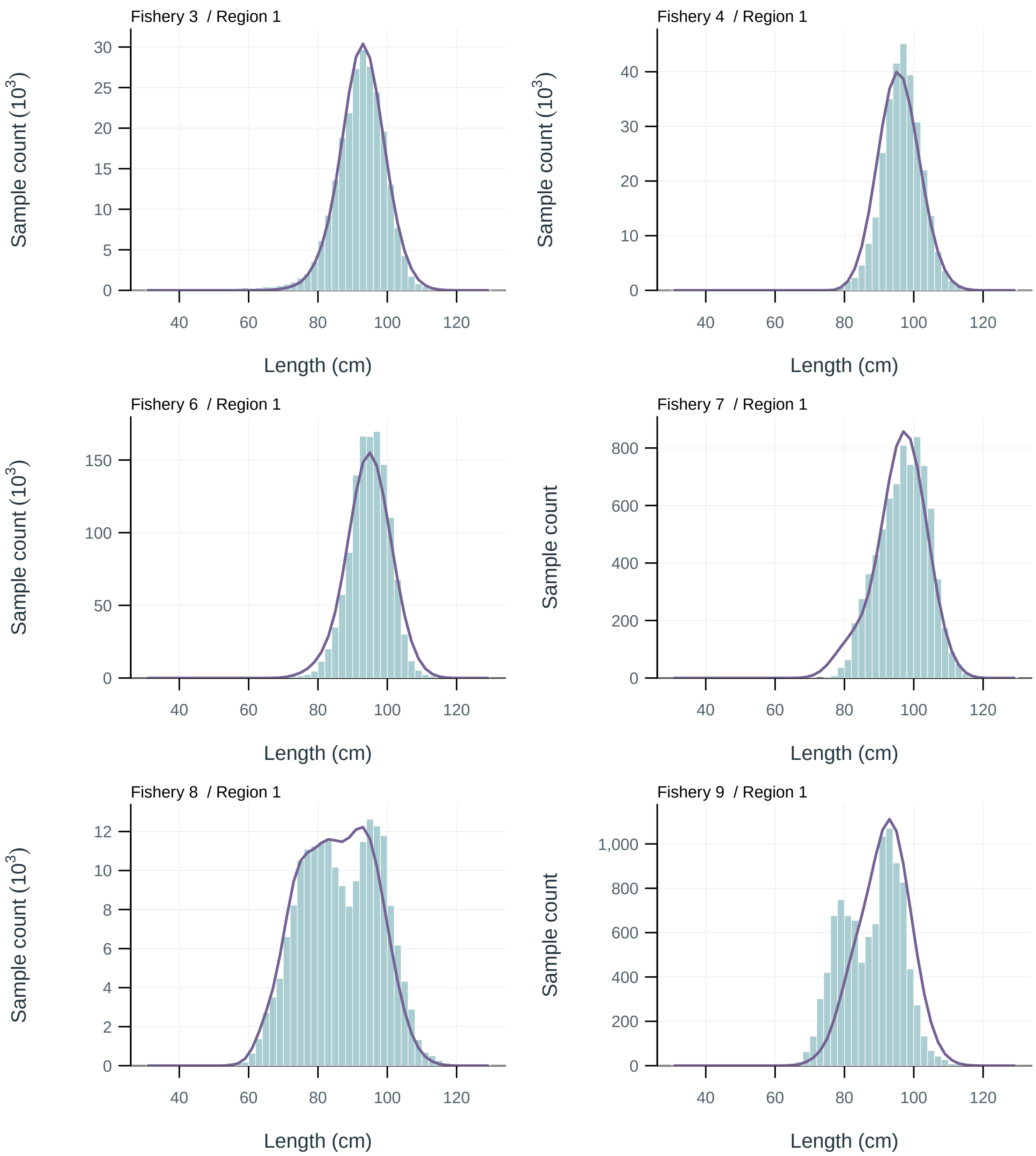

Figure 16: Length-composition fits by fishery. Observed sample counts (bars) and predictions (lines), pooled over years.

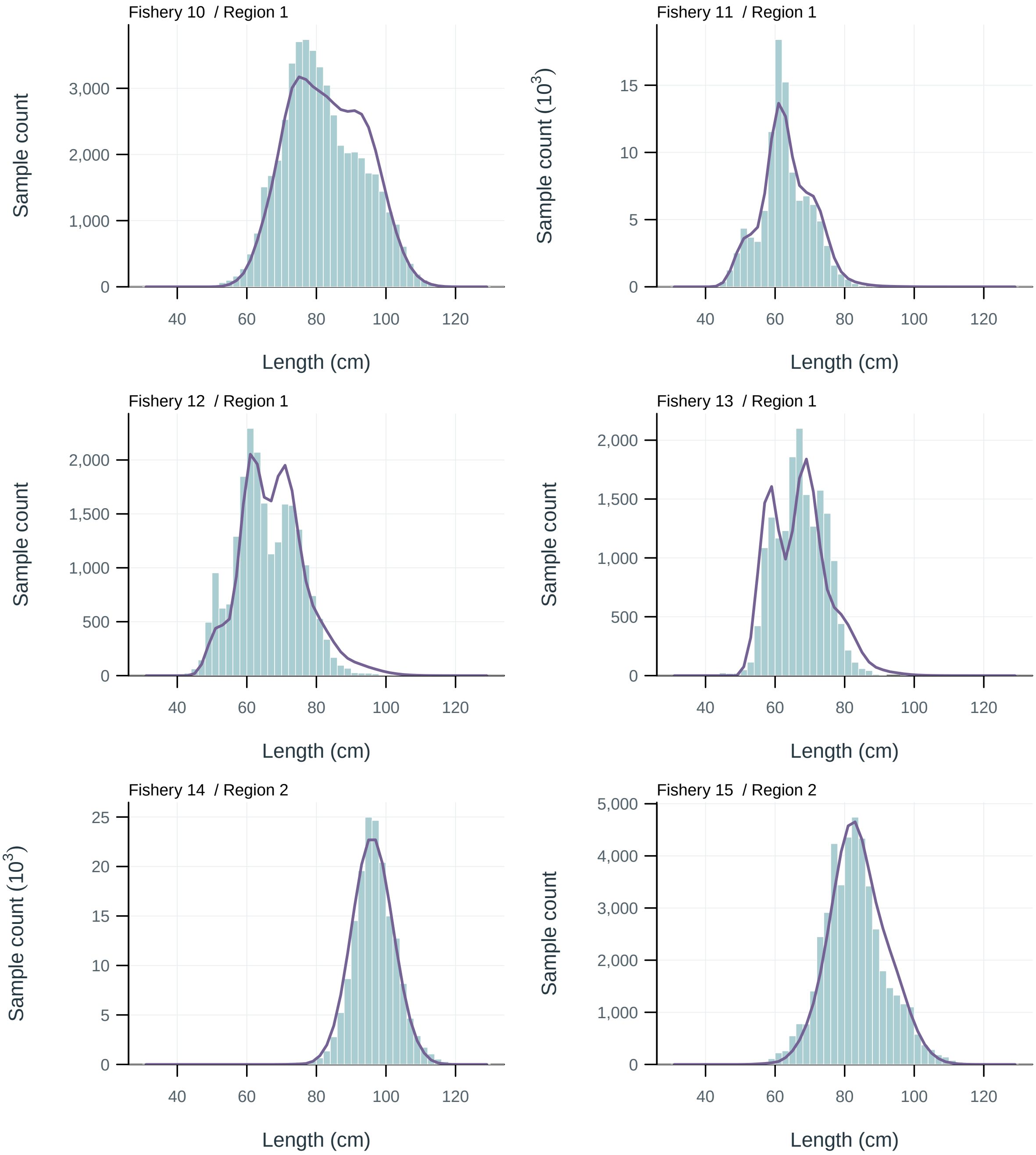

Figure 17: Length-composition fits by fishery. Observed sample counts (bars) and predictions (lines), pooled over years.

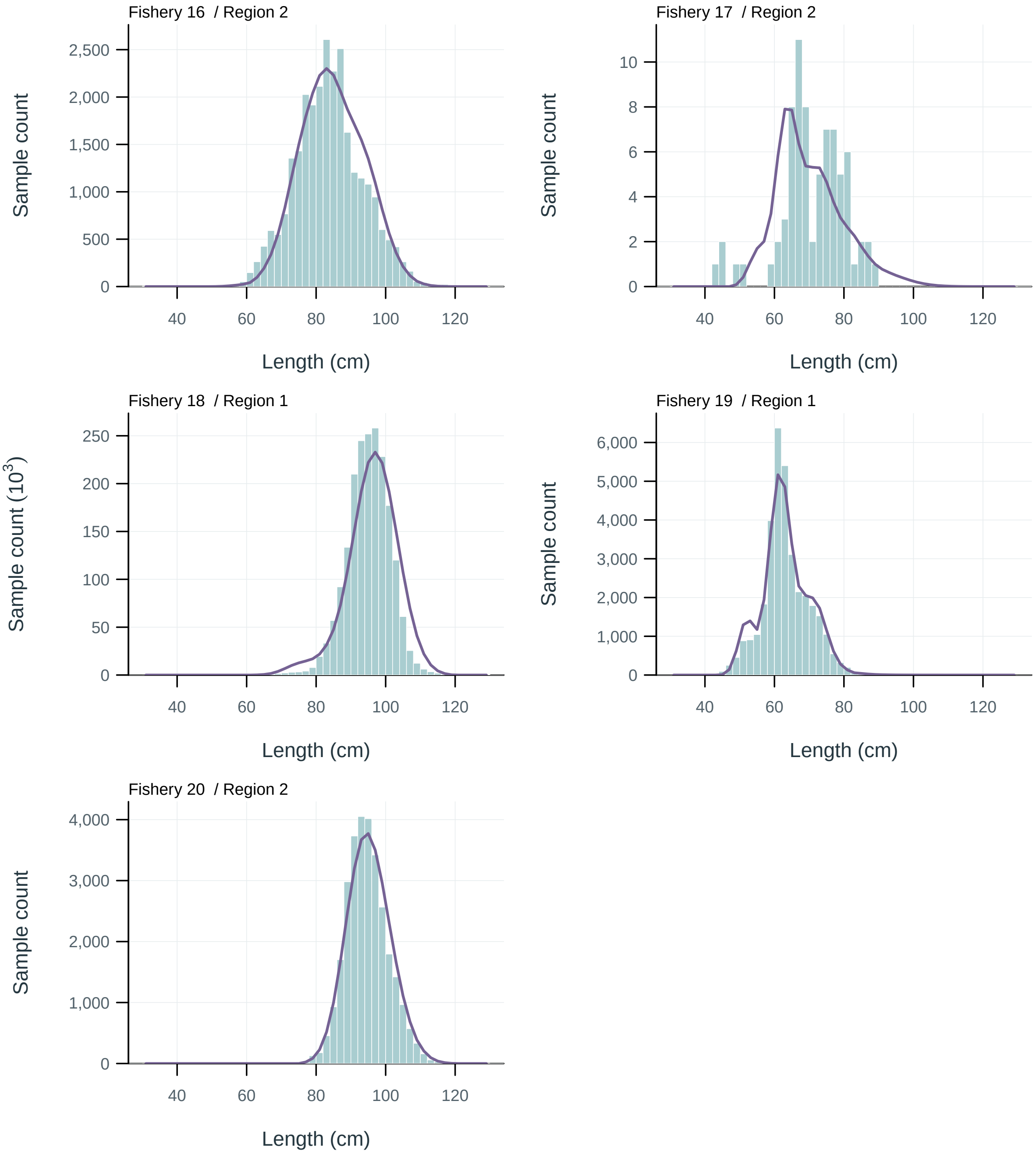

Figure 18: Length-composition fits by fishery. Observed sample counts (bars) and predictions (lines), pooled over years.

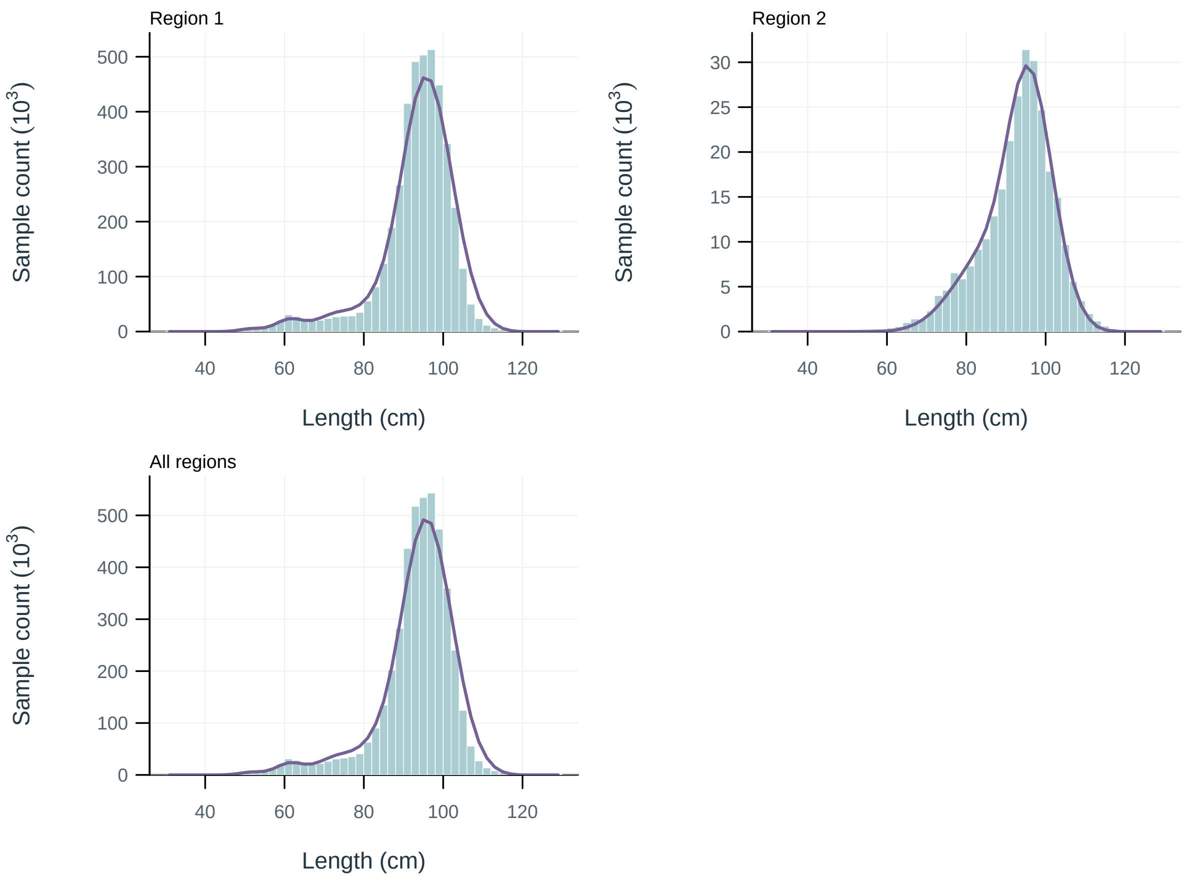

Figure 19: Length-composition fits by region. Observed sample counts (bars) and predictions (lines), pooled over years.

Figure 20: Illustrative output. Observed and predicted length summaries for longline fisheries.

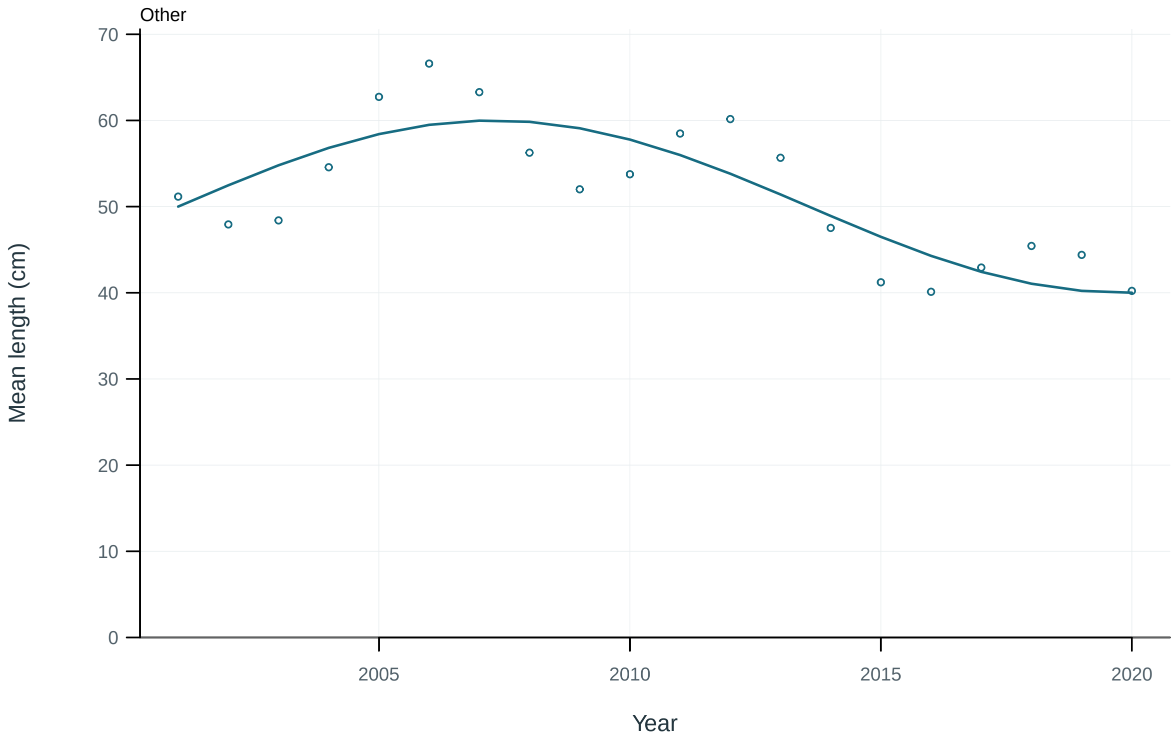

Figure 21: Illustrative output. Observed and predicted length summaries for other fisheries.

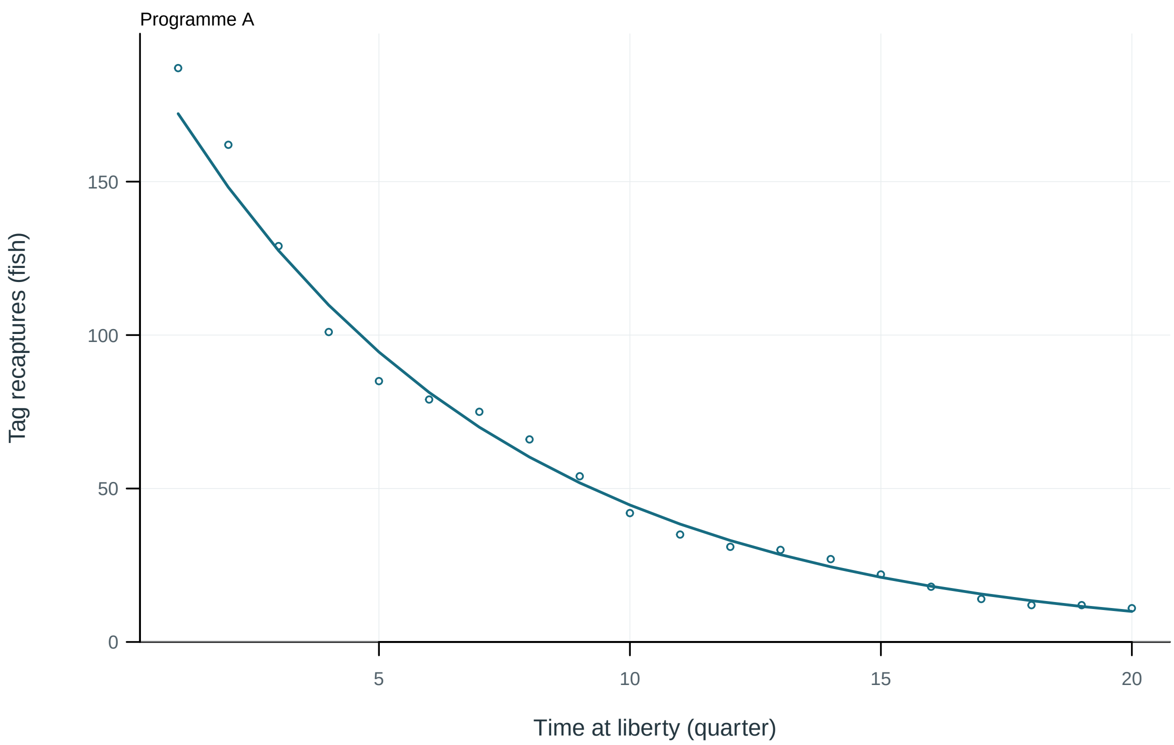

Figure 22: Illustrative output. Observed and predicted tag recaptures by time at liberty.

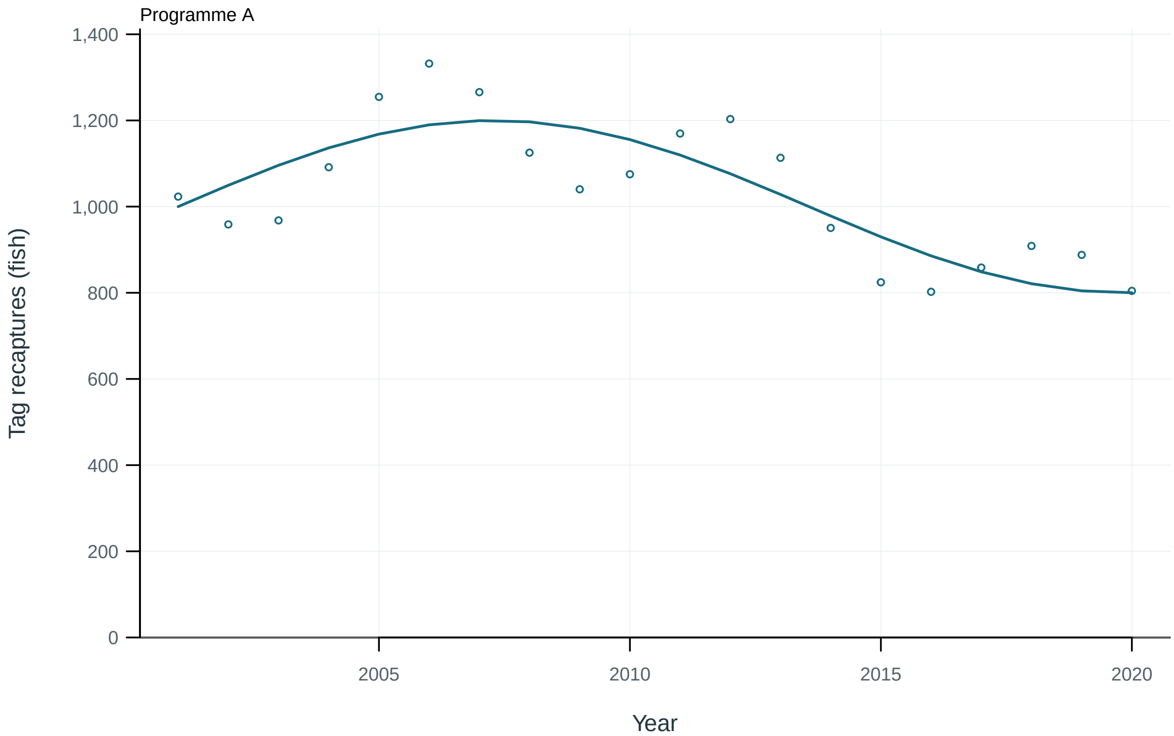

Figure 23: Illustrative output. Observed and predicted tag recaptures through time.

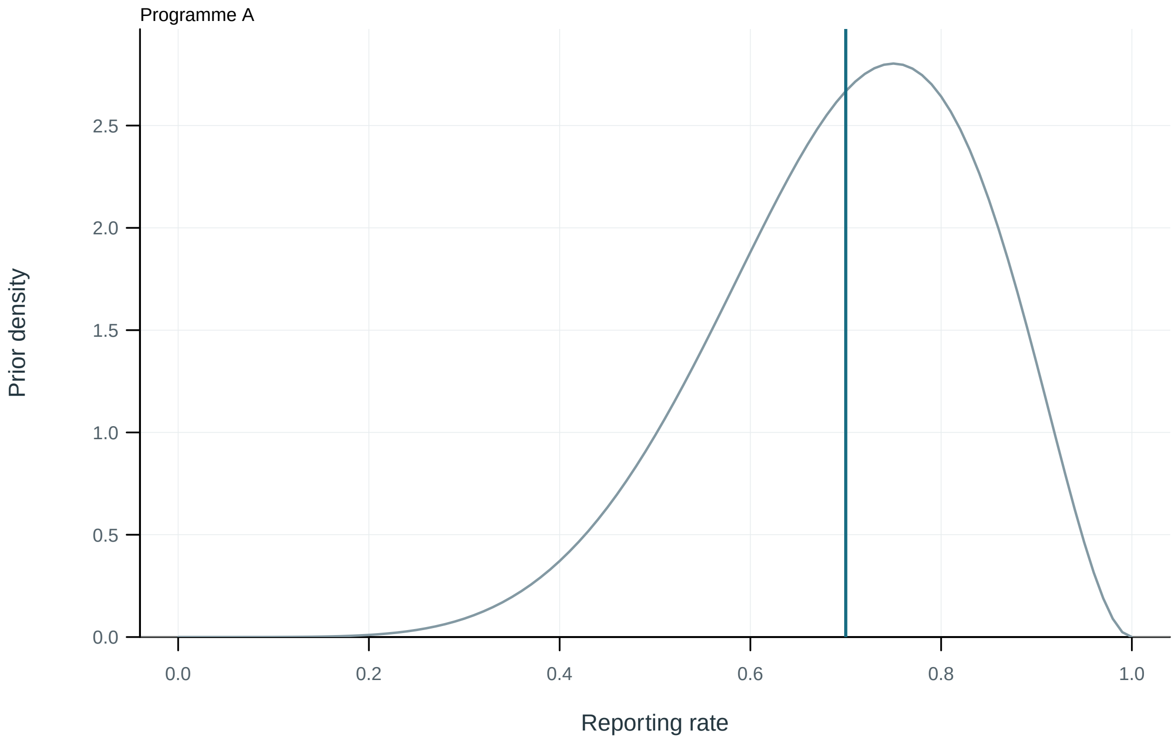

Figure 24: Illustrative output. Estimated tag-reporting rates and supplied priors.

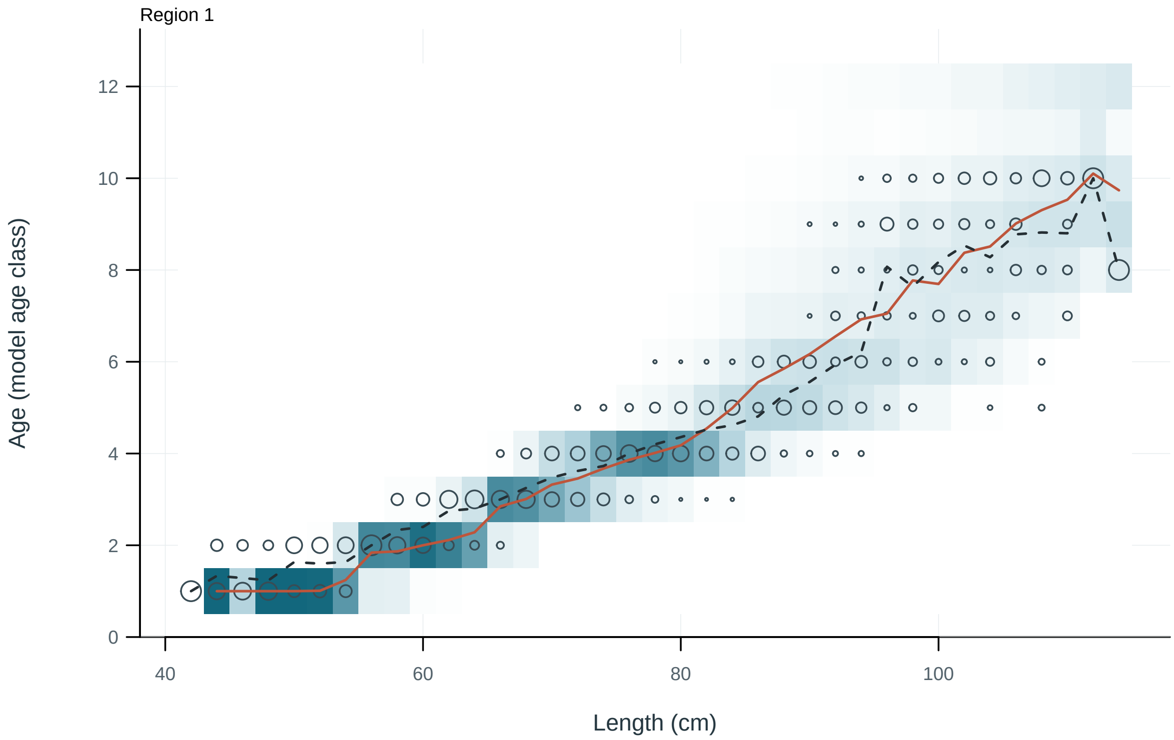

Figure 25: Observed and predicted age distributions conditional on length. Cells: predictions; circles: observations; solid and dashed lines: predicted and observed mean age.

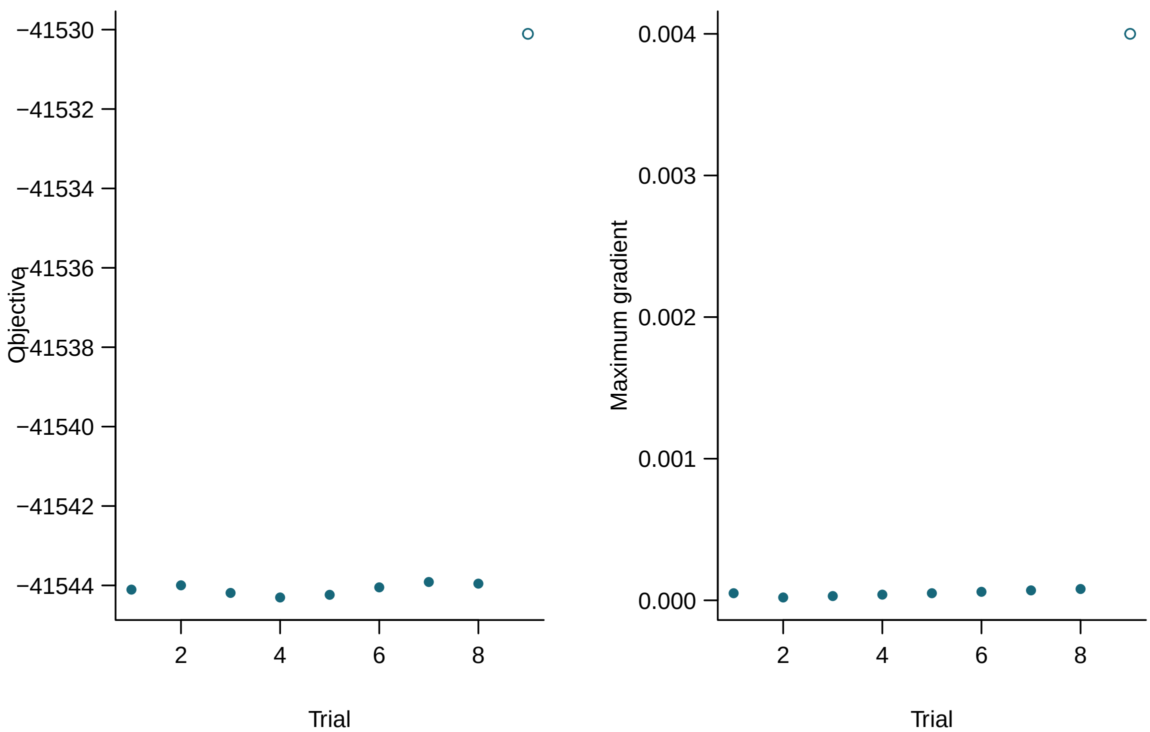

Figure 26: Final objectives and maximum gradients for independent starting values. Filled points meet all recorded numerical checks.

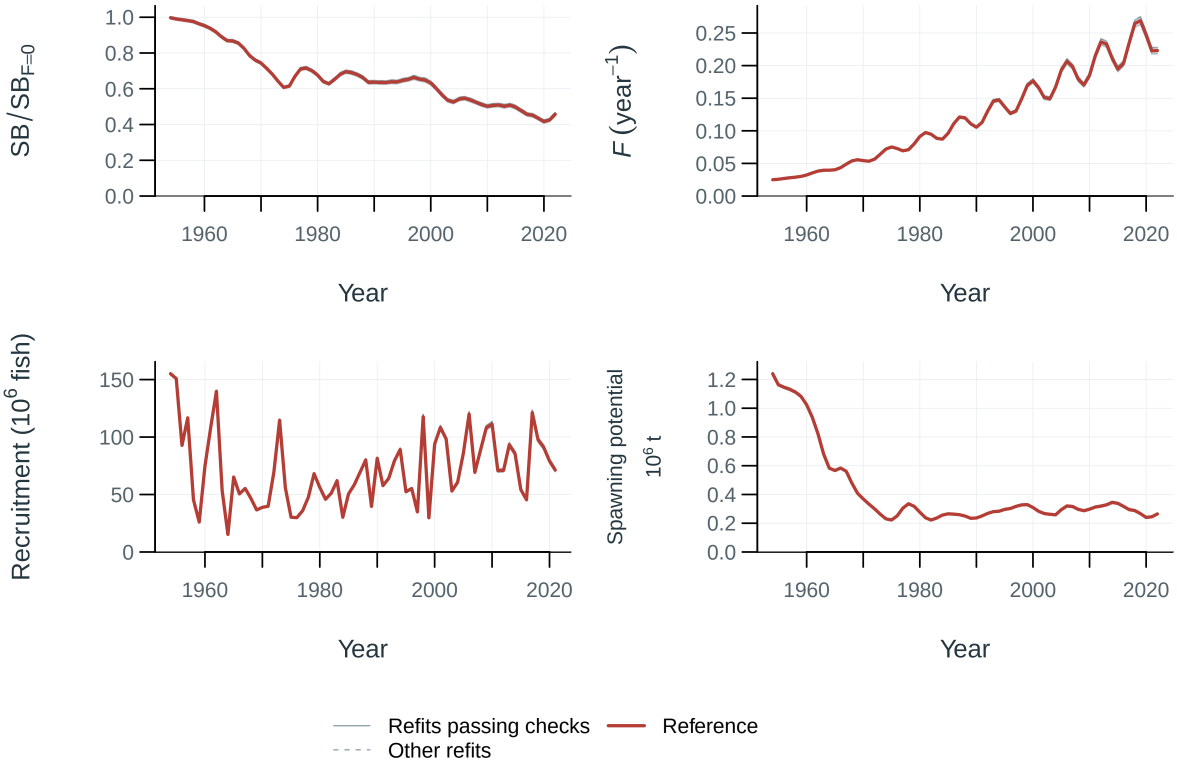

Figure 27: Jitter model trajectories.

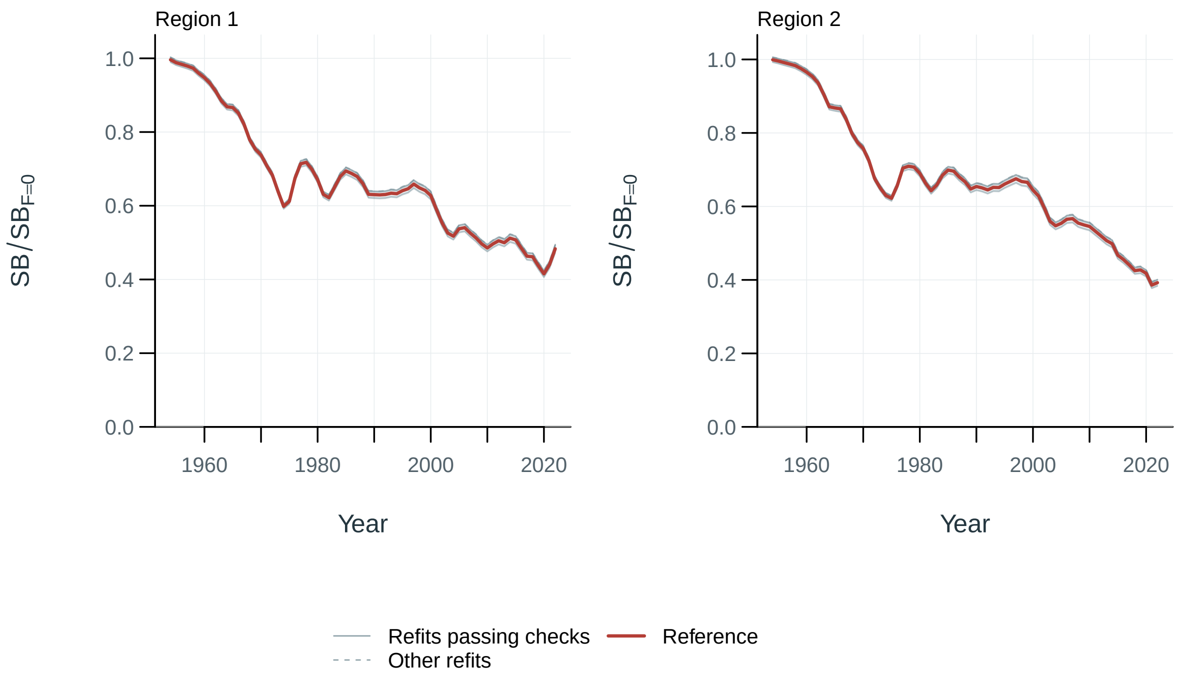

Figure 28: Spawning depletion by model and region, relative to each model's dynamic unfished reference.

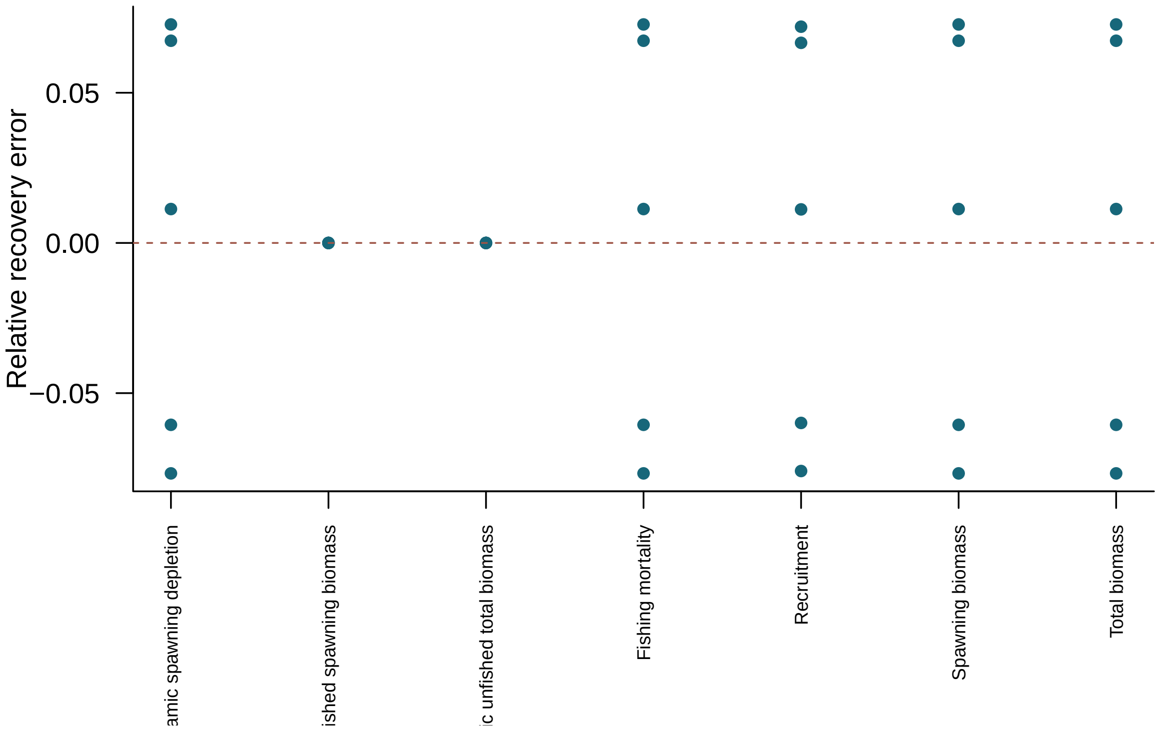

Figure 29: Relative recovery error in terminal quantities for converged simulated-data refits.

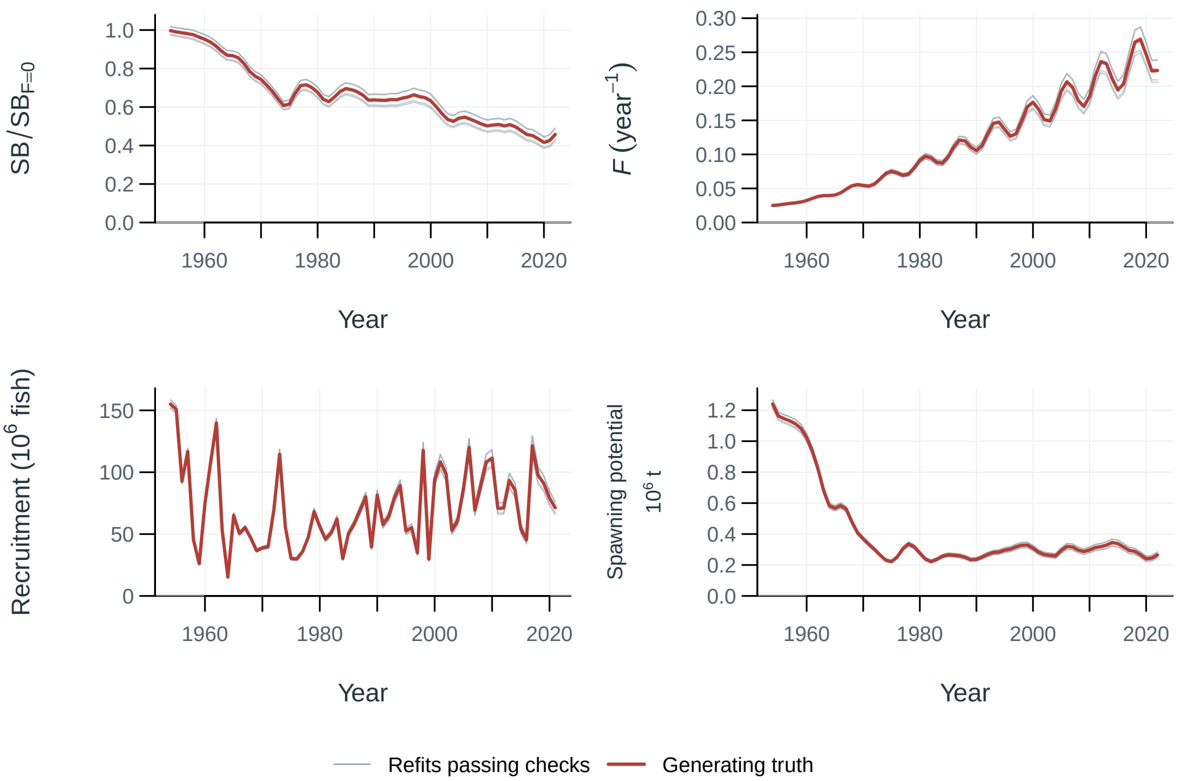

Figure 30: Selftest model trajectories.

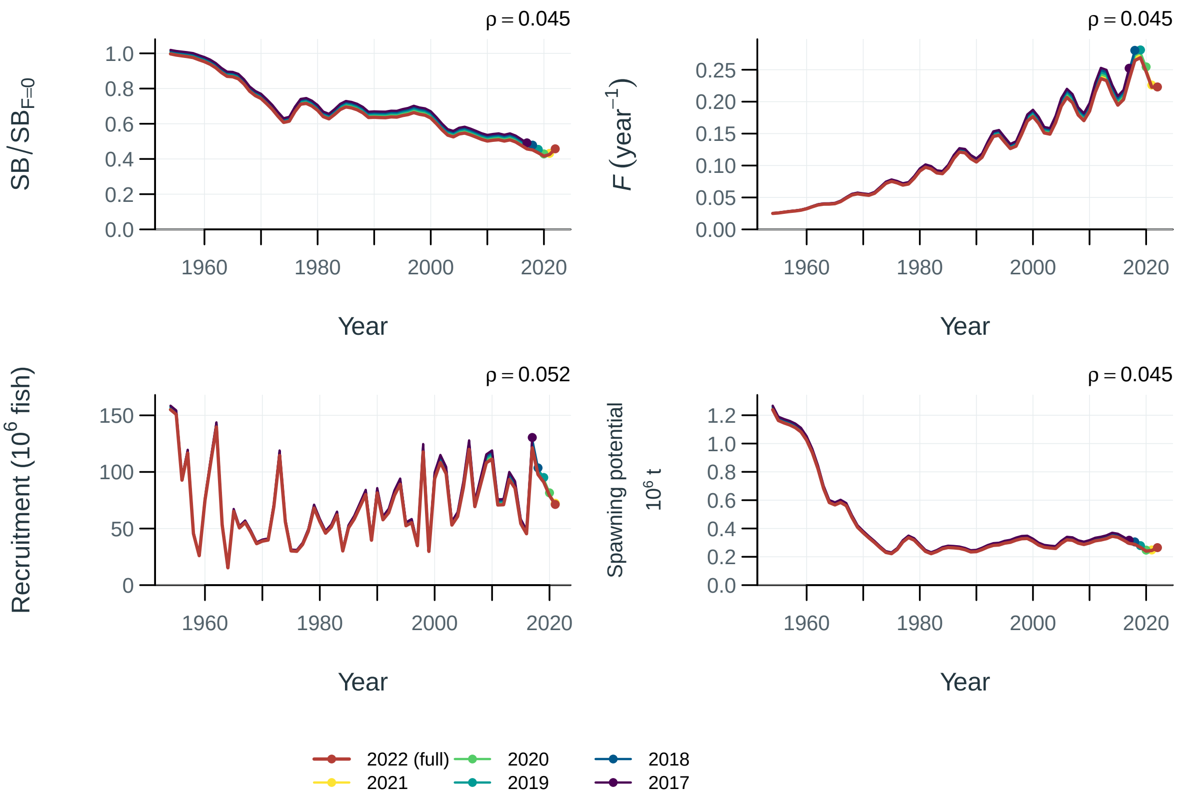

Figure 31: Population trajectories for the full-data fit and retrospective peels. Points mark terminal years; rho is Mohn's mean terminal relative difference for peels passing the recorded numerical checks.

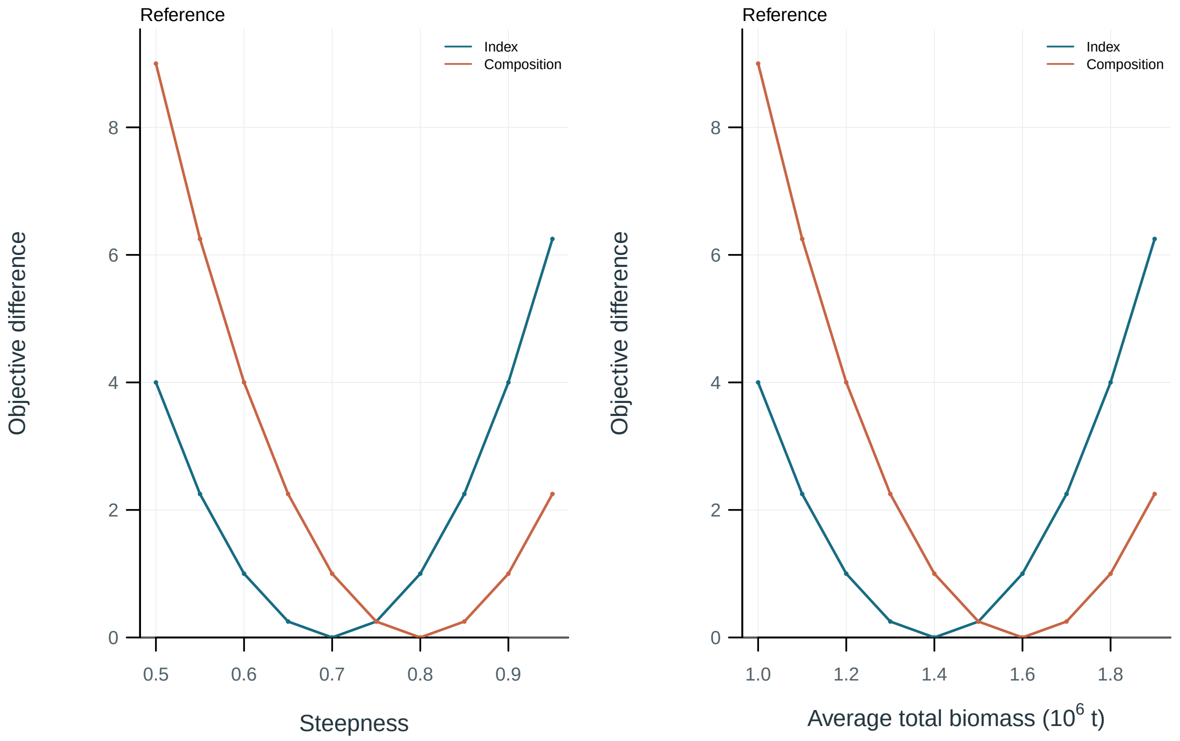

Figure 32: Total objective and component profiles. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

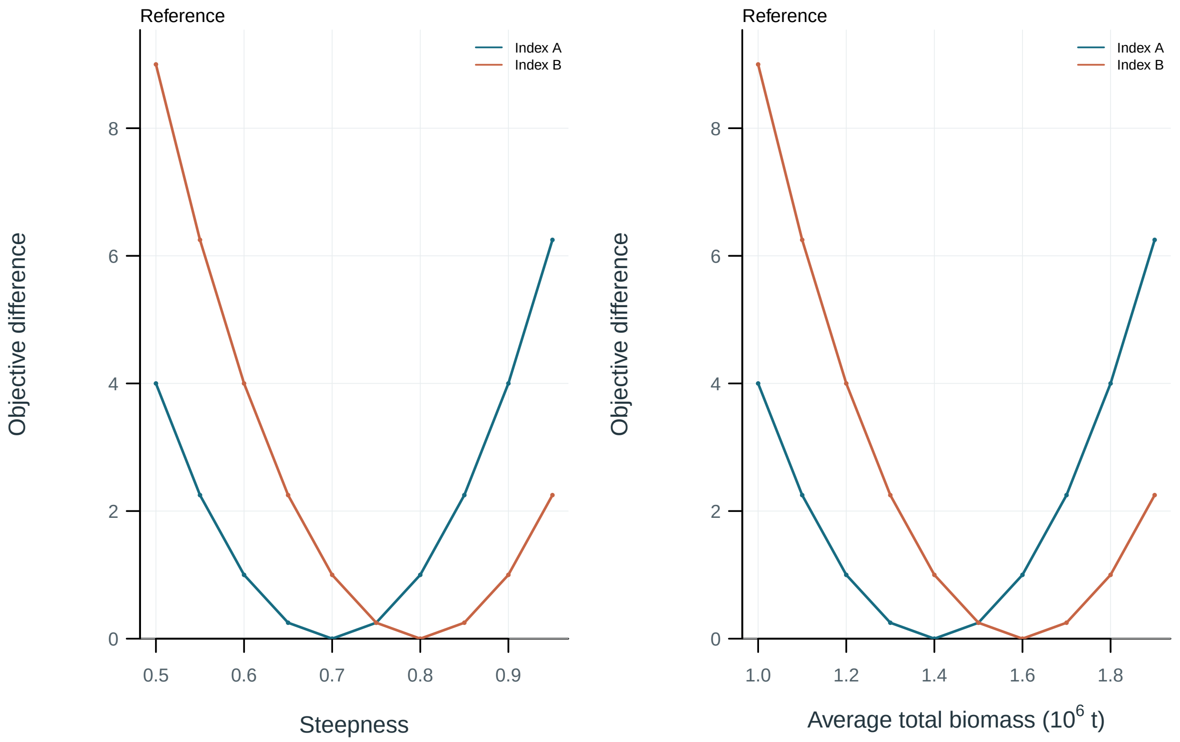

Figure 33: Objective profiles for abundance-index components. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

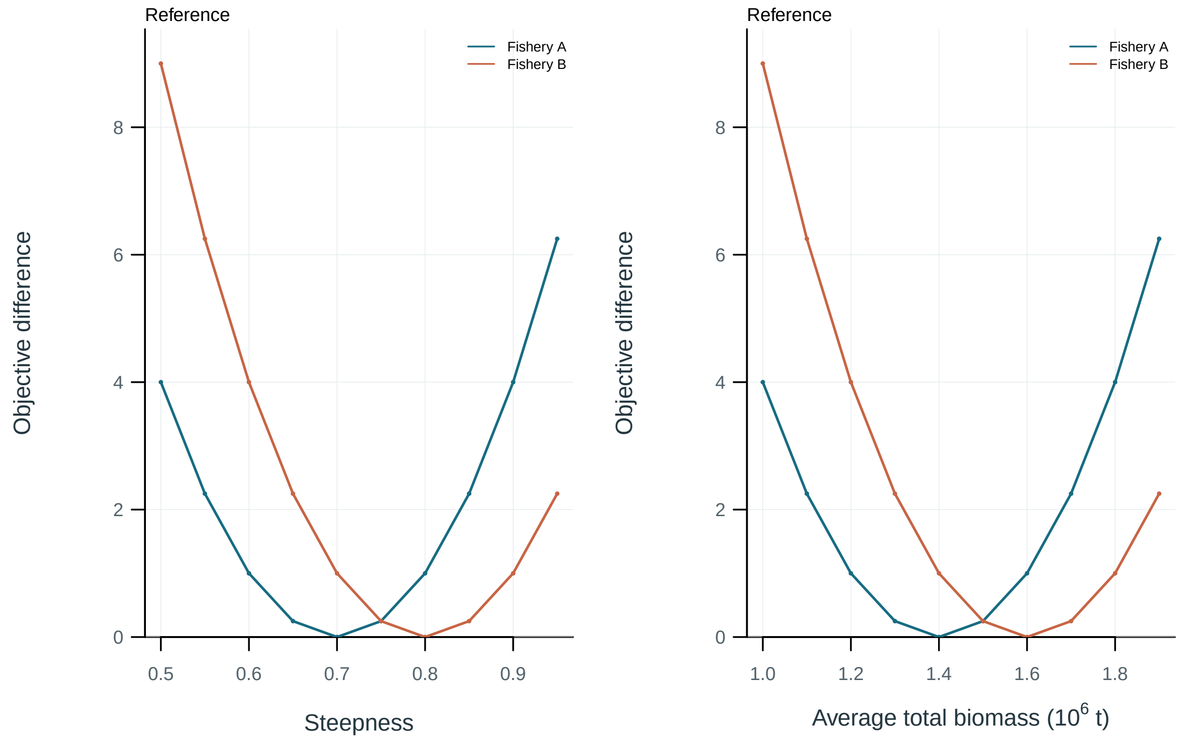

Figure 34: Objective profiles for length-composition components. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

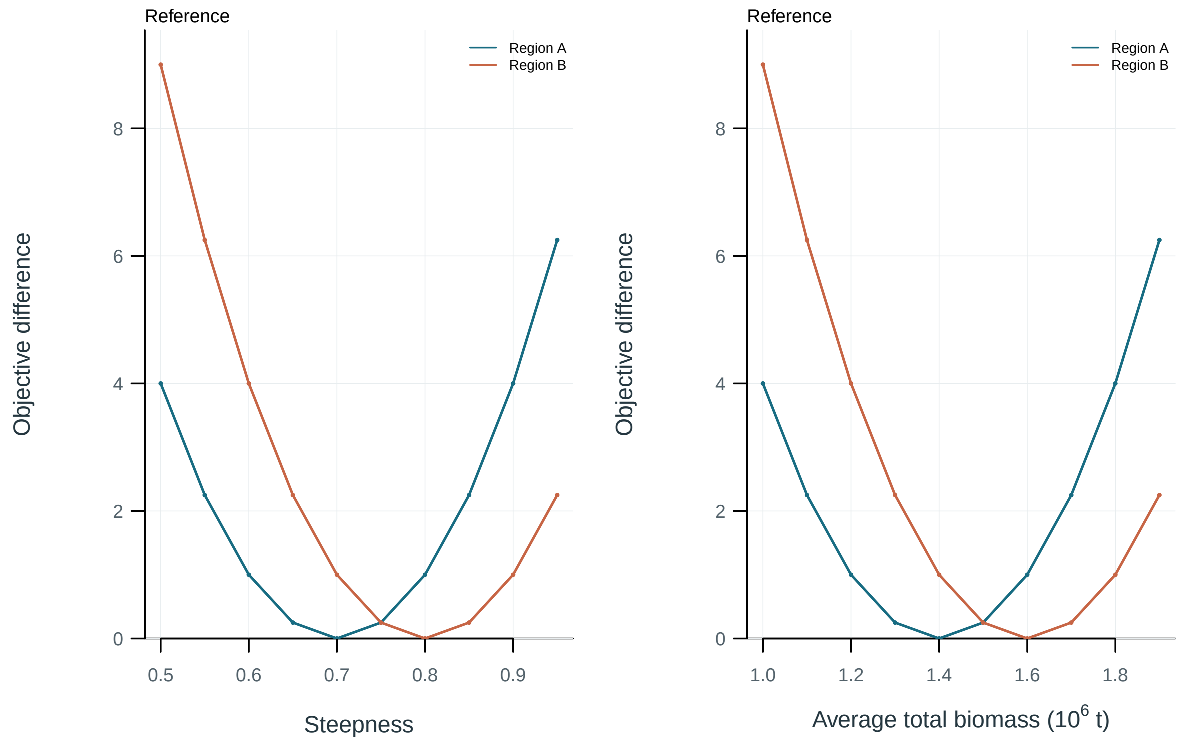

Figure 35: Objective profiles for conditional age-at-length components. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

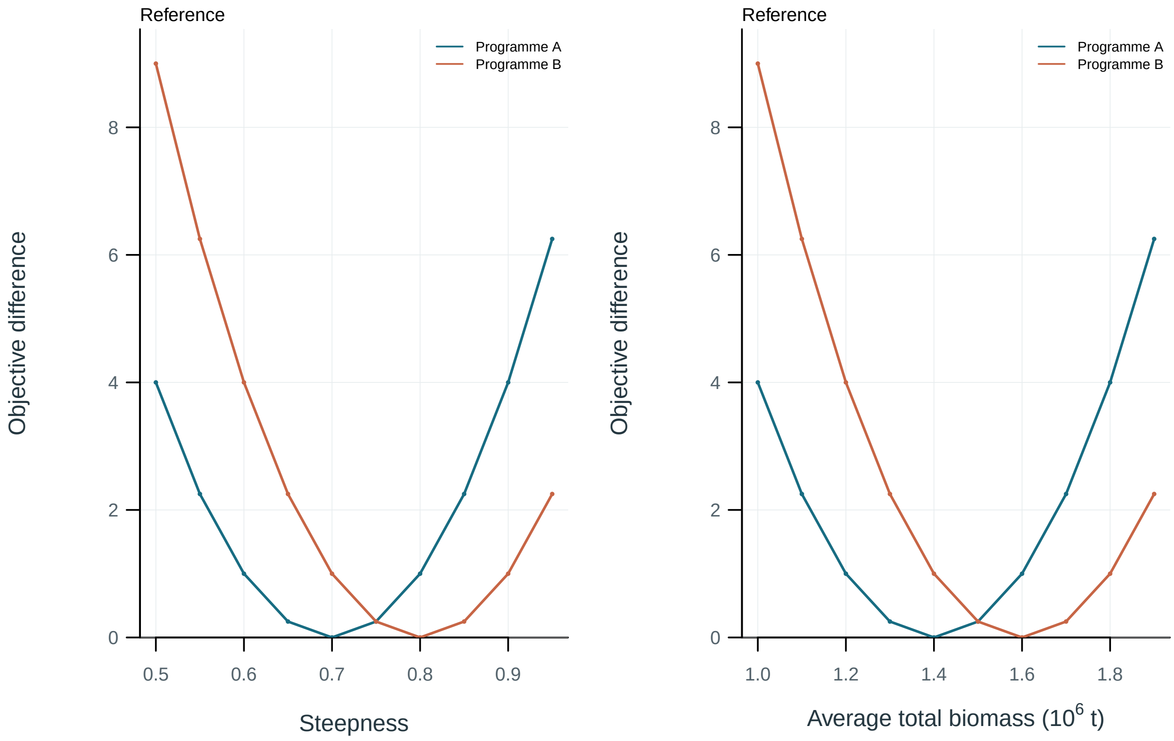

Figure 36: Objective profiles for tagging programmes. Each curve is centered on its own minimum. Profile on the supplied parameter scale.; Total biomass averaged over the declared model periods, 2001-2020.

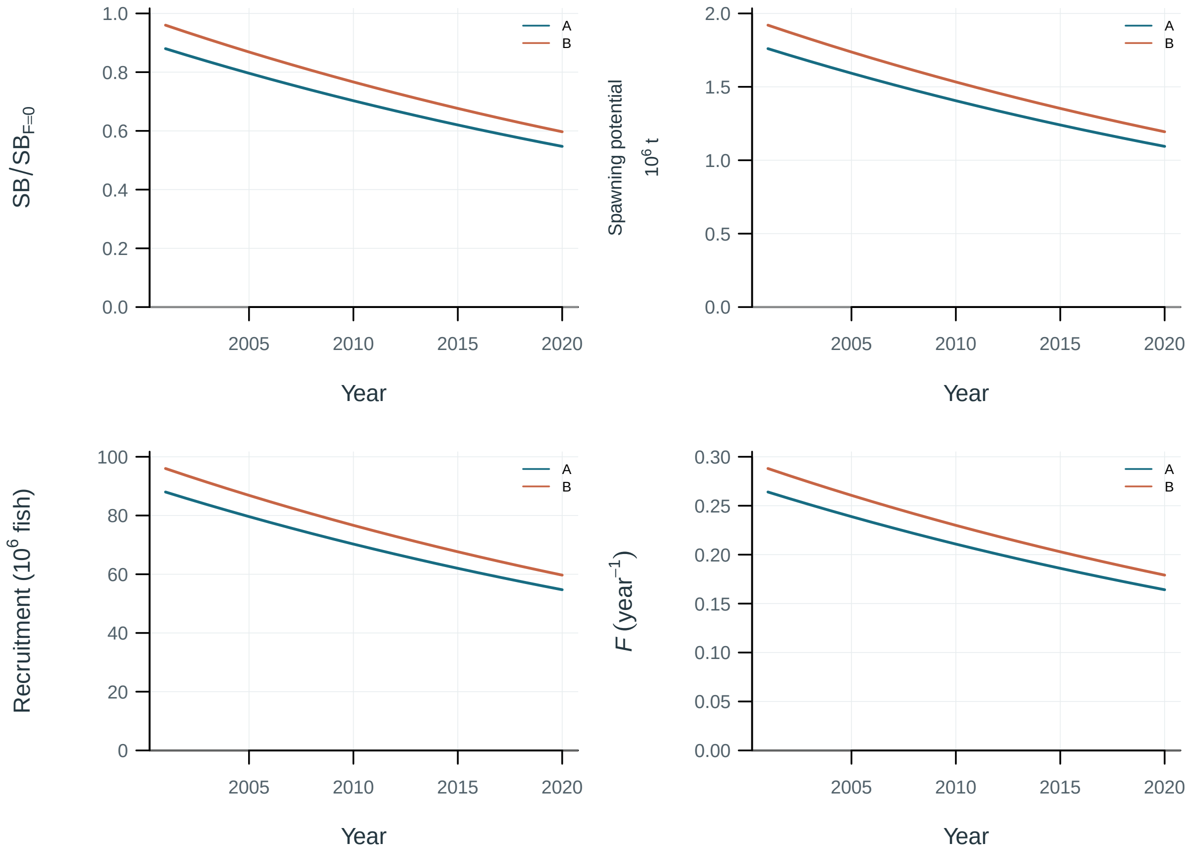

Figure 37: Illustrative output. Diagnostic and production-model trajectories.

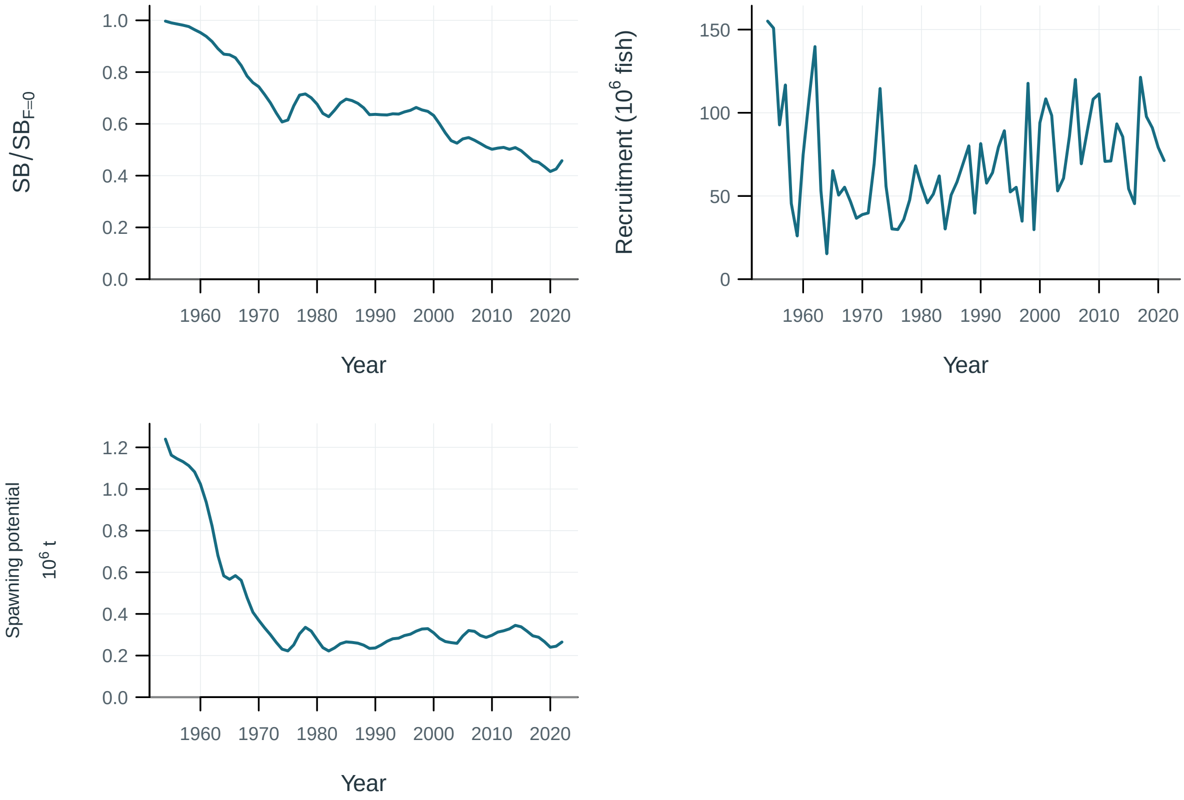

Figure 38: Estimated population trajectories.

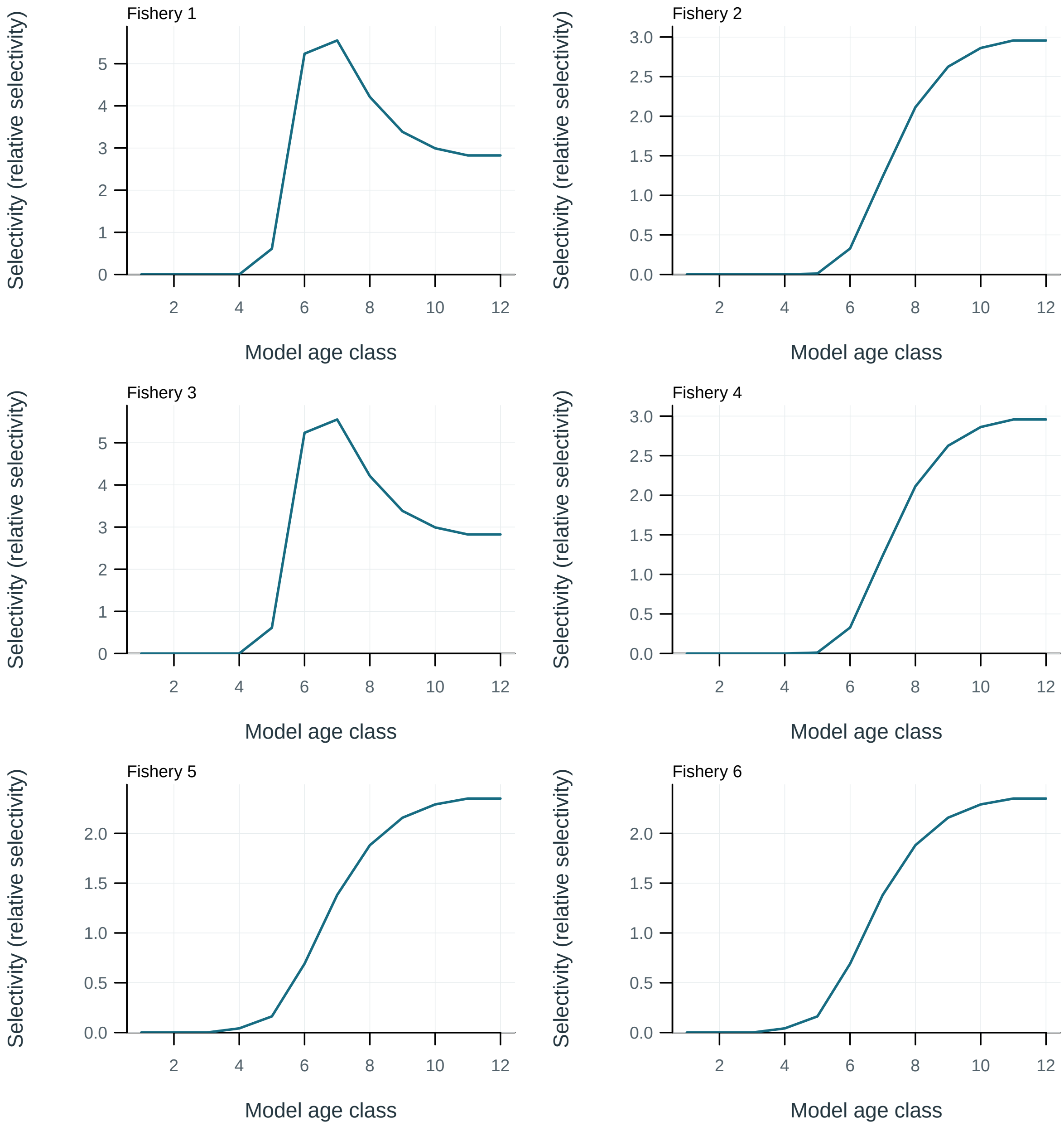

Figure 39: Fishery selectivity at age.

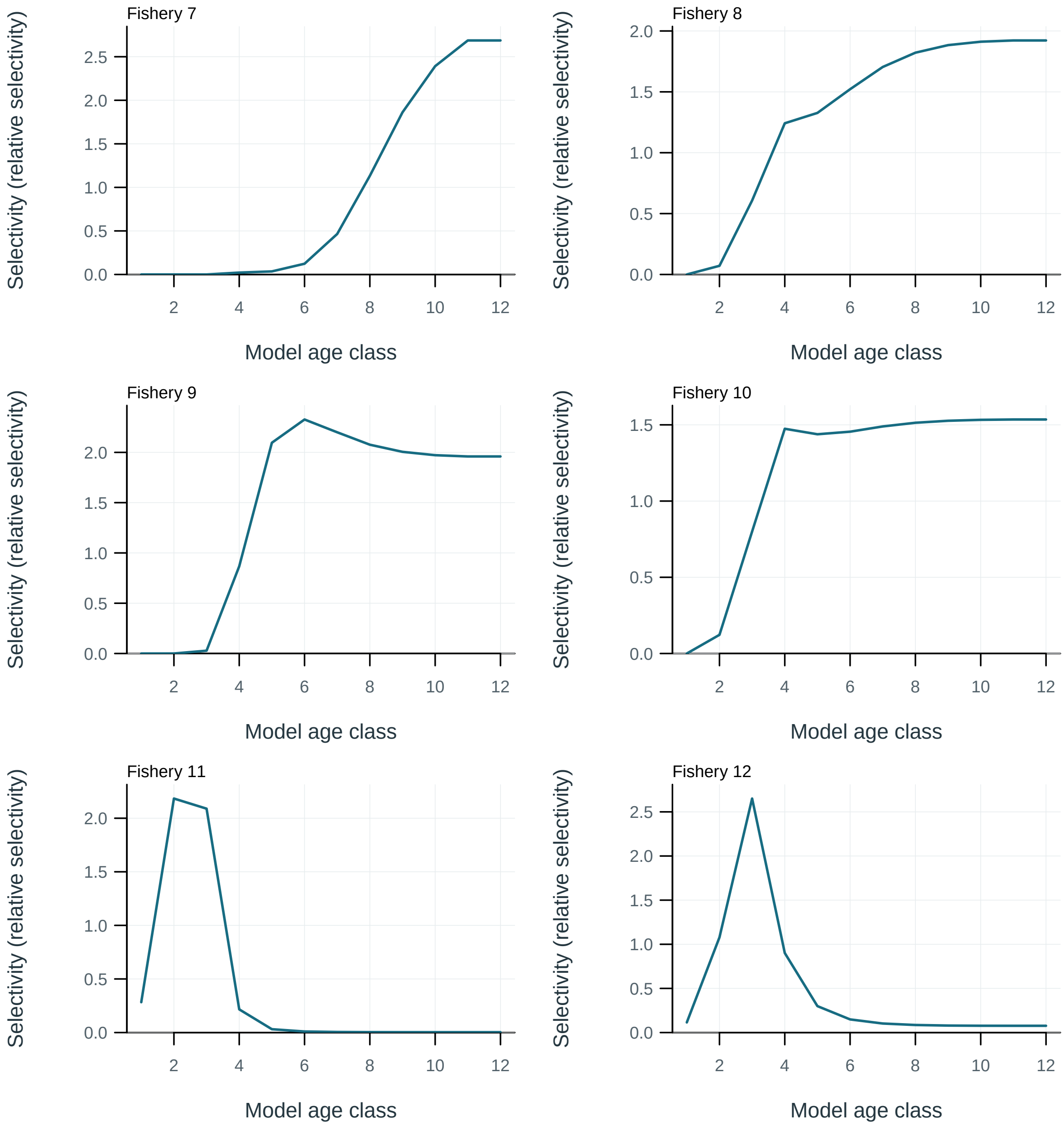

Figure 40: Fishery selectivity at age.

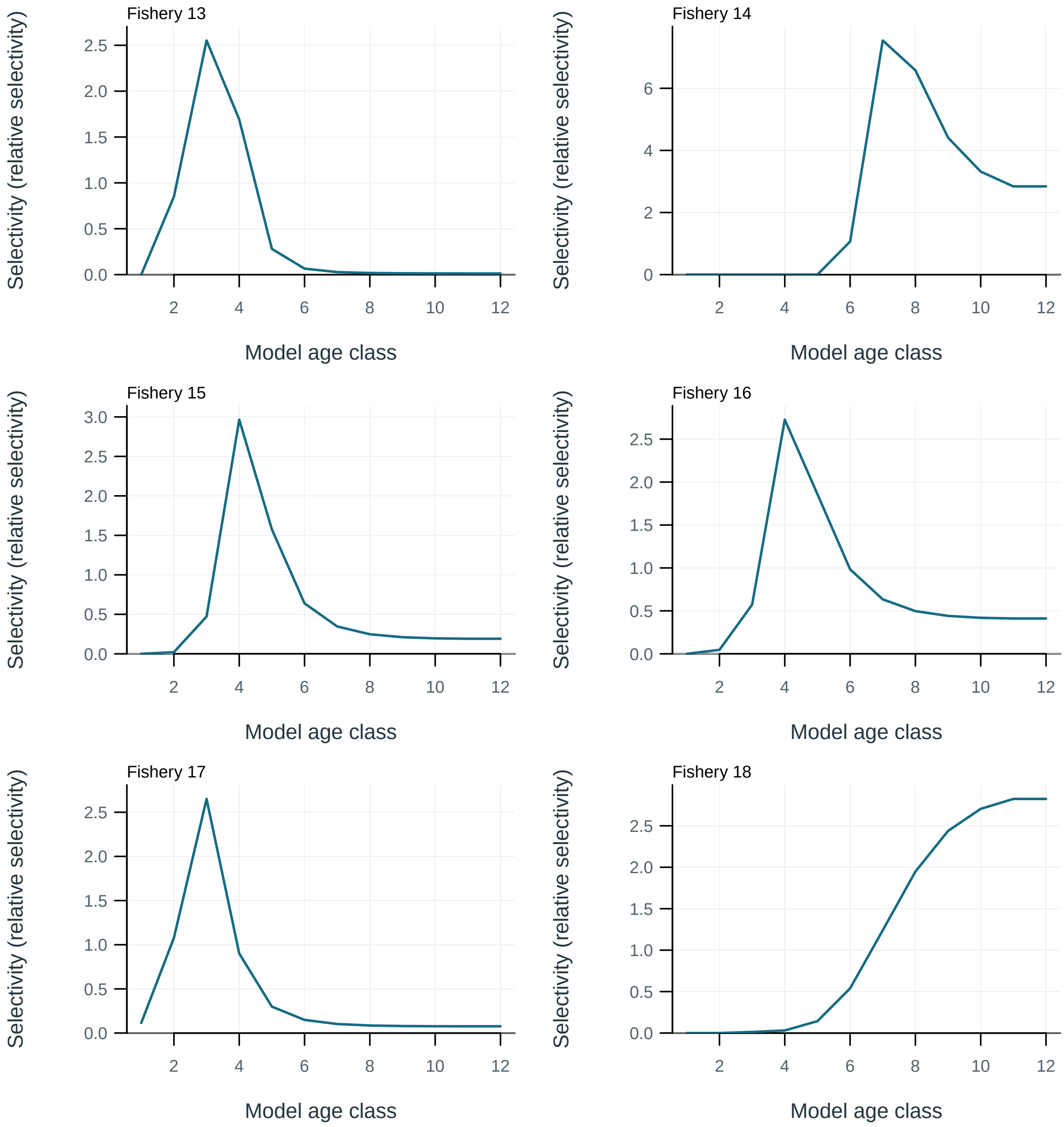

Figure 41: Fishery selectivity at age.

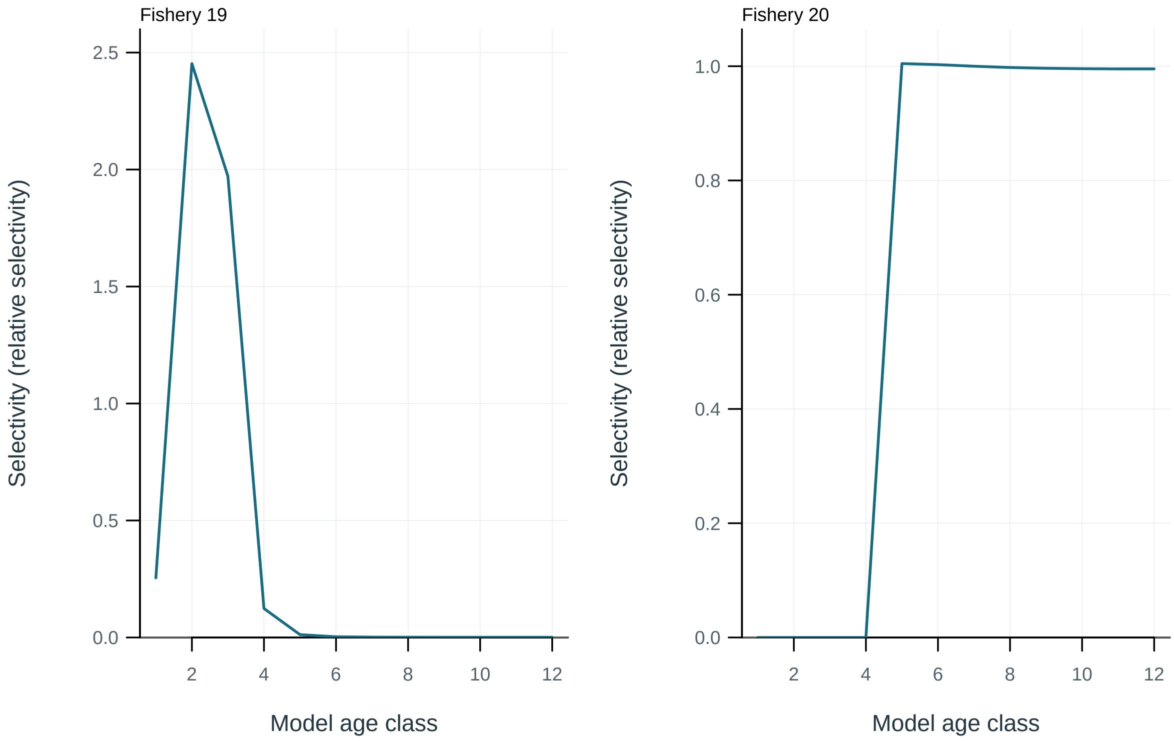

Figure 42: Fishery selectivity at age.

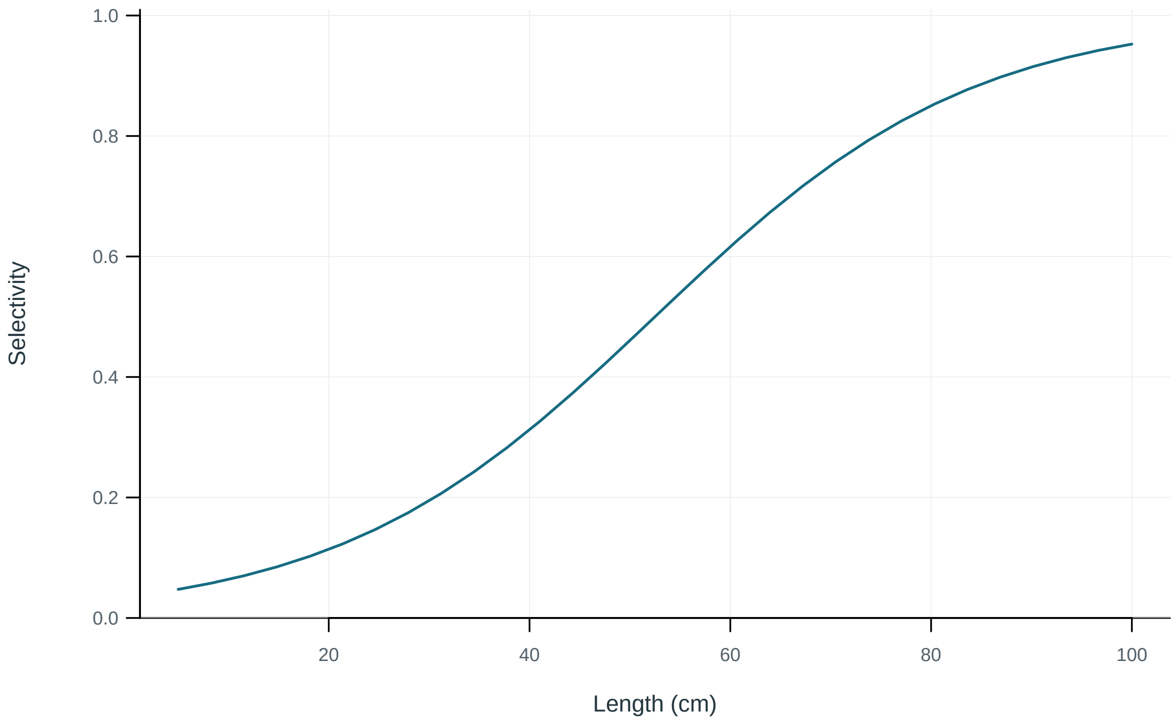

Figure 43: Illustrative output. Selectivity at length by fishery.

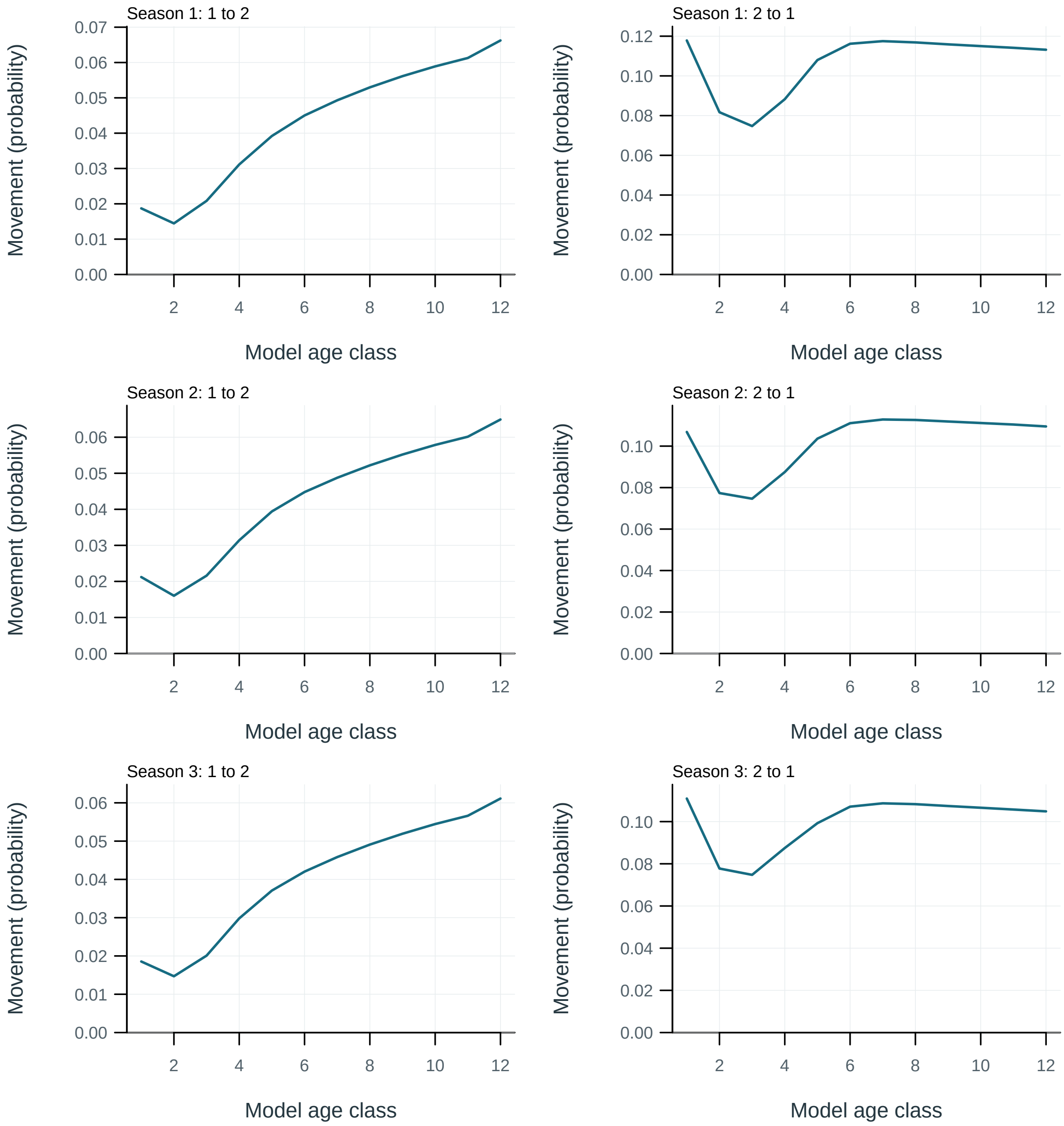

Figure 44: Probability of movement from source to destination region per movement event.

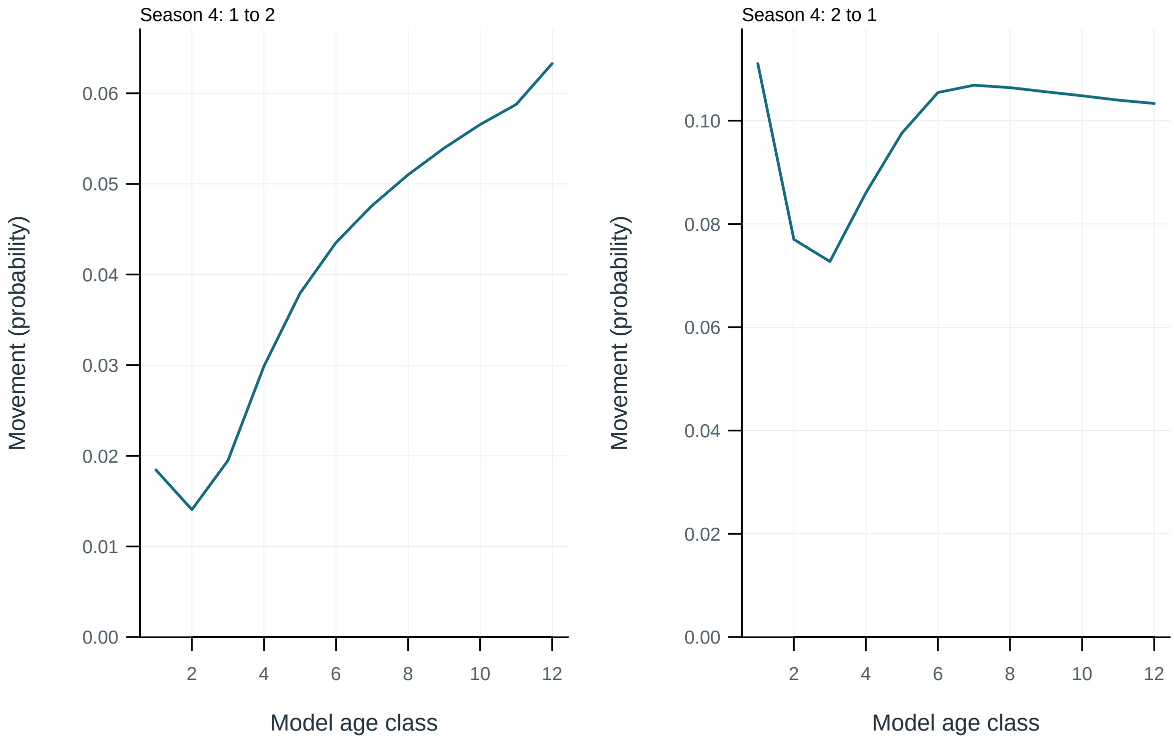

Figure 45: Probability of movement from source to destination region per movement event.

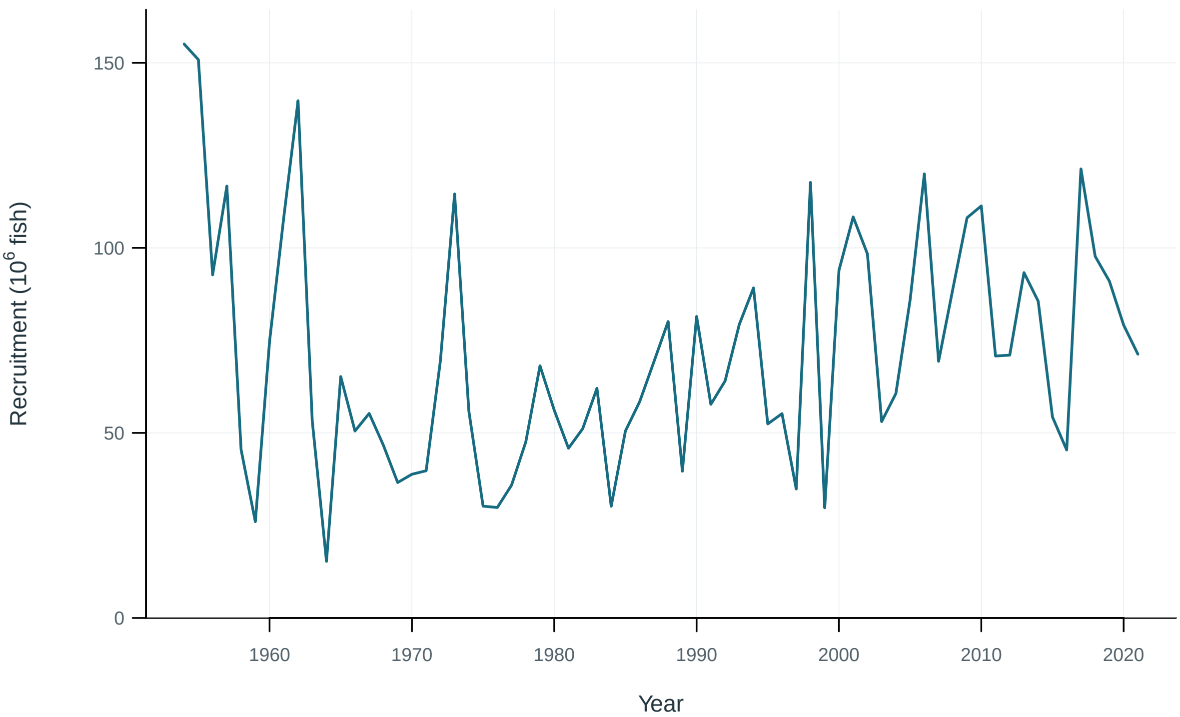

Figure 46: Annual recruitment.

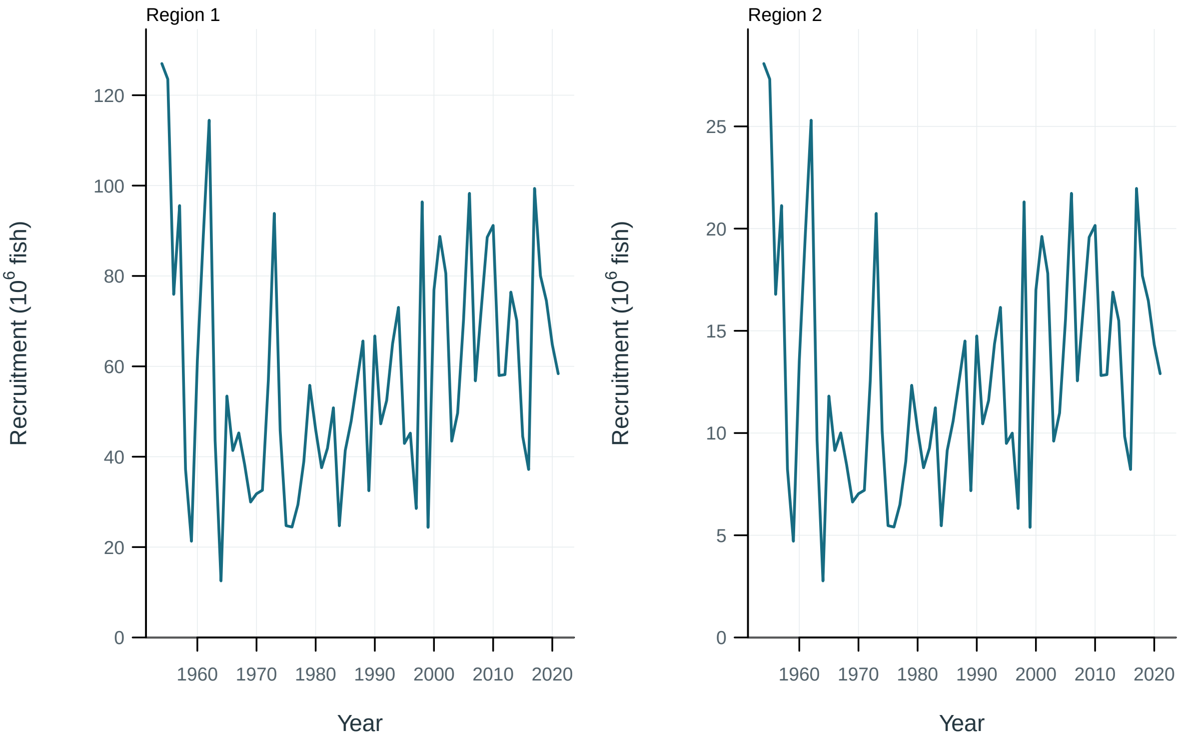

Figure 47: Recruitment by model region.

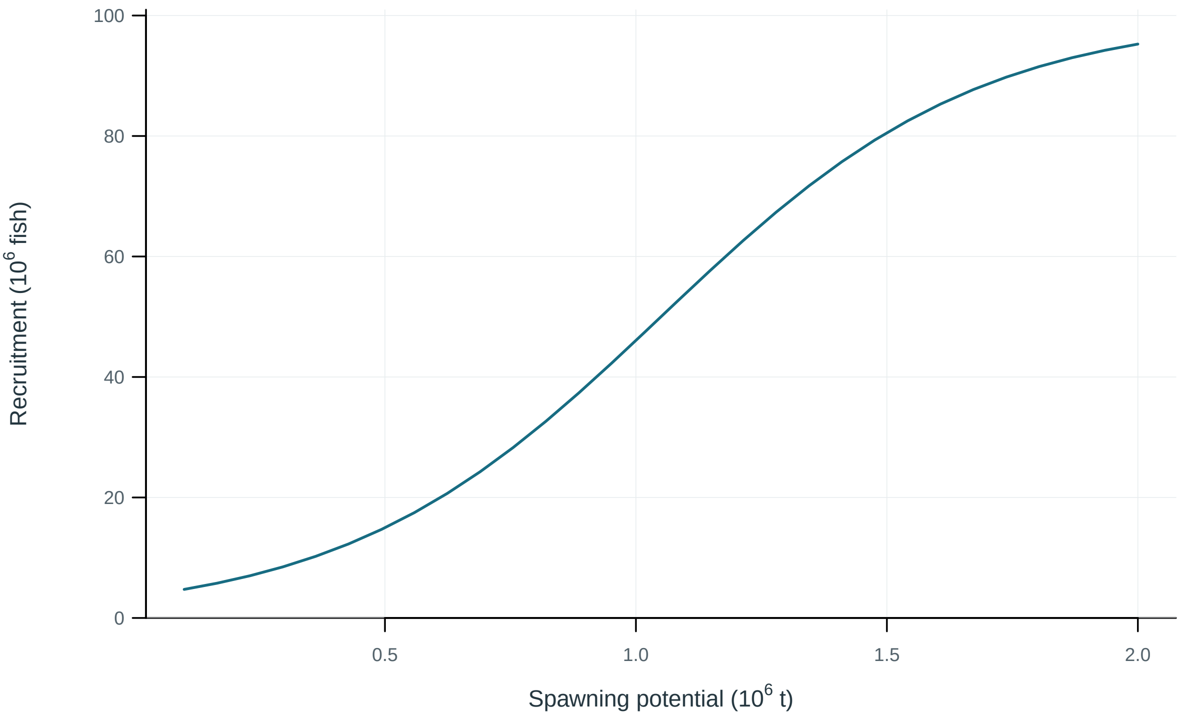

Figure 48: Illustrative output. Stock–recruitment observations and fitted relationship.

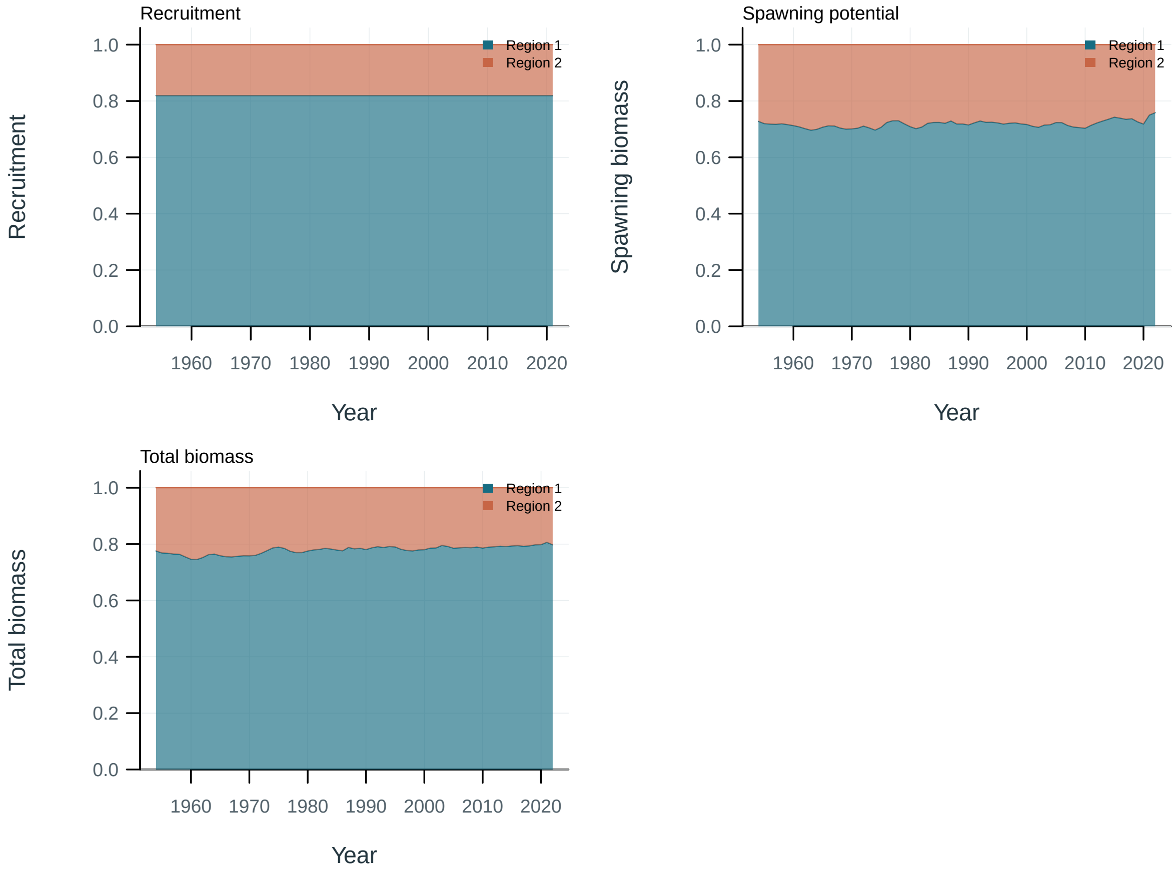

Figure 49: Regional shares of population quantities.

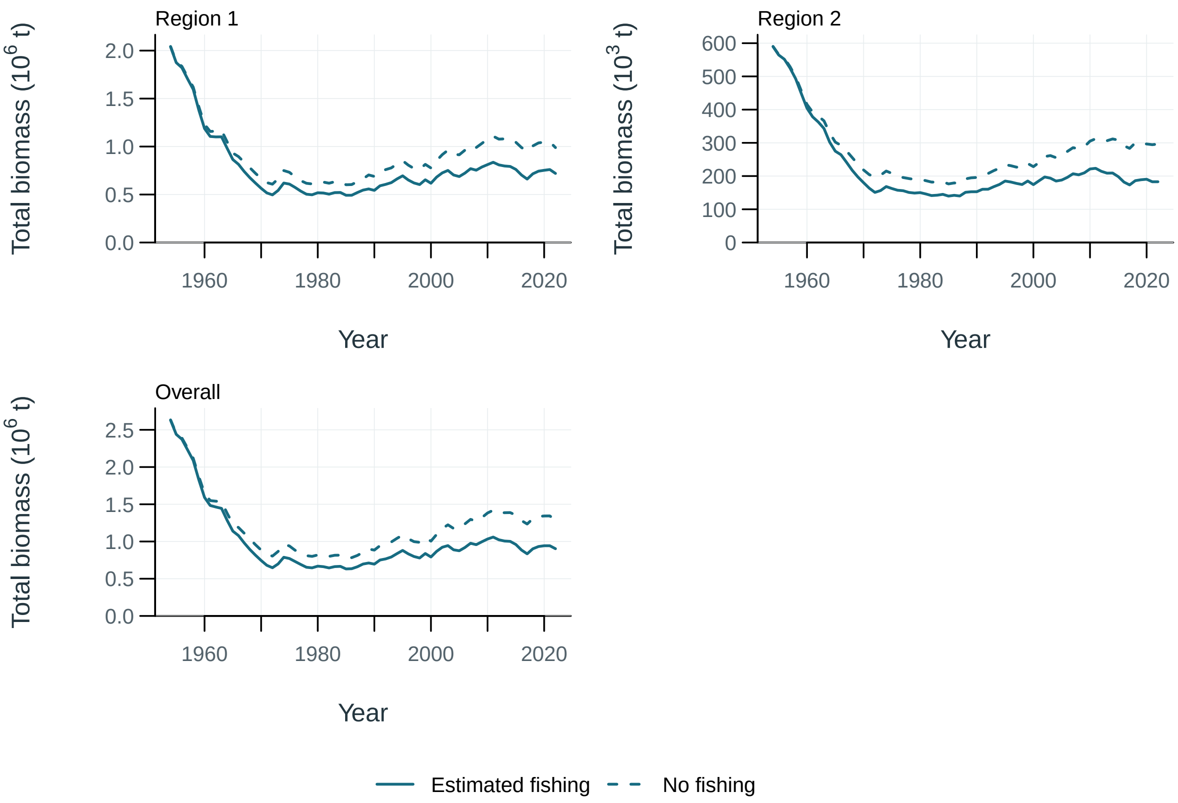

Figure 50: Total biomass under estimated fishing (solid) and the dynamic no-fishing counterfactual (dashed), by region and overall.

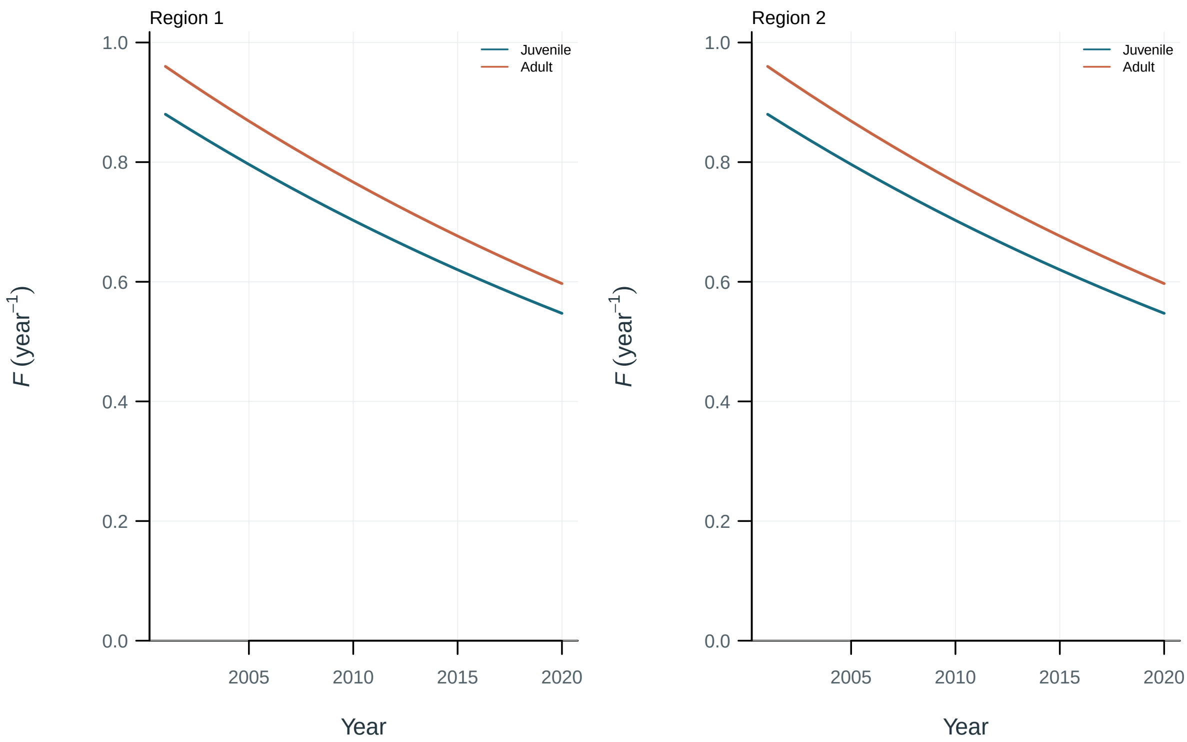

Figure 51: Illustrative output. Fishing mortality by life stage and region.

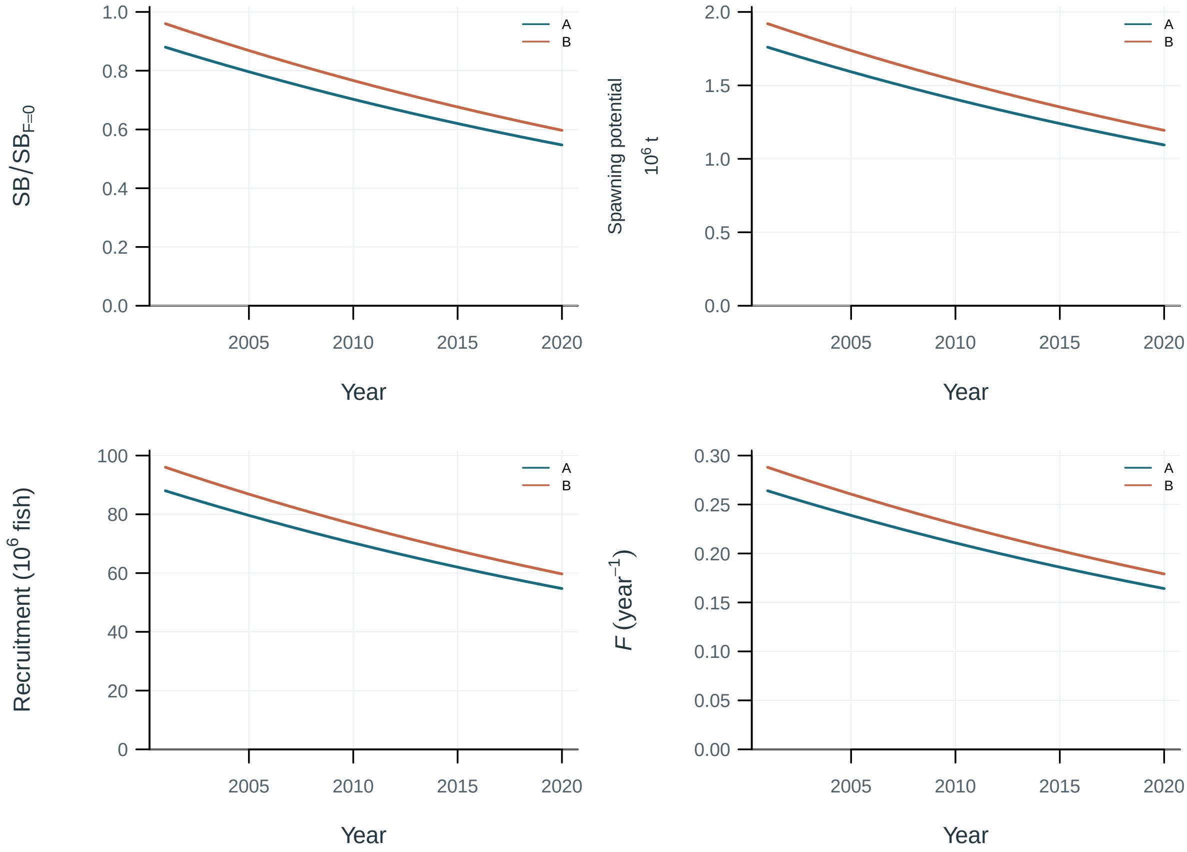

Figure 52: Sensitivity of population trajectories to steepness.

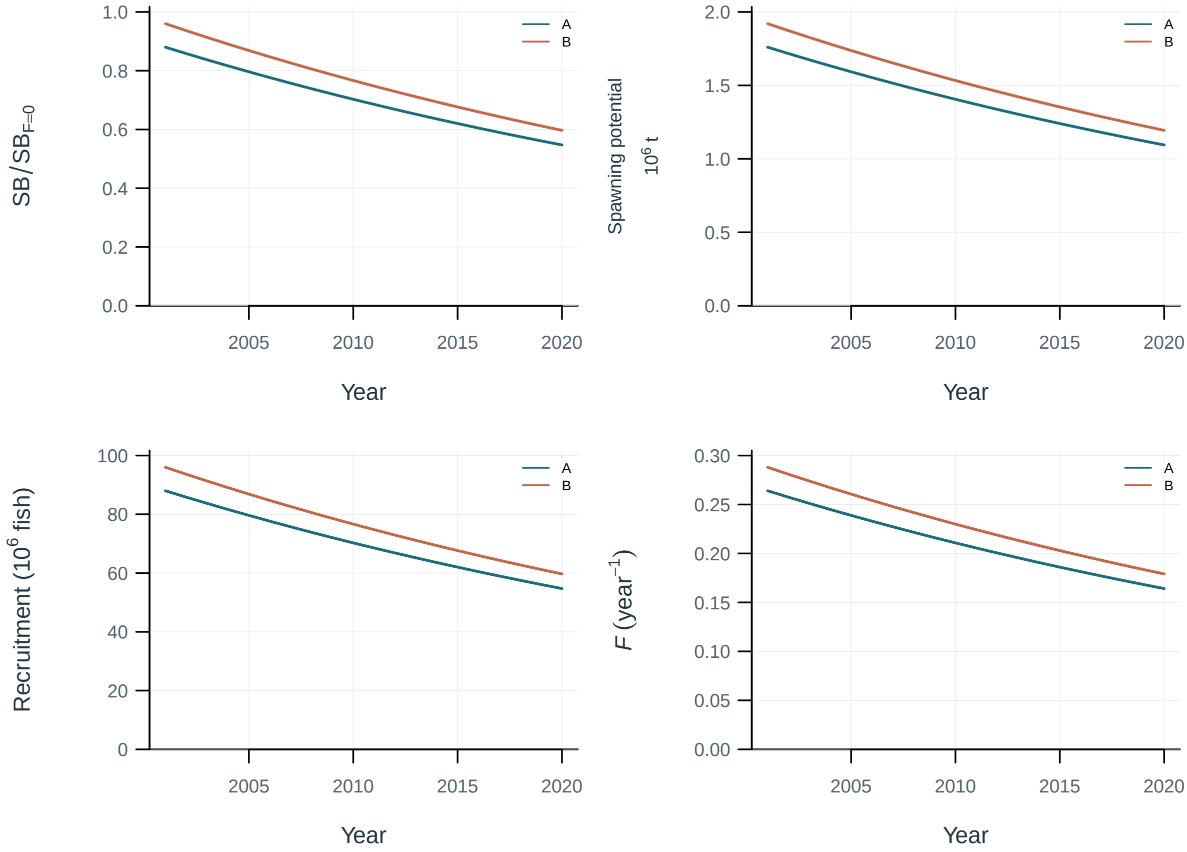

Figure 53: Sensitivity of population trajectories to natural mortality.

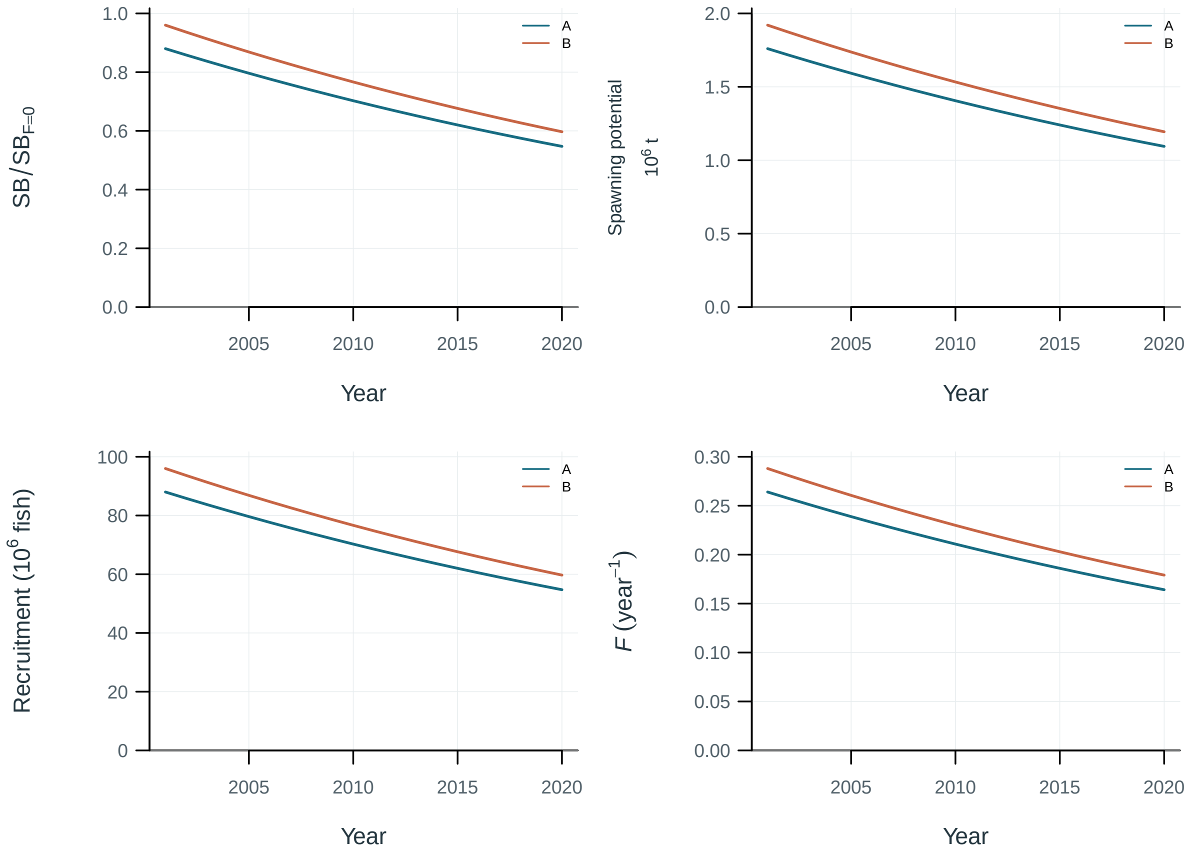

Figure 54: Sensitivity of population trajectories to conditional age-at-length data.

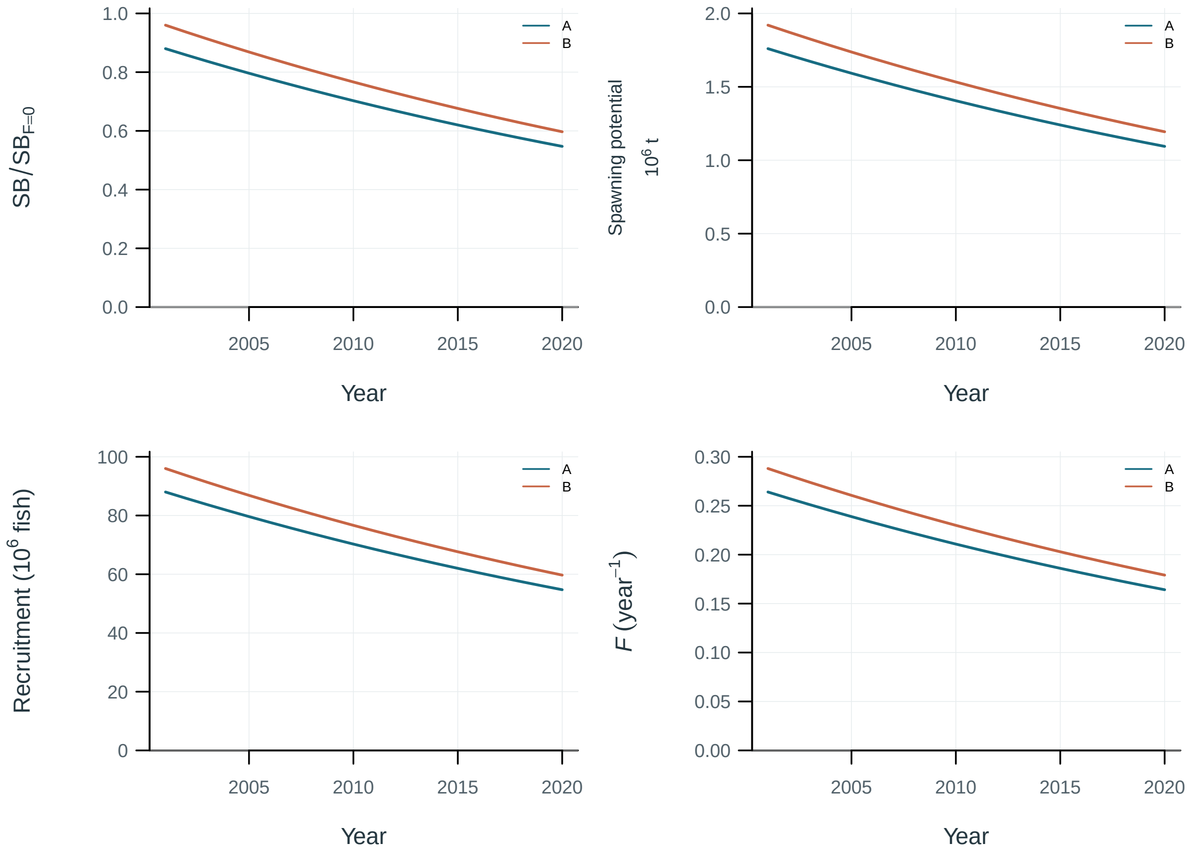

Figure 55: Sensitivity of population trajectories to effort creep.

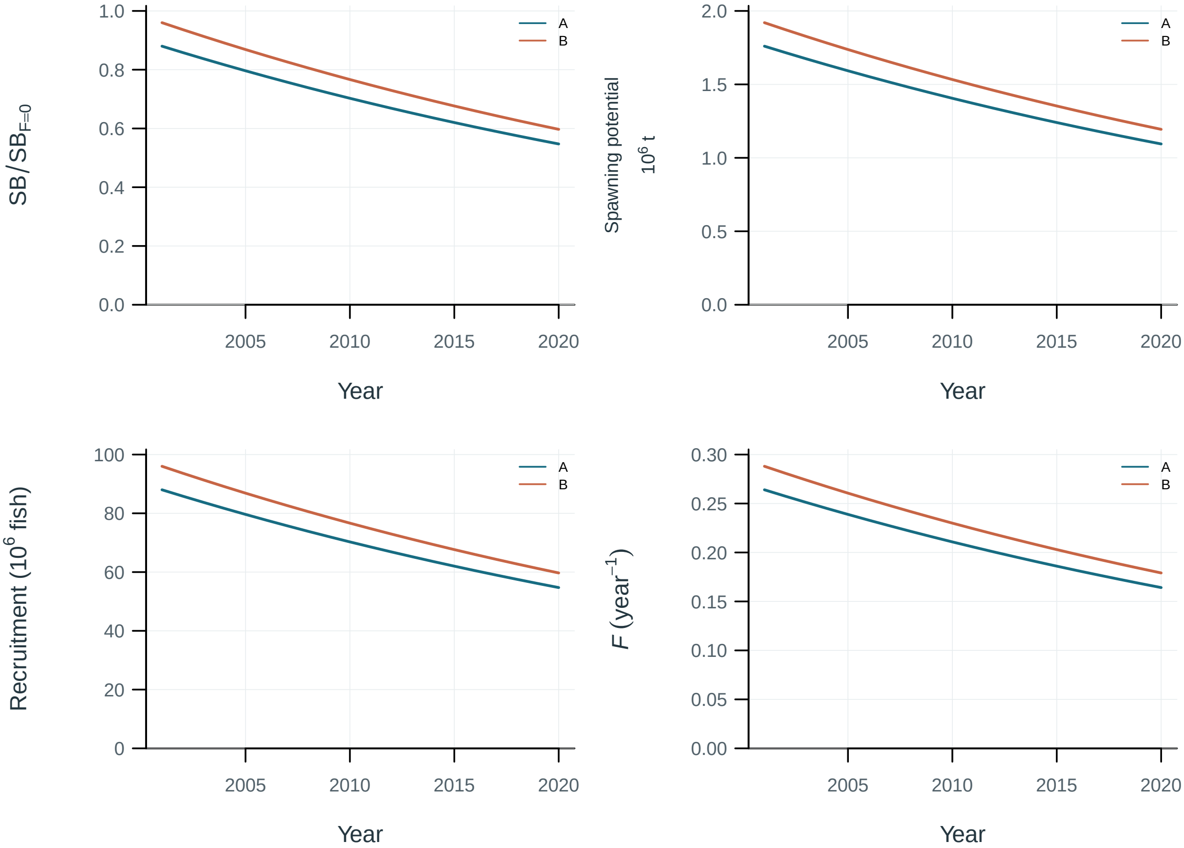

Figure 56: Sensitivity of population trajectories to regional scaling.

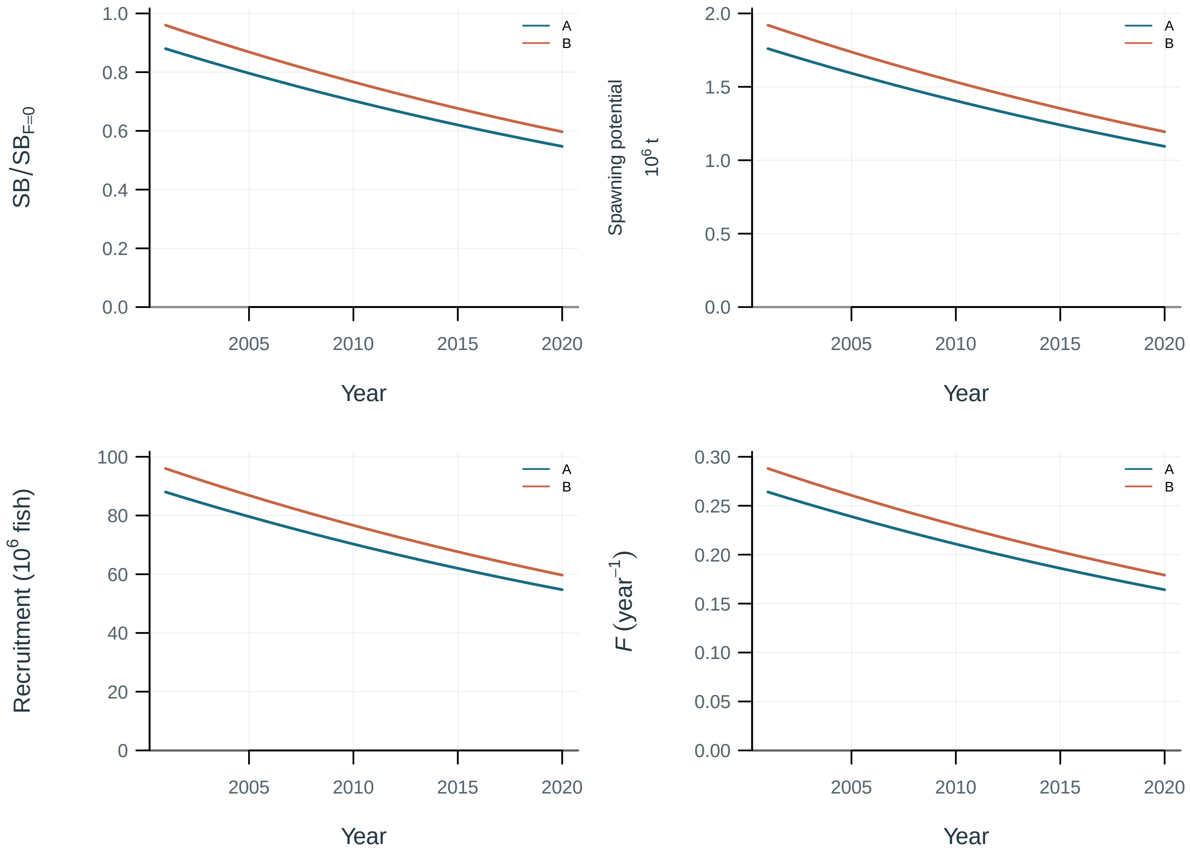

Figure 57: Sensitivity of population trajectories to tag-mixing periods and the K-statistic cutoff.

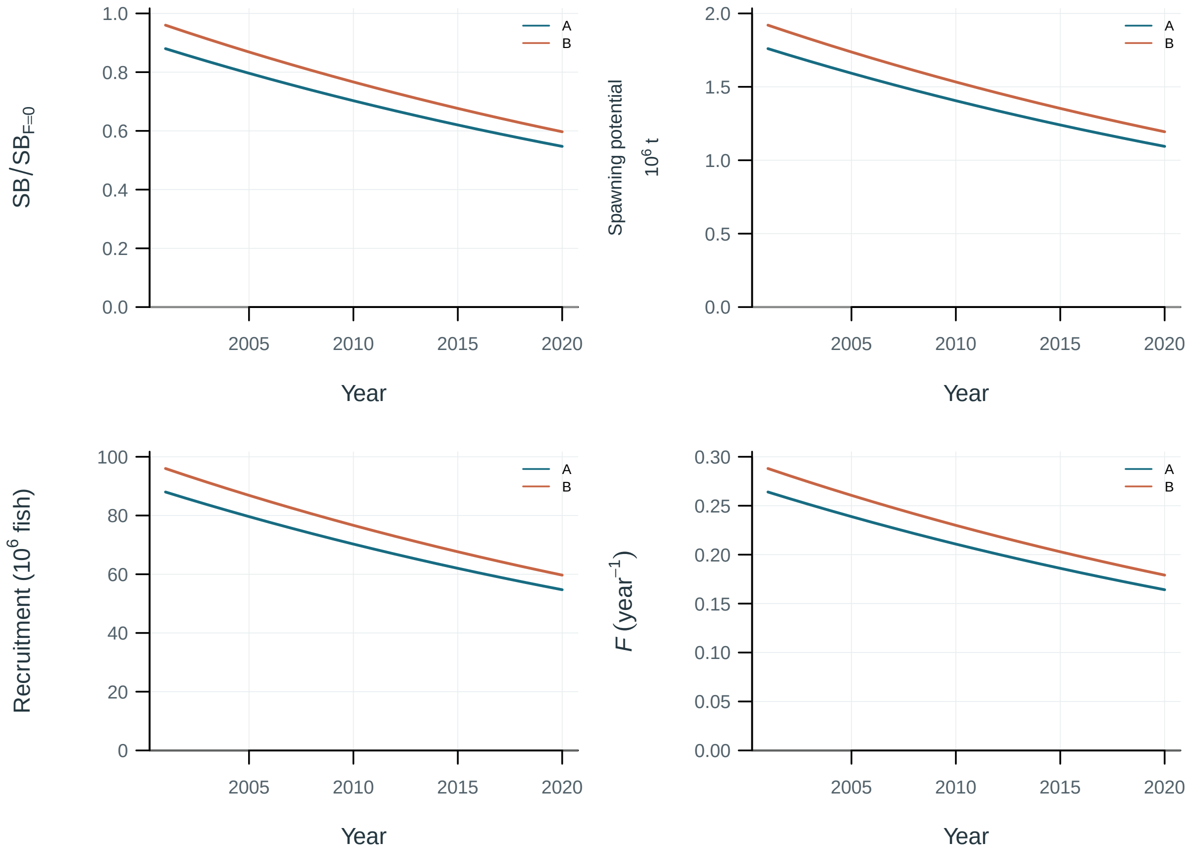

Figure 58: Sensitivity of population trajectories to pre-mixing tag-reporting rates.

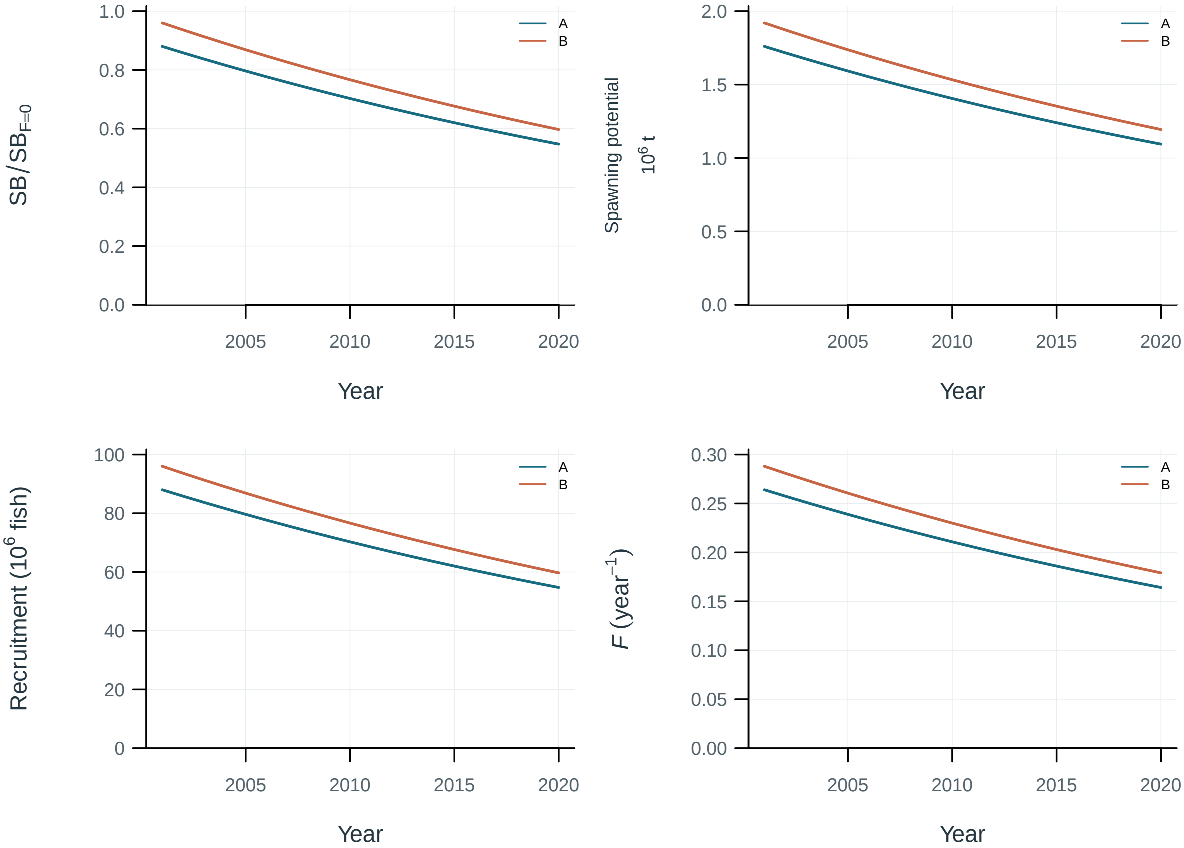

Figure 59: Sensitivity of population trajectories to tag overdispersion.

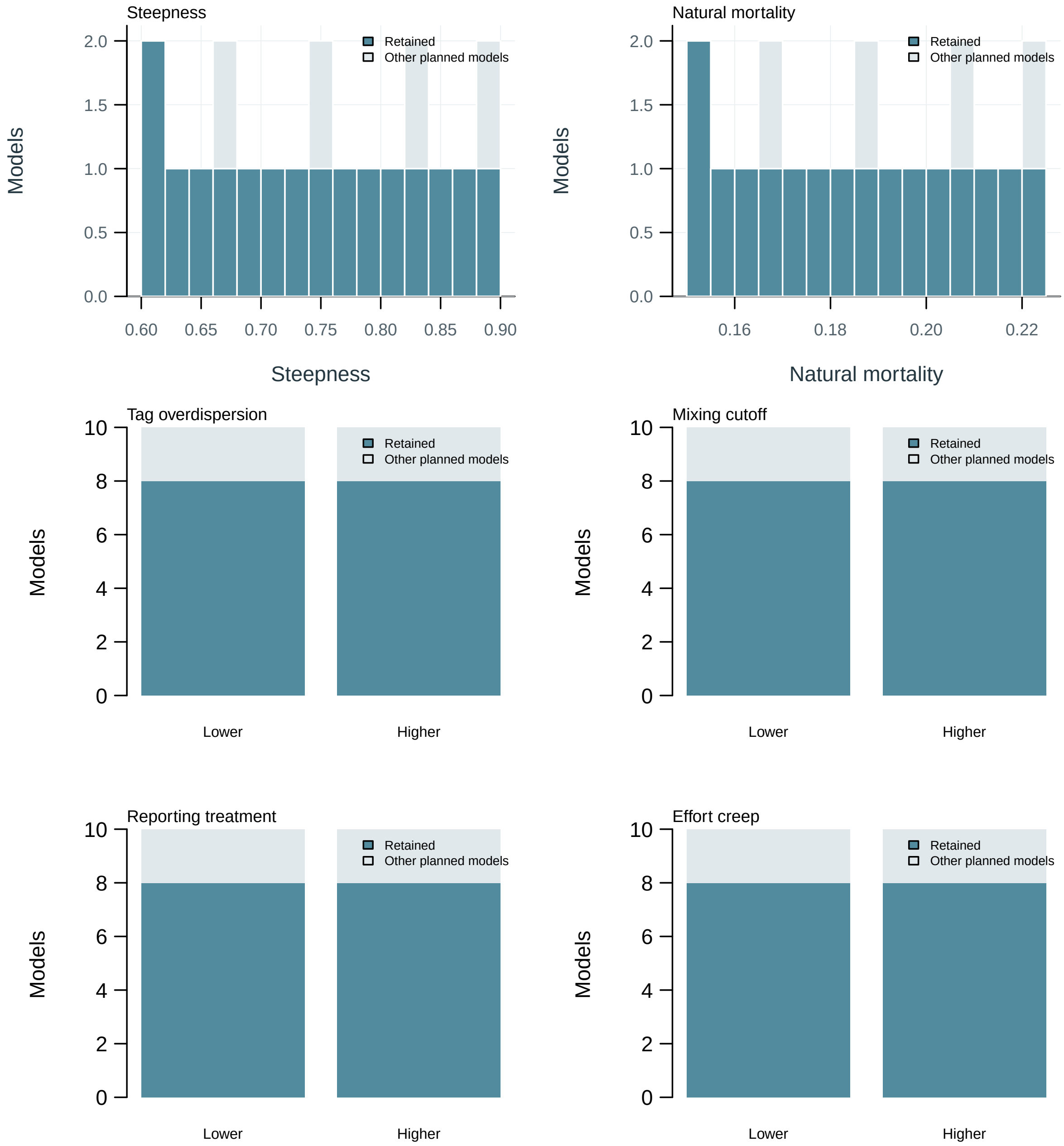

Figure 60: Planned and retained model inputs.

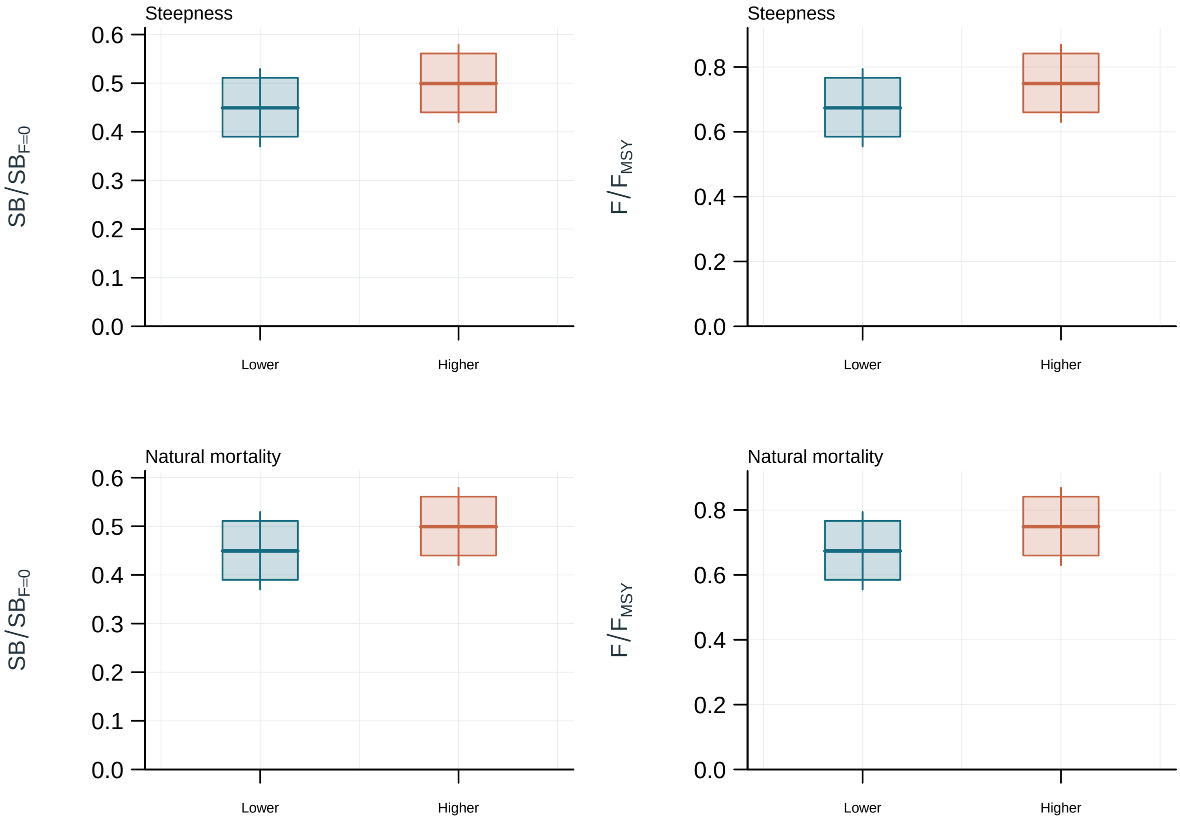

Figure 61: Weighted distributions of management quantities by design setting. Boxes: interquartile range; line: median; whiskers: 2.5th and 97.5th weighted percentiles.

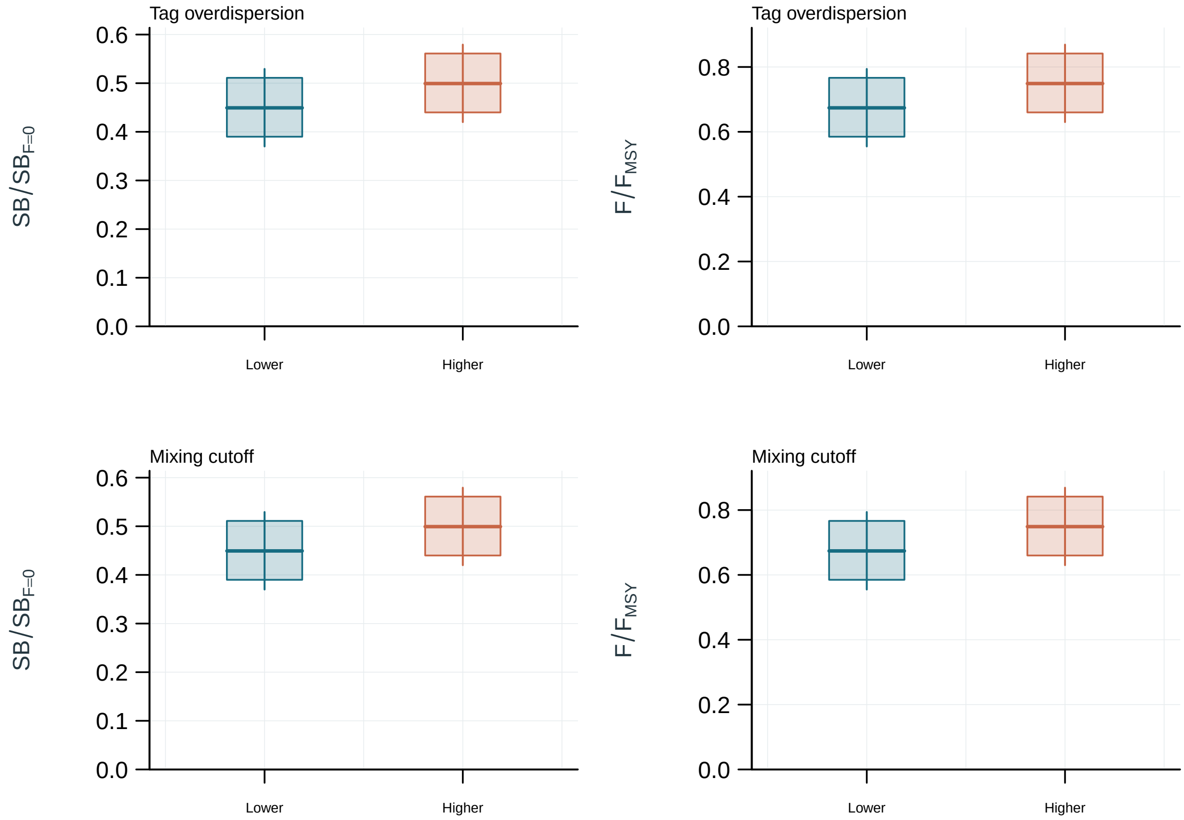

Figure 62: Weighted distributions of management quantities by design setting. Boxes: interquartile range; line: median; whiskers: 2.5th and 97.5th weighted percentiles.

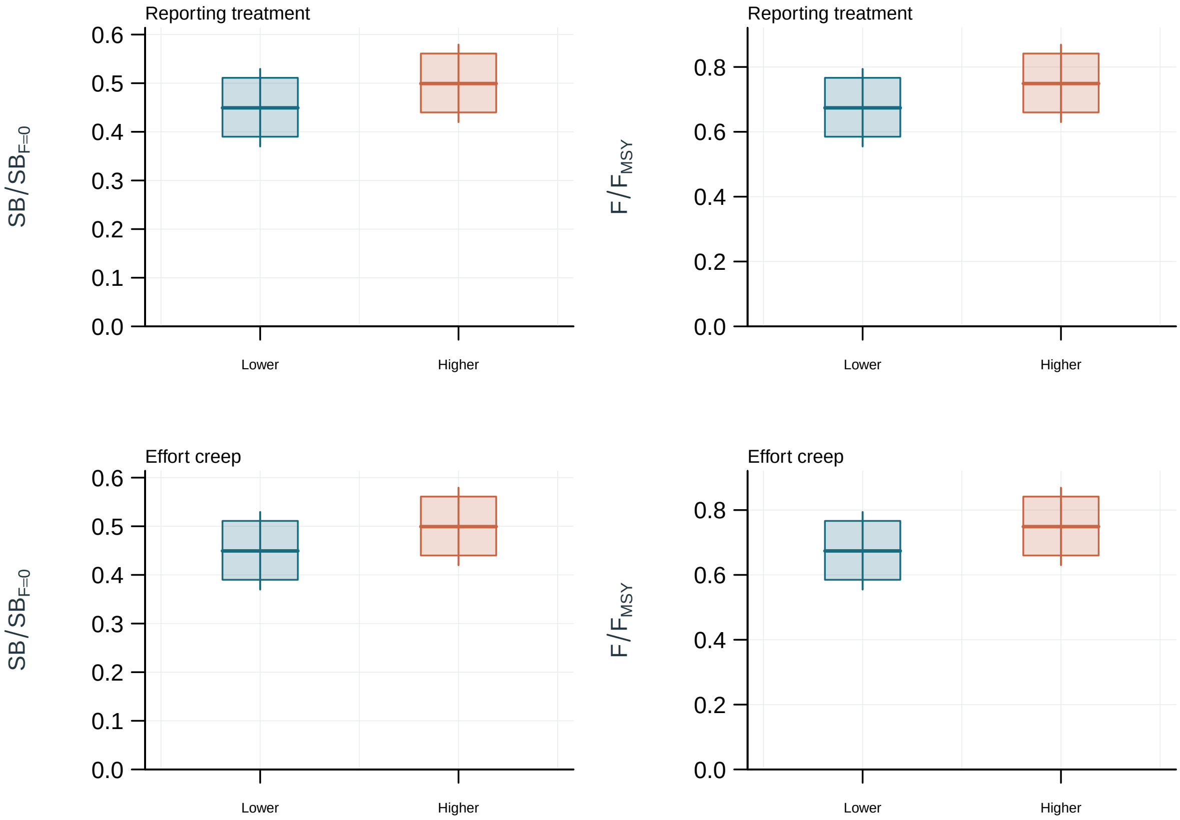

Figure 63: Weighted distributions of management quantities by design setting. Boxes: interquartile range; line: median; whiskers: 2.5th and 97.5th weighted percentiles.

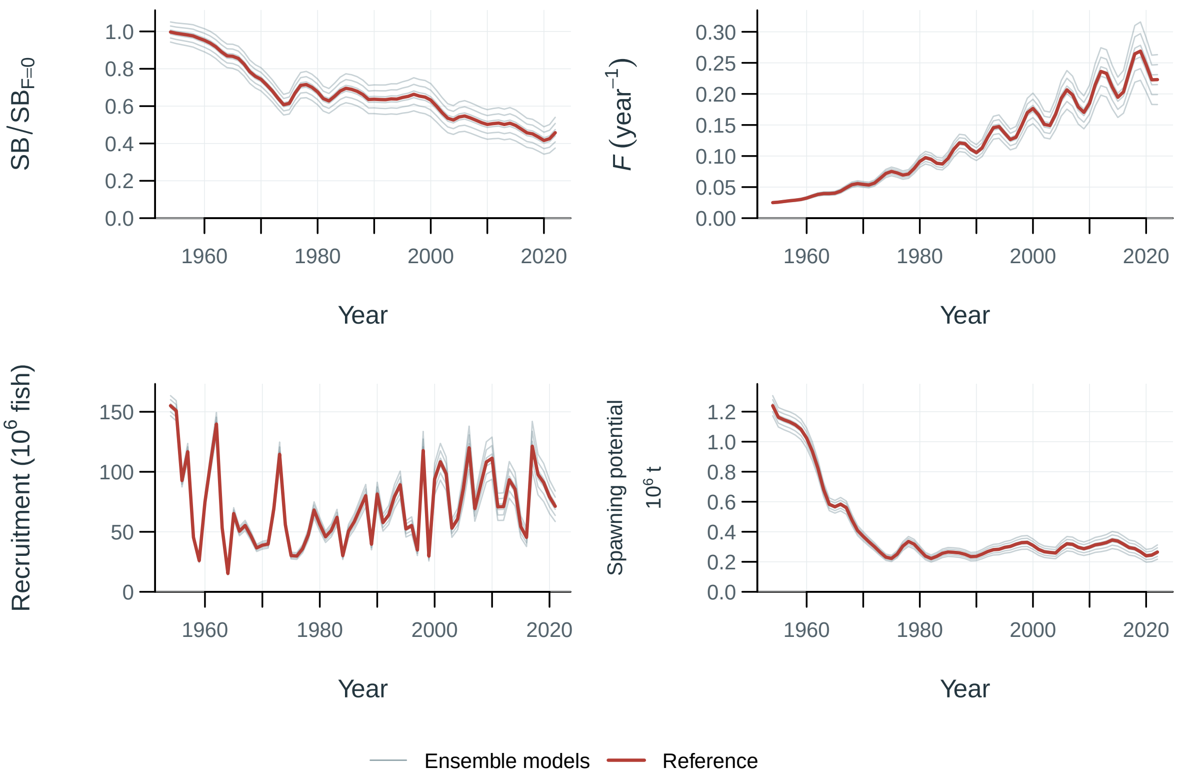

Figure 64: Population trajectories across ensemble models (thin lines) and the reference model (red).

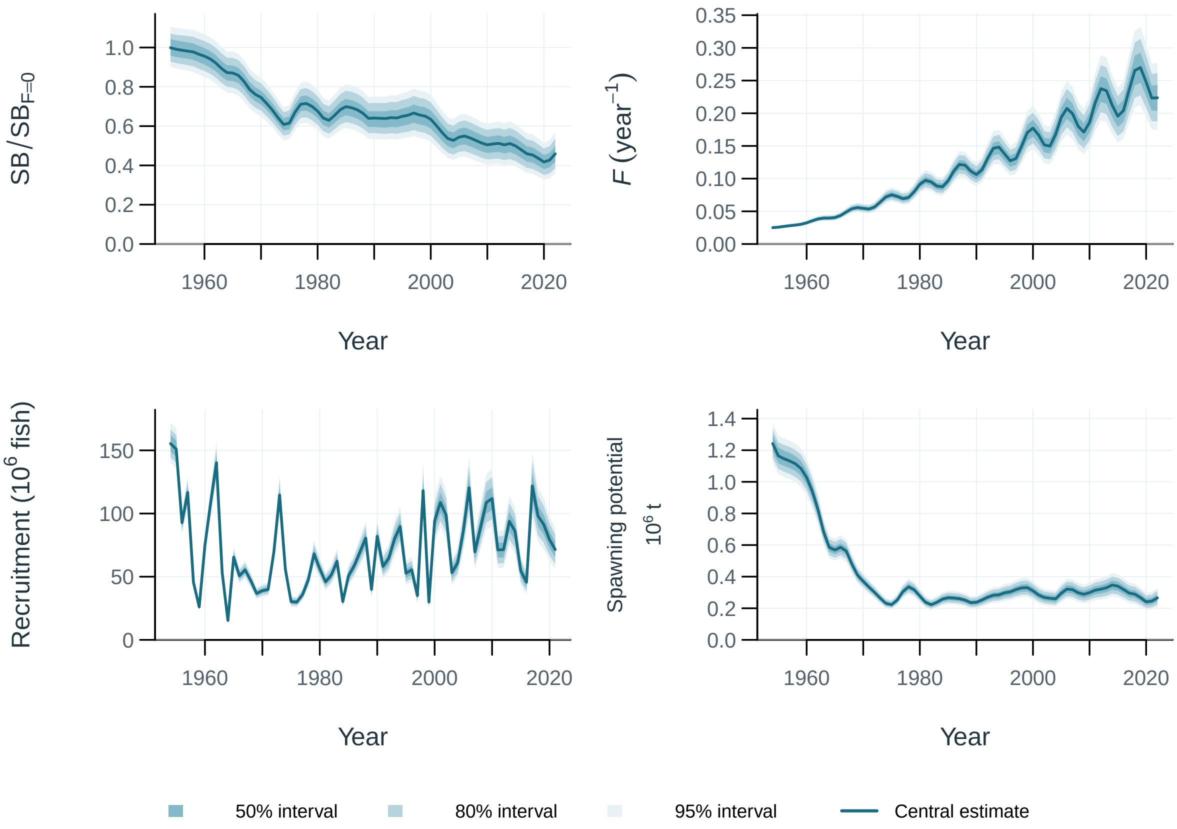

Figure 65: Population trajectories with supplied uncertainty intervals. Central 50, 80, 95% intervals (illustrative model spread and parameter perturbations).

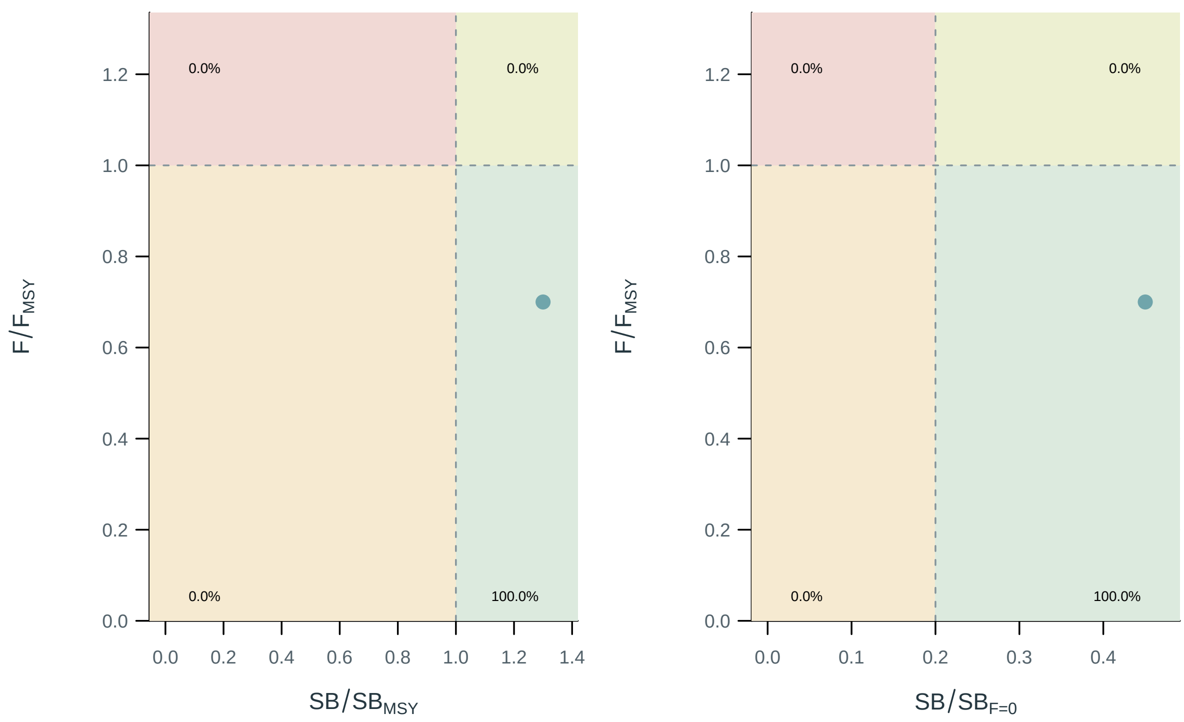

Figure 66: Stock status relative to the supplied spawning and fishing reference points.

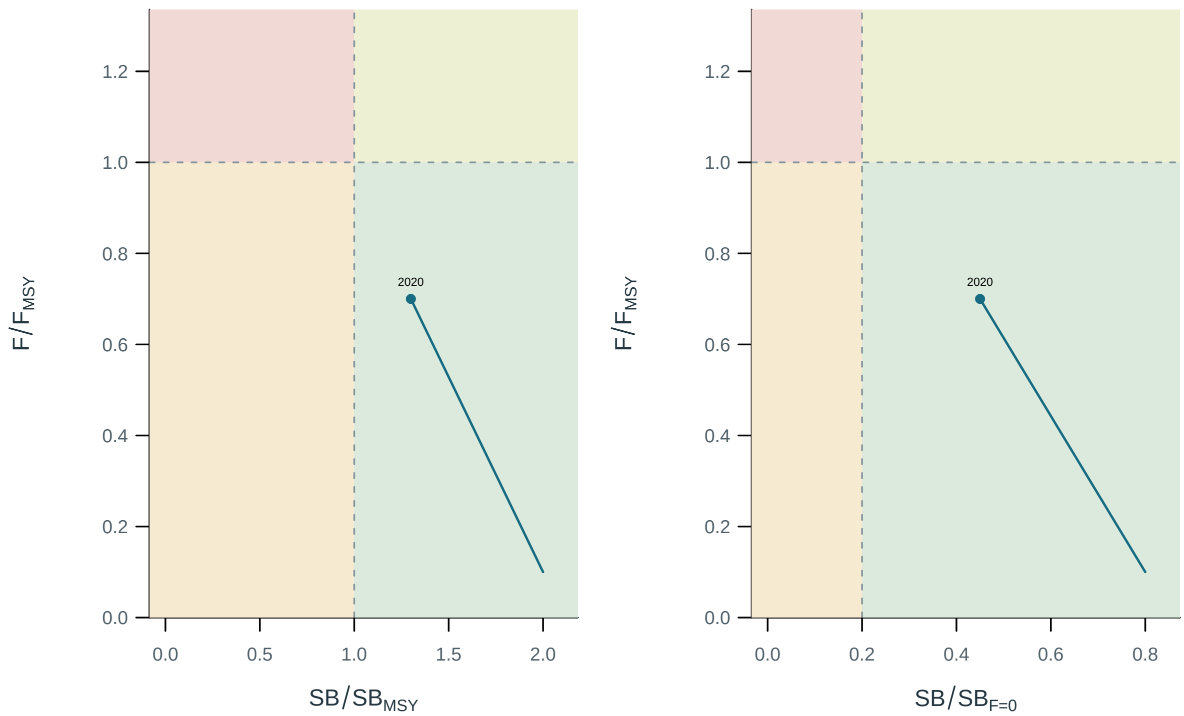

Figure 67: Stock status relative to the supplied spawning and fishing reference points.

Figure 68: Dynamic maximum sustainable yield and catch.

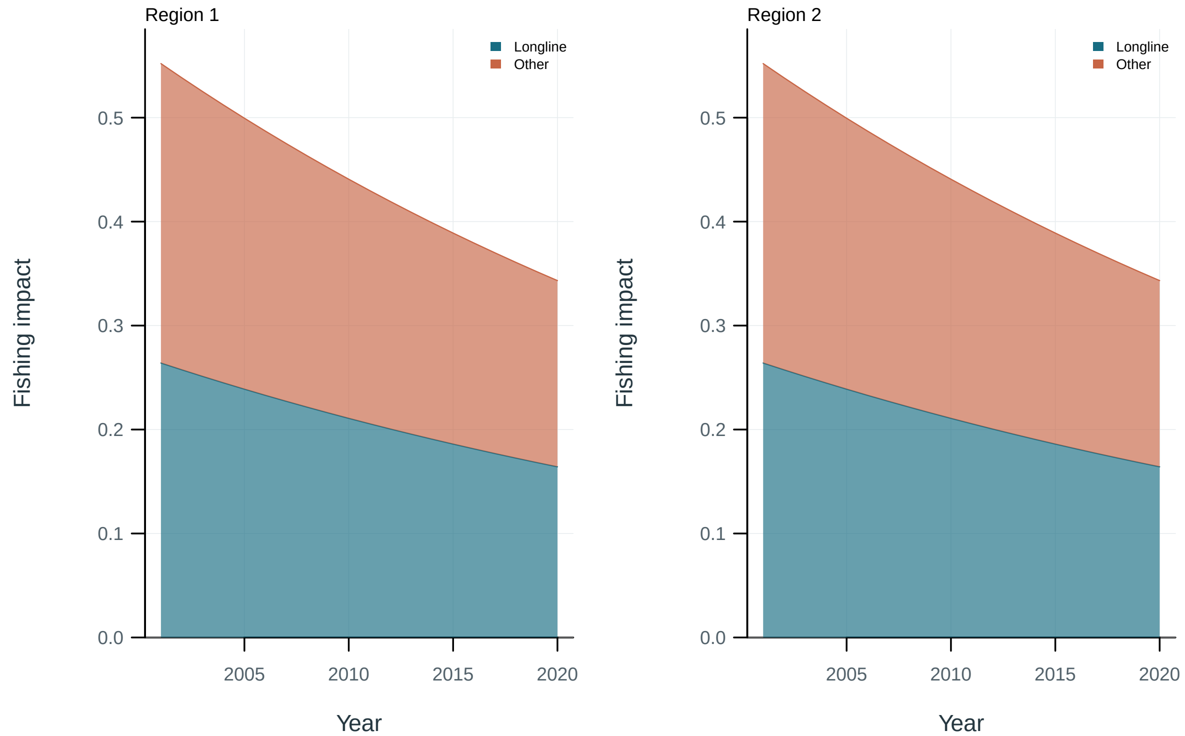

Figure 69: Illustrative output. Contributions of fishing groups to reduced spawning potential.

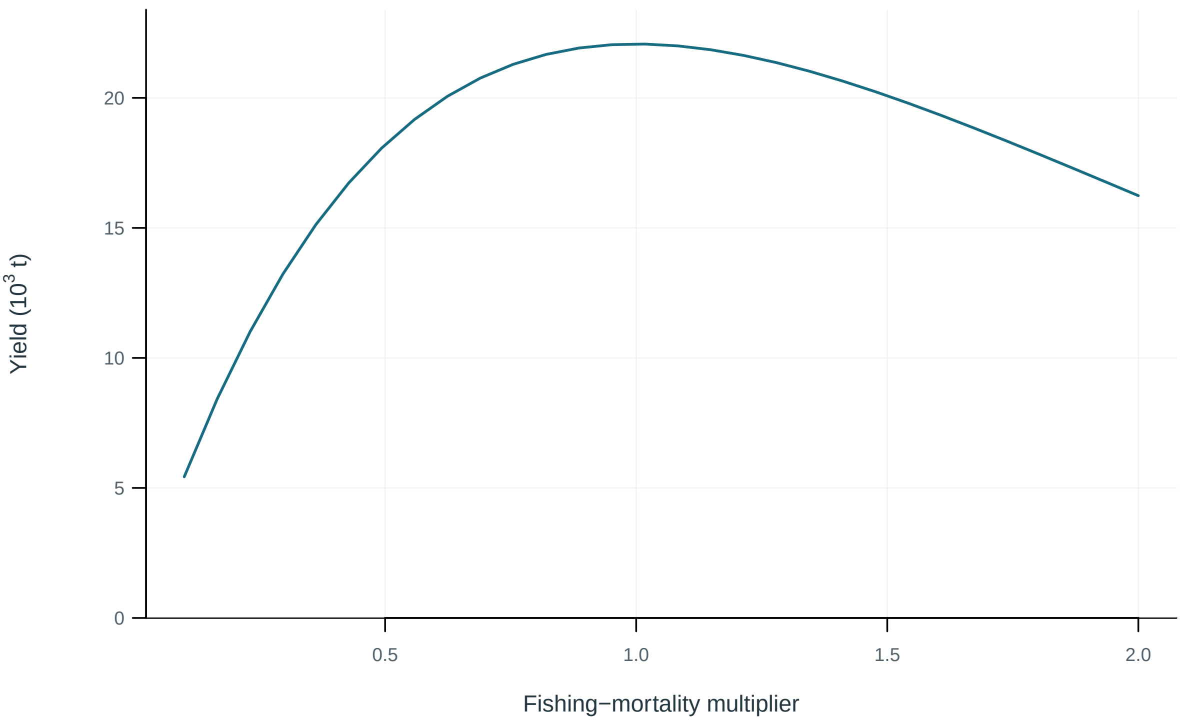

Figure 70: Equilibrium yield across fishing-mortality multipliers.

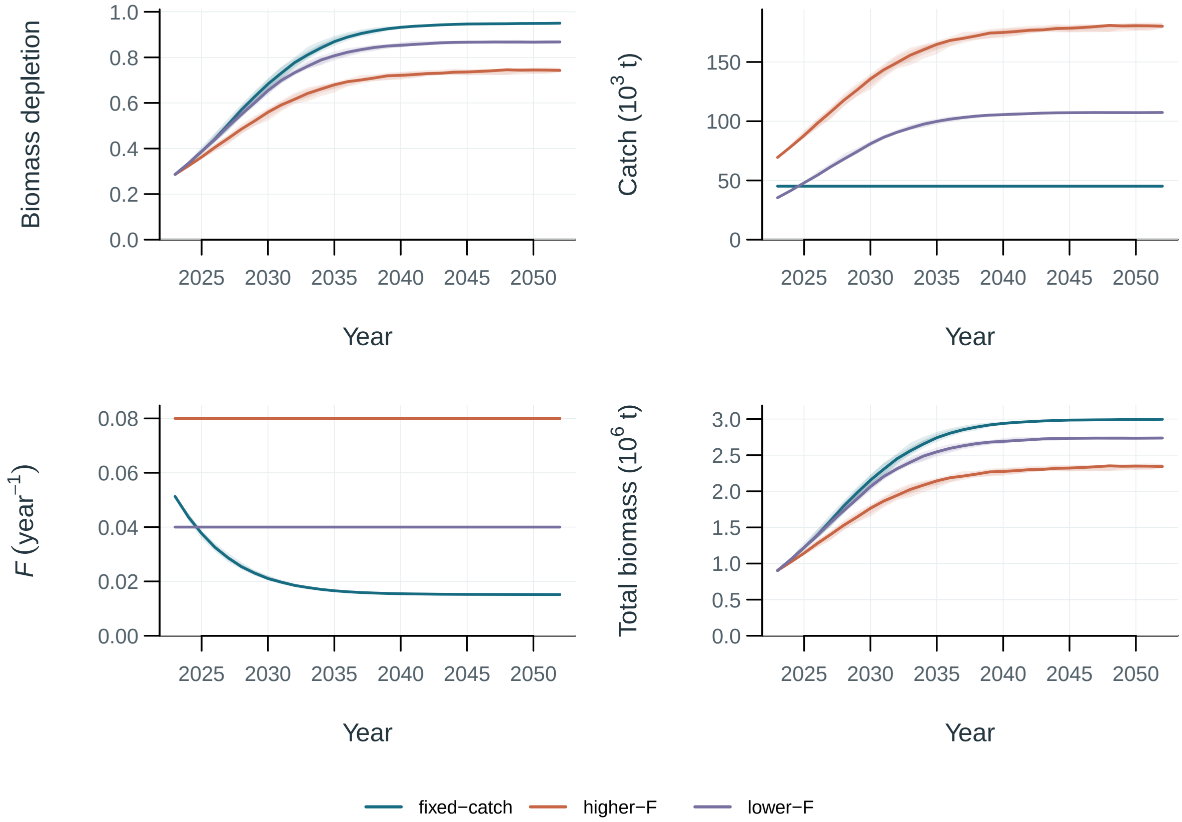

Figure 71: Illustrative surplus-production projections under fixed catch and two F scenarios. Lines are medians; bands span the central 50%, 80% and 95% of production-variability replicates.

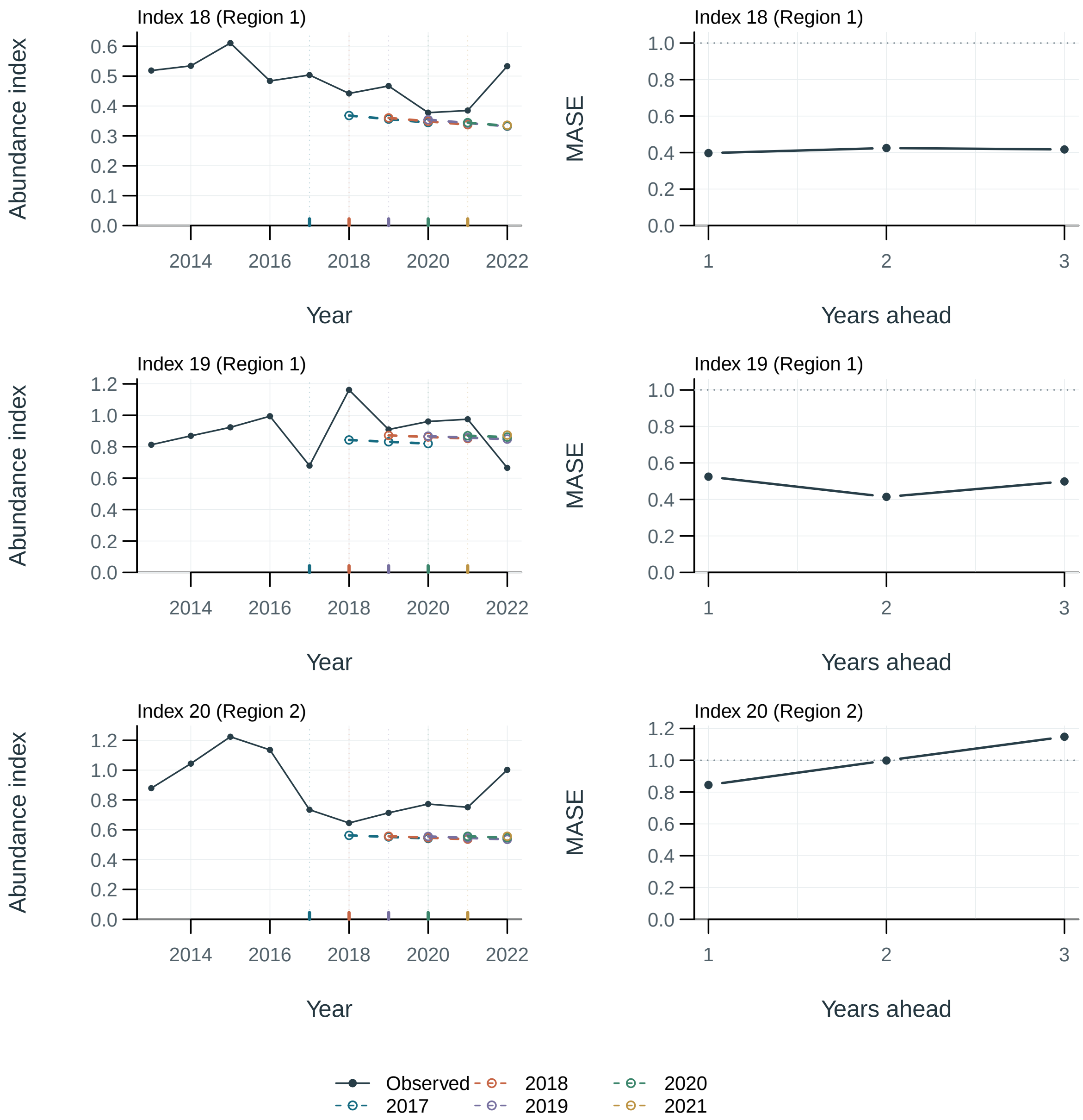

Figure 72: Observations (black) and held-out predictions (coloured dashed lines); vertical marks identify training cutoffs. Right panels show mean absolute scaled error (MASE) by forecast lead. The dotted line marks the training naive-error scale of one; lower values indicate smaller errors.

11 Supplementary outputs

ALB · Diagnostic model interactive comparison

ALB · Diagnostic model figures and tables

ALB · Native MFCL and KIMSA interactive comparison

ALB · Native MFCL and KIMSA figures and tables

ALB · Starting-value trials interactive comparison

ALB · Starting-value trials figures and tables

ALB · Self-test recovery interactive comparison

ALB · Self-test recovery figures and tables

ALB · Retrospective analysis interactive comparison

ALB · Retrospective analysis figures and tables

ALB · Sensitivities interactive comparison

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ALB · Uncertainty ensemble interactive comparison

ALB · Uncertainty ensemble figures and tables

Objective profiles · Display example interactive comparison

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Additional output formats · Display examples interactive comparison

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Hindcasting interactive comparison

Hindcasting figures and tables

Projection scenarios interactive comparison

Projection scenarios figures and tables

12 Appendix: supplementary methods

13 References